

MATHEMATICS 2024

(Science Pre-Engineering Group)

TIME: 3 Hours

(100 Marks)

SECTION 'A' (Multiple Choice Questions)(40)

NOTE: i) This section consists of 20 part questions and all are to be answered. Each part question carries one mark.
 ii) Do not copy the part questions in your answer script. Write only the answer in full against the proper number of the question and its part.
 iii) All notations are used in their usual meanings. The use of Scientific Calculator is allowed.

1. Choose the correct answer for each from the given options:

i) Multiplicative inverse of $(-3, 4)$ is:

* $\left(\frac{3}{25}, \frac{4}{25}\right)$ * $\left(\frac{3}{25}, -\frac{4}{25}\right)$ ✓ * $\left(-\frac{3}{25}, -\frac{4}{25}\right)$ * $(1, 0)$

ii) Real part of $\frac{2+i}{i}$ is:

* -1 * 1 ✓ * $\frac{1}{2}$ * 2

iii) A square matrix A is skew - symmetric if :

* $A^t = A$ * $(\bar{A})^t = A$ * $A^t = -A$ ✓ * $A^t = |A|$

iv) If $A = [-7]$, then $|A| =$:

* 4 * 3 * 7 * -7 ✓

v) The number of non-zero rows in echelon form of matrix is called:

* Order of matrix * Leading rows
 * Leading columns * Rank of a matrix ✓

vi) If three vectors $\vec{a}, \vec{b}$ and $\vec{c}$ are coplanar, then $[\vec{a} \vec{b} \vec{c}] =$:

* 0 ✓ * $\vec{a}$ * $\vec{b}$ * $\vec{c}$

vii) $\hat{k} \times \hat{j} =$:

* $\hat{i}$ ✓ * $-\hat{i}$ * $\hat{j}$ * $-\hat{j}$

viii) No term in G.P is:

* 1 * -1 * 0 ✓ * i

ix) If $a = 3, d = 7$ and $a_n = 59$ then n is equal to :

* 2 * 4 * 9 ✓ * 11

x) $\sum n$ is equal to:

* $\frac{n(n+1)(2n+1)}{6}$ * $\frac{n(n+1)}{2}$ ✓ * $\left[\frac{n(n+1)}{2}\right]^2$ * $\frac{n(n+1)}{4}$

xi) If $n = r$, then ${}^n P_r =$:

* $(n-r)!$ * 1 ✓ * 0 * $n!$

$x + x^2 + x^3 + \dots = ?$
 $\frac{1}{1-x}$
 $\frac{1}{1-x^2}$
 $\frac{1}{1-x^3}$
 $\frac{1}{1-x^4}$

one function is called:
 one function
 one function

$f(x) = |x| - x^2$ is a/an:
 * Even ✓

$\cos \frac{\theta}{2} = ?$
 * $1 + \cos \theta$

$f(x) = 2x + 5$, then find $f(2)$:
 * 11

region of solution in L.P problem is called:

ble region
 region

any triangle ABC, Area of triangle is:

* $abs \sin \gamma$ * $\frac{1}{2} abs \sin \alpha$ * $\frac{1}{2} abs \sin \beta$

period of $\sin \frac{x}{2}$ is:

* 2π ✓ * 3π * 4π

$\sin^{-1} \left(-\frac{1}{2} \right) = ?$

* $\frac{\sqrt{3}}{2}$ * $\frac{2}{\sqrt{3}}$ ✓ * $-\frac{2}{\sqrt{3}}$

SECTION 'B' (Short-Answer Questions) (40)

Attempt any TEN part questions in all. All questions

equal marks. (i.e. 4 marks of each part).

the following complex equation.

$(x, y)(2, 3) = (-4, 7)$

$(x, y)(2, 3) = (-4, 7)$

$\therefore (a, b)(c, d) = (ac - bd, ad + bc)$

$\{(x)(2) - (y)(3), (x)(3) + (y)(2)\} = (-4, 7)$

$2x - 3y, 3x + 2y = -4, 7$

ing corresponding sides we get

$2x - 3y = -4$, $3x + 2y = 7$

$2x - 3y = -4$ --- (i)

$3x + 2y = 7$ --- (ii)

Equation (i)

$2x - 3y = -4$

$$2x = -4 + 3y$$

$$x = \frac{-4 + 3y}{2} \quad \text{--- (iii)}$$

On putting the value of 'x' in equation (ii) we get,

$$3\left(\frac{-4 + 3y}{2}\right) + 2y = 7$$

$$\frac{-12 + 9y + 4y}{2} = 7$$

$$-12 + 13y = 14$$

$$13y = 14 + 12$$

$$13y = 26$$

$$y = \frac{26}{13}$$

$$y = 2$$

On putting the value of 'y' in equation (iii) we get,

$$x = \frac{-4 + 3(2)}{2}$$

$$x = \frac{-4 + 6}{2}$$

$$x = \frac{2}{2}$$

$$x = 1$$

$$x = 1, y = 2 \text{ Answer}$$

ii) If $A = \begin{bmatrix} -2 & 1 & 0 \\ -1 & 4 & 3 \\ 0 & 8 & 5 \end{bmatrix}$ then find $A^2 - 5A + 4I$

SOLUTION:

$$A = \begin{bmatrix} -2 & 1 & 0 \\ -1 & 4 & 3 \\ 0 & 8 & 5 \end{bmatrix}$$

$$A^2 = A.A = \begin{bmatrix} -2 & 1 & 0 \\ -1 & 4 & 3 \\ 0 & 8 & 5 \end{bmatrix} \begin{bmatrix} -2 & 1 & 0 \\ -1 & 4 & 3 \\ 0 & 8 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} (-2)(-2) + (1)(-1) + (0)(0) & (-2)(1) + (1)(4) + (0)(8) & (-2)(0) + (1)(3) \\ (-1)(-2) + (4)(-1) + (3)(0) & (-1)(1) + (4)(4) + (3)(8) & (-1)(0) + (4)(3) \\ (0)(-2) + (8)(-1) + (5)(0) & (0)(1) + (8)(4) + (5)(8) & (0)(0) + (8)(3) \end{bmatrix}$$

$$= \begin{bmatrix} 4 - 1 + 0 & -2 + 4 + 0 & 0 + 3 + 0 \\ 2 - 4 + 0 & -1 + 16 + 24 & 0 + 12 + 15 \\ 0 - 8 + 0 & 0 + 32 + 40 & 0 + 24 + 25 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 3 & 2 & 3 \\ -2 & 39 & 27 \\ -8 & 72 & 49 \end{bmatrix}$$

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$$\begin{aligned}
 &= \begin{vmatrix} 3 & 2 & 3 \\ -2 & 39 & 27 \\ -8 & 72 & 49 \end{vmatrix} - 5 \begin{vmatrix} -2 & 1 & 0 \\ -1 & 4 & 3 \\ 0 & 8 & 5 \end{vmatrix} + 4 \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} \\
 &= \begin{vmatrix} 3 & 2 & 3 \\ -2 & 39 & 27 \\ -8 & 72 & 49 \end{vmatrix} - \begin{vmatrix} -10 & 5 & 0 \\ -5 & 20 & 15 \\ 0 & 40 & 25 \end{vmatrix} + \begin{vmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{vmatrix} \\
 &= \begin{vmatrix} 3+10+4 & 2-5+0 & 3+0+0 \\ -2+5+0 & 39-20+4 & 27-15+0 \\ -8+0+0 & 72-40+0 & 49-25+4 \end{vmatrix} \\
 &= \begin{vmatrix} 17 & -3 & 3 \\ 3 & 23 & 12 \\ -8 & 32 & 28 \end{vmatrix} \text{ Answer}
 \end{aligned}$$

Using properties of determinant prove that:

$$\begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix} = 4abc$$

SOLUTION:

$$\begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix} = 4abc$$

Using L.H.S.:

$$= \begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$$

Subtracting R_2 and R_3 from R_1

$$= \begin{vmatrix} b+c-b-c & a-c-a-c & a-b-a-b \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$$

$$= \begin{vmatrix} 0 & -2c & -2b \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$$

Taking Common -2 from R_1

$$= -2 \begin{vmatrix} 0 & c & b \\ b & c+a & b \\ c & c & a+b \end{vmatrix}$$

Subtracting R_1 from R_2 and R_3

$$= -2 \begin{vmatrix} 0 & c & b \\ b-0 & c+a-c & b-b \\ c-0 & c-c & a+b-b \end{vmatrix}$$

$$= -2 \begin{vmatrix} 0 & c & b \\ b & a & 0 \\ c & 0 & a \end{vmatrix}$$

Expanding by R_1

$$\begin{aligned}
 &= -2 \left\{ 0 \begin{vmatrix} a & 0 \\ 0 & a \end{vmatrix} - c \begin{vmatrix} b & 0 \\ c & a \end{vmatrix} + b \begin{vmatrix} b & a \\ c & 0 \end{vmatrix} \right\} \\
 &= -2 \{ 0(a \times a - 0) - c(b \times a - 0 \times c) + b(b \times 0 - a \times c) \} \\
 &= -2 \{ 0(a^2) - c(ab) + b(-ac) \} \\
 &= -2(0 - abc - abc) \\
 &= -2(-2abc) \\
 &= 4abc
 \end{aligned}$$

L.H.S = R.H.S Hence proved

iv) Calculate the projection of $\vec{b}$ along $\vec{a}$ when $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$ and $3\hat{i} - \hat{j} - 5\hat{k}$

SOLUTION:

$$\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$$

$$\vec{b} = 3\hat{i} - \hat{j} - 5\hat{k}$$

$$\begin{aligned}
 \therefore \text{Projection of } \vec{b} \text{ along } \vec{a} &= \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|} \\
 \frac{\vec{a} \cdot \vec{b}}{|\vec{a}|} &= \frac{(\hat{i} - 2\hat{j} + 3\hat{k})(3\hat{i} - \hat{j} - 5\hat{k})}{\sqrt{(1)^2 + (-2)^2 + (3)^2}} \\
 &= \frac{(1 \times 3)(\hat{i}^2) + (-2 \times -1)(\hat{j}^2) + (3 \times -5)(\hat{k}^2)}{\sqrt{1 + 4 + 9}} \\
 &= \frac{(3)(1) + (2)(1) + (-15)(1)}{\sqrt{14}} \\
 &= \frac{3 + 2 - 15}{\sqrt{14}}
 \end{aligned}$$

$$\text{Projection of } \vec{b} \text{ along } \vec{a} = \frac{-10}{\sqrt{14}}$$

$$\begin{aligned}
 \therefore \text{Projection of } \vec{a} \text{ along } \vec{b} &= \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} \\
 \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} &= \frac{(\hat{i} - 2\hat{j} + 3\hat{k})(3\hat{i} - \hat{j} - 5\hat{k})}{\sqrt{(3)^2 + (-1)^2 + (-5)^2}} \\
 &= \frac{(1 \times 3)(\hat{i}^2) + (-2 \times -1)(\hat{j}^2) + (3 \times -5)(\hat{k}^2)}{\sqrt{9 + 1 + 25}} \\
 &= \frac{(3)(1) + (2)(1) + (-15)(1)}{\sqrt{35}} \\
 &= \frac{3 + 2 - 15}{\sqrt{35}}
 \end{aligned}$$

Projection of $\bar{a}$ along $\bar{b} = \frac{-10}{\sqrt{35}}$ Answer
 the Arithmetic, Geometric and Harmonic means between 1 and 9.
 verify that $AH = G^2$

QUESTION:
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$$\begin{aligned} \therefore A.M. = A &= \frac{a+b}{2} \\ A &= \frac{1+9}{2} \\ A &= \frac{10}{2} = 5 \\ \therefore H.M. = H &= \frac{2ab}{a+b} \\ H &= \frac{2(1)(9)}{1+9} \\ H &= \frac{18}{10} = \frac{9}{5} \\ \therefore G.M. = G &= \sqrt{ab} \\ G &= \sqrt{1 \times 9} \\ G &= \sqrt{9} = 3 \end{aligned}$$

VERIFICATION

$$\begin{aligned} AH &= G^2 \\ 5 \times \frac{9}{5} &= (3)^2 \\ 9 &= 9 \\ L.H.S. &= R.H.S \\ \text{Hence verified} \end{aligned}$$

Find the sum of the series $3 + 6 + 21 + 96 + 471 + \dots$ to n

QUESTION:

$$\begin{aligned} &3 + 6 + 21 + 96 + 471 + \dots \text{ to } n \text{ terms} \\ &6 + 21 + 96 + 471 + \dots \text{ to } n \text{ terms} \end{aligned} \quad \left| \begin{array}{l} \text{R.W} \\ a_1 = a = 3, a_2 = 6, a_3 = 21. \end{array} \right.$$

Here, differences of consecutive terms are in G.P.
i.e. 3, 15, 75 ... is in G.P.

Since the differences of consecutive terms are in G.P. so we use method of differences

$$\begin{aligned} S_n &= 3 + 6 + 21 + 96 + 471 + \dots \text{to } n \text{ term} \\ S_n &= \quad 3 + 6 + 21 + 96 + \dots a_n \\ \hline 0 &= 3 + \{3 + 15 + 75 + 375 + \dots \text{to } (n-1)\} - a_n \\ a_n &= 3 + \{3 + 15 + 75 + 375 + \dots \text{to } (n-1)\} \end{aligned}$$

But

$$\{3 + 15 + 75 + 375 + \dots \text{to } (n-1)\} = S_n$$

Therefore

$$a_n = 3 + S_n$$

$$\therefore S_n = \frac{a(r^n - 1)}{r - 1}$$

$$\therefore a_n = 3 + \frac{a(r^{n-1} - 1)}{r - 1}$$

In this case $a = 3$, $r = \frac{15}{3} = 5$ and number of terms = n

$$a_n = 3 + \frac{3(5^{n-1} - 1)}{5 - 1}$$

$$a_n = 3 + \frac{3 \cdot 5^{n-1} - 3}{4}$$

$$a_n = \frac{12 + 3 \cdot 5^{n-1} - 3}{4}$$

$$a_n = \frac{3}{4}(5^{n-1} + 3)$$

$$a_n = \frac{3}{4}(5^n \cdot 5^{-1} + 3)$$

$$a_n = \frac{3}{4}\left(\frac{5^n}{5} + 3\right)$$

$$a_n = \frac{3}{4}\left(\frac{5^n + 15}{5}\right)$$

$$a_n = \frac{3}{20}(5^n + 15)$$

For sum of the series,

$$S_n = \frac{3}{20} \left| \sum_{1}^n 5^n + 15 \sum_{1}^n n^0 \right|$$

$$S_n = \frac{3}{20} [(5^1 + 5^2 + 5^3 + \dots + 5^n) + 15n]$$

Geometrical Progression for $5^1 + 5^2 + 5^3 + \dots + 5^n$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$= \frac{5(5^n - 1)}{5 - 1}$$

$$S_n = \frac{5(5^n - 1)}{4}$$

$$S_n = \frac{3}{20} \left[\frac{5(5^n - 1)}{4} + 15n \right]$$

$$S_n = \frac{3}{20} \left[\frac{5(5^n - 1) + 60n}{4} \right]$$

$$S_n = \frac{15}{20} \left[\frac{(5^n - 1) + 12n}{4} \right]$$

$$S_n = \frac{3}{4} \left[\frac{(5^n - 1) + 12n}{4} \right]$$

$$S_n = \frac{3(5^n - 1) + 36n}{16}$$

$$S_n = \frac{36n + 3(5^n - 1)}{16} \quad \text{Answer}$$

How many words can be formed by 3 vowels and 4 consonants out of 5 vowels and 7 consonants?

SOLUTION:

of vowels = 5

of consonants = 7

of letters in a word = 7

Word formed by 3 vowels out of 5 is $\binom{5}{3}$ and 4 consonants out of 7 is

Also there is no limits of arrangements of letters therefore,

Product Principle, the required number of words are:

$$\begin{aligned} \text{Total No. of words} &= \binom{5}{3} \cdot \binom{7}{4} \cdot 5! = {}^5C_3 \times {}^7C_4 \times 7! \\ &= 10 \times 35 \times 5040 \end{aligned}$$

$$\text{Total No. of words} = 1764000 \text{ words Answer}$$

How many arrangements can be formed using all the letters of the word 'MATHEMATICS'?

SOLUTION:

Total Number of Letter in MATHEMATICS = 11

Total Number of Letter M = 2

Total Number of Letter A = 2

Total Number of Letter T = 2

Total Number of Letter H = 1

Total Number of Letter E = 1

Total Number of Letter I = 1

Total Number of Letter C = 1

Total Number of Letter S = 1

$$\therefore P = \binom{n}{r, s, t} = \frac{n!}{r! s! t!}$$

$$= \frac{11!}{2! \times 2! \times 2! \times 1! \times 1! \times 1! \times 1! \times 1! \times 1! \times 1! \times 1!}$$

$$= \frac{11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{2 \times 2 \times 2 \times 1 \times 1 \times 1 \times 1 \times 1 \times 1 \times 1 \times 1}$$

$$= \frac{19958400}{4}$$

Total Number of Arrangements = 4989600

viii) Using the principle of mathematical induction for every integer n, prove that:

$$\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

SOLUTION:

$$\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$$

Condition I:

For n = 1

$$\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots + \frac{1}{1(1+1)} = \frac{1}{1+1}$$

$$\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots + \frac{1}{2} = \frac{1}{2}$$

So, P(n) is true for n = 1

Condition II:

Assuming the result to be true for n = k, we get the hypothesis

$$\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots + \frac{1}{k(k+1)} = \frac{k}{k+1}$$

R.W.

Adding $\frac{1}{(k+1)(k+2)}$ On both Sides

$$\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots$$

last term $\frac{1}{(k+1)(k+2)}$

Put n = k + 1

$$= \frac{k}{(k+1)(k+1)}$$

$$\frac{1}{(k+1)(k+2)} + \frac{1}{(k+1)(k+2)}$$

$$= \frac{1}{k+1} + \frac{1}{(k+1)(k+2)}$$

$$= \frac{1}{k+1} \left[k + \frac{1}{k+2} \right]$$

$$= \frac{1}{k+1} \left[\frac{k^2 + 2k + 1}{k+2} \right]$$

$$= \frac{1}{k+1} \left[\frac{(k+1)^2}{k+2} \right]$$

$$= \frac{(k+1)}{(k+2)}$$

i.e. $p(n)$ is true for $n = k+1$

by the principle of mathematical induction $P(n)$ is true for all natural numbers n . Answer

and the middle term in the expansion of $\left(\frac{a}{b} - \frac{b}{a}\right)^{18}$

SOLUTION:

$$(A+B)^n = \left(\frac{a}{b} - \frac{b}{a}\right)^{18}$$

$$\text{Here } A = \frac{a}{b}, \quad B = -\frac{b}{a}, \quad n = 18$$

$n = 18$ is an even. So, there is only one middle term, that is T_{10} term.

$$\therefore T_{r+1} = {}^nC_r A^{n-r} B^r$$

$$\text{Let } r+1 = 10 \Rightarrow r = 10-1 \Rightarrow r = 9$$

$$T_{9+1} = {}^{18}C_9 \left(\frac{a}{b}\right)^{18-9} \left(-\frac{b}{a}\right)^9$$

$$T_{10} = \frac{18!}{9!(18-9)!} \left(\frac{a}{b}\right)^9 \left(-\frac{b}{a}\right)^9$$

$$= \frac{18 \times 17 \times 16 \times 15 \times 14 \times 13 \times 12 \times 11 \times 10 \times 9!}{9! 9!} \left(-\frac{a^9}{b^9} \times \frac{b^9}{a^9}\right)$$

$$= \frac{1.76432256 \times 10^{10}}{362880} (-1)$$

$$T_{10} = -48620 \text{ Answer}$$

find the term independent of 'x' in $\left(2x + \frac{1}{x^2}\right)^9$

SOLUTION:

$$\text{Let } (A + B)^n = \left(2x + \frac{1}{x^2}\right)^9$$

$$\therefore A = 2x, \quad B = \frac{1}{x^2}, \quad n = 9$$

$$T_{r+1} = {}^nC_r A^{n-r} B^r$$

$$T_{r+1} = {}^9C_r (2x)^{9-r} \cdot \left(\frac{1}{x^2}\right)^r$$

$$T_{r+1} = {}^9C_r (2)^{9-r} \cdot (x)^{9-r} \cdot \left(\frac{1}{x^2}\right)^r$$

$$T_{r+1} = {}^9C_r (2)^{9-r} \cdot x^{9-r} \cdot x^{-2r}$$

$$T_{r+1} = {}^9C_r (2)^{9-r} \cdot x^{-2r} \cdot x^{9-r}$$

$$T_{r+1} = {}^9C_r (2)^{9-r} \cdot x^{9-3r} \dots (1)$$

If the term independent of 'x' then power of "x" will be equal zero

$$9 - 3r = 0$$

$$3r = 9$$

$$r = 3$$

Substitute the value of "r" in equation (1) we get

$$T_{3+1} = \frac{{}^9C_3 (2)^{9-3} \cdot x^{9-3(3)}}{9!}$$

$$T_4 = \frac{9!}{3!(9-3)!} \cdot (2)^6 \cdot x^0$$

$$T_4 = \frac{9 \times 8 \times 7 \times 6!}{3! 6!} \cdot (64) \cdot (1)$$

$$T_4 = \frac{32256}{6}$$

$$T_4 = \frac{32256}{6}$$

$$T_4 = 5376$$

Hence the 9th independent term of "x" is 5376.

x) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by $f(x) = \sqrt{x+7}; x \geq -7$
the value of $f^{-1}(x)$

SOLUTION:

$$f(x) = \sqrt{x+7}$$

Let

$$f(x) = \sqrt{x+7} = y$$

$$\sqrt{x+7} = y$$

On squaring both sides, we get

$$x+7 = y^2$$

$$x = y^2 - 7$$

Replacing x by $f^{-1}(y)$, we get

$$f^{-1}(y) = y^2 - 7$$

and $f^{-1}(x)$, replace y by x , i.e.

$$f^{-1}(x) = x^2 - 7$$

There is no variable in the denominator,
the domain of $f(x) = \{x | x \in \mathbb{R}\}$

$$f(x) = x + 7$$

$$f(-7) = -7 + 7 = 0$$

$$f(x) \geq 0 \forall x \geq 7$$

$$\therefore \text{Range} = \{y | y \in \mathbb{R}, \wedge y \geq -7\} = \mathbb{R}^+ \text{ Answer}$$

If $A + B + C = 180^\circ$ then prove that $\cos\left(\frac{B+C}{2}\right) = \sin\frac{A}{2}$

SOLUTION:

Given that

$$A + B + C = 180^\circ$$

$$A = 180^\circ - (B + C)$$

Divide equation by '2'

$$\frac{A}{2} = \frac{180^\circ}{2} - \frac{(B + C)}{2}$$

$$\frac{A}{2} = 90^\circ - \left(\frac{B + C}{2}\right)$$

Apply $\sin$ on both sides

$$\sin\frac{A}{2} = \sin\left\{90^\circ - \left(\frac{B + C}{2}\right)\right\}$$

$$\because \sin(90^\circ - \theta) = \cos \theta$$

$$\sin\frac{A}{2} = \cos\left(\frac{B + C}{2}\right)$$

Hence Proved

Find the smallest angle in $\triangle ABC$ when $a = 25\text{cm}$, $b = 18\text{cm}$, $c = 21\text{cm}$

SOLUTION:

$$a = 25\text{ cm}, \quad b = 18\text{ cm}, \quad c = 21\text{ cm}$$

Let α, β and γ be the three angles of $\triangle ABC$ then,
know that the smallest angle is opposite to the smallest sides of the triangle, in this case, the smallest side is 'b', therefore

Using Law of cosine

$$b^2 = a^2 + c^2 - 2ac \cos \beta$$

$$2ac \cos \beta = a^2 + c^2 - b^2$$

$$\cos \beta = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\cos \beta = \frac{(25)^2 + (21)^2 - (18)^2}{2(25)(21)}$$

$$\cos \beta = \frac{625 + 441 - 324}{1050}$$

$$\cos \beta = \frac{1050}{1050}$$

$$\cos \beta = 0.7066$$

$$\beta = \cos^{-1}(0.7066)$$

$$\beta = 45.0356^\circ \text{ Answer}$$

The smallest angle in ΔABC is β Answer

xiii) Prove that $\frac{1}{r^2} + \frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} = \frac{a^2 + b^2 + c^2}{2}$

SOLUTION:

$$\frac{1}{r^2} + \frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} = \frac{a^2 + b^2 + c^2}{2}$$

Taking L.H.S.:

$$\begin{aligned} &= \frac{1}{r^2} + \frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} \\ &= \frac{1}{\left(\frac{A}{s}\right)^2} + \frac{1}{\left(\frac{A}{s-a}\right)^2} + \frac{1}{\left(\frac{A}{s-b}\right)^2} + \frac{1}{\left(\frac{A}{s-c}\right)^2} \\ &= \frac{s^2}{A^2} + \frac{(s-a)^2}{A^2} + \frac{(s-b)^2}{A^2} + \frac{(s-c)^2}{A^2} \\ &= \frac{s^2 + (s-a)^2 + (s-b)^2 + (s-c)^2}{A^2} \\ &= \frac{s^2 + s^2 - 2as + a^2 + s^2 - 2bs + b^2 + s^2 - 2cs + c^2}{A^2} \\ &= \frac{4s^2 + a^2 + b^2 + c^2 - 2as - 2bs - 2cs}{A^2} \\ &= \frac{4s^2 + a^2 + b^2 + c^2 - 2s(a+b+c)}{A^2} \\ &\therefore s = \frac{a+b+c}{2} \Rightarrow 2s = a+b+c \end{aligned}$$

$$\frac{4s^2 + a^2 + c^2 + b^2 - 2s(2s)}{2}$$

$$= \frac{4s^2 - 4s^2 + a^2 + c^2 + b^2}{2}$$

$$= \frac{a^2 + c^2 + b^2}{2}$$

L.H.S = R.H.S

Hence Proved

Solve the equation $\cos \theta - \theta = 0$ graphically for interval $-\frac{\pi}{2} \leq \theta \leq$

SOLUTION:

$$\cos \theta - \theta = 0$$

$$\cos \theta = \theta \quad \text{--- (i)}$$

$$\text{Let } \cos \theta = y = \theta \quad \text{--- (ii)}$$

$$y = \cos \theta$$

Put $\theta = -\frac{\pi}{2}$

$$y = \cos\left(-\frac{\pi}{2}\right)$$

$$y = 0$$

Put $\theta = -\frac{\pi}{4}$

$$y = \cos\left(-\frac{\pi}{4}\right)$$

$$y = \frac{1}{\sqrt{2}} \approx 0.71$$

$$\theta = 0$$

$$y = \cos(0)$$

$$y = 1$$

Put $\theta = \frac{\pi}{4}$

$$y = \cos\left(\frac{\pi}{4}\right)$$

$$y = \frac{1}{\sqrt{2}} \approx 0.71$$

Put $\theta = \frac{\pi}{2}$

$$y = \cos\left(\frac{\pi}{2}\right)$$

$$y = 0$$

θ	$-\frac{\pi}{2}$	$-\frac{\pi}{4}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$
$\cos \theta$	0	0.71	1	0.71	0
$y = \theta$	0	0.71	1	0.71	0

Put $\theta = -\frac{\pi}{2}$

$$y = -1.57$$

Put $\theta = \frac{\pi}{4}$

$$y = 0.71$$

$$\text{Put } \theta = -\frac{\pi}{4}$$

$$y = -0.78$$

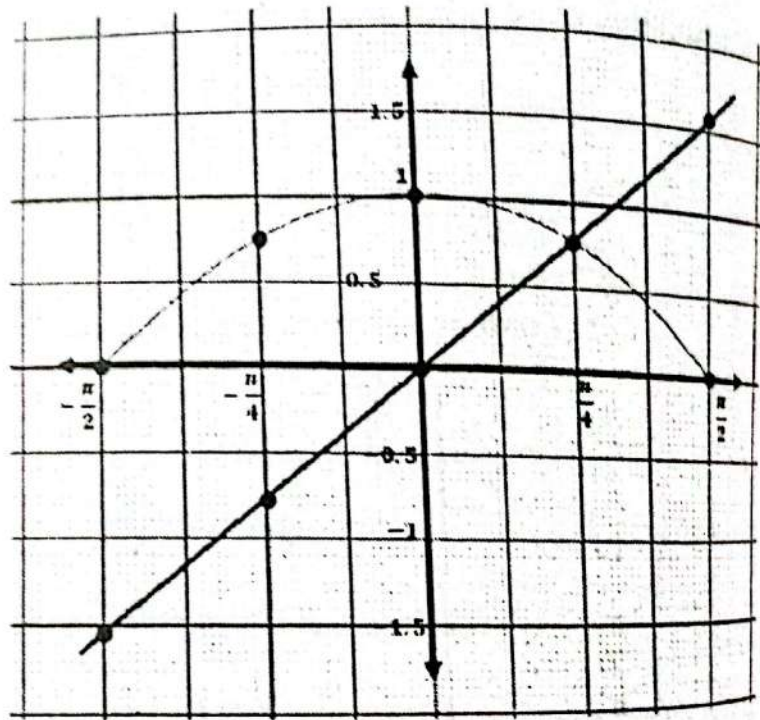
$$\text{Put } \theta = 0$$

$$y = 0$$

$$\text{FOR } y = -\frac{x}{\pi}$$

θ	$-\frac{\pi}{2}$	$-\frac{\pi}{4}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$
$y = \theta$	-1.57	-0.78	0	0.78	1.57

GRAPH



The point of intersection of these equation is $\frac{\pi}{4}$. Thus solution is

$$\theta \approx \frac{\pi}{4}$$

$$\text{Solution} \approx \frac{\pi}{4}$$

xiv) (OR) Show that $2\tan^{-1}\left(\frac{2}{3}\right) = \cos^{-1}\left(\frac{5}{13}\right)$

SOLUTION:

$$2\tan^{-1}\left(\frac{2}{3}\right) = \cos^{-1}\left(\frac{5}{13}\right)$$

Taking L.H.S.:

$$= 2\tan^{-1}\left(\frac{2}{3}\right)$$

$$= \tan^{-1}\left(\frac{2}{3}\right) + \tan^{-1}\left(\frac{2}{3}\right)$$

$$\therefore \tan^{-1} A + \tan^{-1} B = \tan^{-1} \left(\frac{A+B}{1-AB} \right)$$

$$\tan^{-1} \left\{ \frac{\frac{2}{3} + \frac{2}{3}}{1 - \left(\frac{2}{3}\right)\left(\frac{2}{3}\right)} \right\}$$

$$\tan^{-1} \left(\frac{\frac{2+2}{3}}{1 - \frac{4}{9}} \right)$$

$$= \tan^{-1} \left(\frac{\frac{4}{3}}{\frac{9-4}{9}} \right)$$

$$= \tan^{-1} \left(\frac{4}{\frac{3}{5}} \right)$$

$$= \tan^{-1} \left(\frac{4}{3} \times \frac{5}{1} \right)$$

$$= \tan^{-1} \left(\frac{12}{5} \right)$$

$$x = \tan^{-1} \left(\frac{12}{5} \right)$$

$$\tan x = \frac{12}{5}$$

$$\sqrt{\tan^2 x} = \sqrt{\left(\frac{12}{5}\right)^2}$$

$$\sqrt{\tan^2 x + 1} = \sqrt{\frac{144}{25} + 1}$$

$$\sqrt{\tan^2 x + 1} = \sqrt{\frac{144 + 25}{25}}$$

$$\therefore \tan^2 \theta + 1 = \sec^2 \theta$$

$$\sqrt{\sec^2 x} = \sqrt{\frac{169}{25}}$$

$$\sec x = \frac{13}{5}$$

$$\frac{1}{\cos x} = \frac{13}{5}$$

$$\cos x = \frac{5}{13}$$

$$x = \cos^{-1}\left(\frac{5}{13}\right)$$

$$\text{But, } x = \tan^{-1}\left(\frac{12}{5}\right) = 2\tan^{-1}\left(\frac{2}{3}\right)$$

$$2\tan^{-1}\left(\frac{2}{3}\right) = \cos^{-1}\left(\frac{5}{13}\right)$$

Hence Proved

L.H.S. = R.H.S.

xv) Find the point of intersection of the following functions graphically
 $f(x) = x + 4$ and $g(x) = x^2 - 6x + 10$

SOLUTION:

$$f(x) = x + 4 \text{ and } g(x) = x^2 - 6x + 10$$

Let

$$y = x + 4$$

and

$$y = x^2 - 6x + 10$$

$$y = x + 4$$

$$\text{Put } x = 1$$

$$y = 1 + 4$$

$$y = 5$$

$$\text{Put } x = 2$$

$$y = 2 + 4$$

$$y = 6$$

$$\text{Put } x = 3$$

$$y = 3 + 4$$

$$y = 7$$

$$\text{Put } x = 6$$

$$y = 4 + 6$$

$$y = 10$$

x	1	2	3	6
y	5	6	7	10

$$y = x^2 - 6x + 10$$

$$\text{Put } x = 1$$

$$y = (1)^2 - 6(1) + 10$$

$$y = 1 - 6 + 10$$

$$y = 5$$

$$\text{Put } x = 2$$

$$y = (2)^2 - 6(2) + 10$$

$$y = 4 - 12 + 10$$

$$y = 2$$

$$\text{Put } x = 3$$

$$y = (3)^2 - 6(3) + 10$$

$$y = 9 - 18 + 10$$

$$y = 1$$

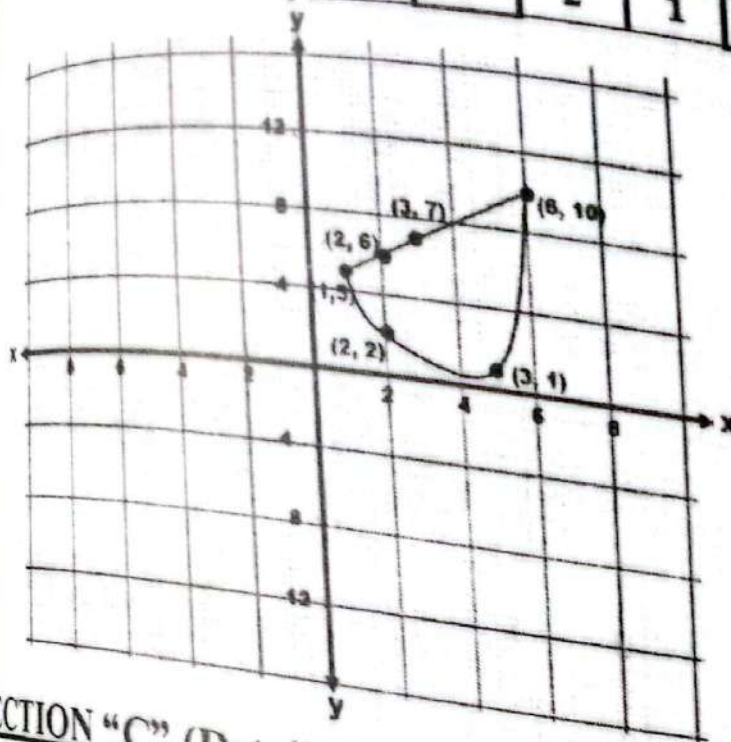
$$\text{Put } x = 6$$

$$y = (6)^2 - 6(6) + 10$$

$$y = 36 - 36 + 10$$

$$y = 10$$

x	1	2	3	6
y	5	2	1	10



SECTION "C" (Detailed Answer Questions)(40)

Attempt any FIVE question from this section.
Questions carry equal marks.

the non-homogenous system of linear equations using Gauss
elimination method:

$$-x + y + z = 0, x + 2y = 5, -3x + 2y - z = -2$$

SOLUTION:

$$-x + y + z = 0$$

$$x + 2y = 5$$

$$-3x + 2y - z = -2$$

Form the system of the linear equation into matrix form of $AX = B$

$$A = \begin{bmatrix} -1 & 1 & 1 \\ 1 & 2 & 0 \\ -3 & 2 & -1 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 0 \\ 5 \\ -2 \end{bmatrix}$$

Form the augmented matrix by including the constant elements as
column in the coefficient matrix

$$\text{Augmented matrix} = A_B = \begin{bmatrix} -1 & 1 & 1 & 0 \\ 1 & 2 & 0 & 5 \\ -3 & 2 & -1 & -2 \end{bmatrix}$$

Converting augmented matrix into echelon form by using elementary row operations.

$$\begin{aligned} A_B &\sim \begin{bmatrix} 1 & -1 & -1 & 0 \\ 1 & 2 & 0 & 5 \\ -3 & 2 & -1 & -2 \end{bmatrix} \text{ by } (-1)R_1 \\ &\sim \begin{bmatrix} 1 & -1 & -1 & 0 \\ 1-1 & 2+1 & 0+1 & 5 \\ -3+3 & 2-3 & -1-3 & -2 \end{bmatrix} \text{ by } R_2 + (-1)R_1 \text{ and } R_3 + (-1)R_1 \\ &\sim \begin{bmatrix} 1 & -1 & -1 & 0 \\ 0 & 3 & 1 & 5 \\ 0 & -1 & -4 & -2 \end{bmatrix} \text{ by } \left(\frac{1}{3}\right)R_2 \\ &\sim \begin{bmatrix} 1 & -1 & -1 & 0 \\ 0 & 1 & \frac{1}{3} & \frac{5}{3} \\ 0 & -1+1 & -4+\frac{1}{3} & -2+\frac{5}{3} \end{bmatrix} \text{ by } R_3 + R_2 \\ &\sim \begin{bmatrix} 1 & -1 & -1 & 0 \\ 0 & 1 & \frac{1}{3} & \frac{5}{3} \\ 0 & 0 & -\frac{11}{3} \left(-\frac{3}{11}\right) & -\frac{1}{3} \left(-\frac{3}{11}\right) \end{bmatrix} \text{ by } \left(-\frac{3}{11}\right)R_3 \\ &\sim \begin{bmatrix} 1 & -1 & -1 & 0 \\ 0 & 1 & \frac{1}{3} & \frac{5}{3} \\ 0 & 0 & 1 & \frac{1}{11} \end{bmatrix} \end{aligned}$$

From the above matrix, we get equivalent matrix equation

$$\begin{aligned} x - y - z &= 0 \text{ --- (i)} \\ 0x + y + \frac{1}{3}z &= \frac{5}{3} \Rightarrow y + \frac{1}{3}z = \frac{5}{3} \text{ --- (ii)} \\ 0x + 0y + z &= \frac{1}{11} \Rightarrow z = \frac{1}{11} \text{ --- (iii)} \end{aligned}$$

Putting the value of 'z' in equation (ii), we get

$$\begin{aligned} y + \frac{1}{3} \left(\frac{1}{11}\right) &= \frac{5}{3} \\ \frac{33y + 1}{33} &= \frac{5}{3} \Rightarrow 33y + 1 = \frac{5 \times 33}{3} \end{aligned}$$

$$33y = 55 - 1 \Rightarrow y = \frac{54}{33}$$

$$y = \frac{18}{11}$$

Putting the value of 'y' and 'z' in equation (i), we get

$$x - \left(\frac{1}{11}\right) - \left(\frac{18}{11}\right) = 0$$

$$\frac{11x - 1 - 18}{11} = 0 \Rightarrow 11x - 19 = 0$$

$$x = \frac{19}{11}$$

$x = \frac{19}{11}$, $y = \frac{18}{11}$, and $z = \frac{1}{11}$ is the required solution Answer

The 3rd term and the 7th term of a H.P are $\frac{2}{9}$ and $\frac{2}{25}$ respectively, the sequence.

SOLUTION:

Ans:

$$\frac{2}{9}$$

$$\frac{2}{25}$$

$$\frac{2}{9}$$

$$\frac{2}{25}$$

terms in A.P are

$$\frac{9}{2}$$

$$a_3 = \frac{2}{9} \text{ and } a_7 = \frac{2}{25}$$

$$\frac{9}{2}$$

$$\frac{2}{25}$$

$$\frac{9}{2}$$

$$\frac{2}{25}$$

$$\frac{9}{2}$$

$$\frac{2}{25}$$

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$$\frac{9}{2}$$

$$\frac{2}{25}$$

$$\frac{9}{2}$$

$$\frac{2}{25}$$

$$\therefore a_n = a + (n - 1)d$$

$$a_3 = a + (3 - 1)d$$

$$\frac{9}{2}$$

$$\frac{2}{9} = a + 2d$$

$$\frac{9}{2}$$

$$a = \frac{9}{2} - 2d \dots (1)$$

$$a_7 = a + (7 - 1)d$$

$$\frac{25}{2}$$

$$\frac{2}{25} = a + (6)d$$

$$\frac{25}{2}$$

$$a = \frac{25}{2} - 6d \dots (2)$$

Comparing equation (1) and (2), we get

$$\frac{9}{2}$$

$$- 2d$$

$$=$$

$$\frac{25}{2}$$

$$- 6d$$

$$=$$

$$\frac{25}{2}$$

$$- 6d$$

$$6d - 2d = \frac{25}{2} - \frac{9}{2}$$

$$4d = \frac{25 - 9}{2}$$

$$d = \frac{16}{8}$$

$$d = 2$$

On putting the value of "d" in equation (1) we get

$$a = \frac{9}{2} - 2(2)$$

$$a = \frac{9 - 8}{2}$$

$$a = \frac{1}{2}$$

Now

$$\therefore a_n = a + (n - 1)d$$

$$a_2 = a + d$$

$$a_2 = \frac{1}{2} + 2 = \frac{1 + 4}{2} = \frac{5}{2}$$

$$a_4 = a + 3d$$

$$a_4 = \frac{1}{2} + 3(2) = \frac{1 + 12}{2} = \frac{13}{2}$$

hence the required H.P. is $\frac{1}{2}, \frac{5}{2}, \frac{13}{2}, \dots$

5. In a three child family, find the probability of:

(a) Three boys

(b) Exactly two boys

(c) At least one girl

SOLUTION:

No. of Family = 3

Possible outcomes of Boy = $O(B) = 3$

Possible outcomes Girl = $O(G) = 3$

Total Possible outcomes = ?

$$\therefore \text{Fundamental Principle} = O(A) \cdot O(B) = O(A \times B)$$

$$\text{Total No. of Outcomes} = O(B \times G)$$

$$= \{BBB, BBG, BGB, BGG, GBB, GBG, GGB, GGG\}$$

Hence

$$\text{Total Possible outcomes} = O(S) = 8$$

$$\therefore (1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots$$

By comparing above expression with equation with given series,

$$nx = \frac{1}{4} \dots \dots (i)$$

Squaring both sides

$$(nx)^2 = \left(\frac{1}{4}\right)^2$$

$$n^2x^2 = \frac{1}{16} \dots \dots (ii)$$

And

$$\frac{n(n-1)}{2!}x^2 = \frac{1 \cdot 3}{4 \cdot 6}$$

$$\frac{n(n-1)}{2 \times 1}x^2 = \frac{3}{24}$$

$$n(n-1)x^2 = \frac{1}{4} \dots \dots (iii)$$

Equation (iii) divided by (ii)

$$\frac{n(n-1)x^2}{n^2x^2} = \frac{\frac{1}{4}}{\frac{1}{16}}$$

$$\frac{n-1}{n} = \frac{1}{4} \times 16$$

$$\frac{n-1}{n} = 4$$

$$n-1 = 4n$$

$$4n - n = -1$$

$$n = -\frac{1}{3}$$

Putting the value of "n" in equation (i) we get

$$\left(-\frac{1}{3}\right)x = \frac{1}{4}$$

$$x = -\frac{3}{4}$$

By substituting the value of n and x in $(1+x)^n$

We have

$$(1+x)^n = \left(1 - \frac{3}{4}\right)^{-\frac{1}{3}} = \left(\frac{4-3}{4}\right)^{-\frac{1}{3}}$$

$$= \left(\frac{1}{4}\right)^{\frac{1}{3}} = (4)^{\frac{1}{3}}$$

$= \sqrt[3]{4}$ Hence shown Answer

Find the volume of the tetrahedron whose vertices are
A(2, 1, 8), B(3, 2, 9), C(2, 1, 4) and D(3, 3, 10)

SOLUTION:

A(2, 1, 8)

B(3, 2, 9)

C(2, 1, 4)

D(3, 3, 10)

$$\therefore \text{Volume of tetrahedron} = \frac{1}{6} [\vec{AB} \quad \vec{AC} \quad \vec{AD}]$$

$$\vec{AB} = B - A = (3, 2, 9) - (2, 1, 8) = 3 - 2, 2 - 1, 9 - 8$$

$$\vec{AB} = (1, 1, 1)$$

$$\vec{AC} = C - A = (2, 1, 4) - (2, 1, 8) = 2 - 2, 1 - 1, 4 - 8$$

$$\vec{AC} = (0, 0, -4)$$

$$\vec{AD} = D - A = (3, 3, 10) - (2, 1, 8) = 3 - 2, 3 - 1, 10 - 8$$

$$\vec{AD} = (1, 2, 2)$$

$$\frac{1}{6} [\vec{AB} \quad \vec{AC} \quad \vec{AD}] = \frac{1}{6} \begin{vmatrix} 1 & 1 & 1 \\ 0 & 0 & -4 \\ 1 & 2 & 2 \end{vmatrix}$$

$$= \frac{1}{6} \left\{ 1 \begin{vmatrix} 0 & -4 \\ 2 & 2 \end{vmatrix} - 1 \begin{vmatrix} 0 & -4 \\ 1 & 2 \end{vmatrix} + 1 \begin{vmatrix} 0 & 0 \\ 1 & 2 \end{vmatrix} \right\}$$

$$= \frac{1}{6} [1\{(0)(2) - (-4)(2)\} - 1\{(0)(2) - (-4)(1)\} + 1\{(0)(2) - (0)(1)\}]$$

$$= \frac{1}{6} \{1(0 + 8) - 1(0 + 4) + 1(0 - 0)\}$$

$$= \frac{1}{6} \{1(8) - 1(4) + 1(0)\}$$

$$= \frac{1}{6} (8 - 4 + 0)$$

$$= \frac{1}{6} (4)$$

Volume of tetrahedron = $\frac{2}{3}$ Unit cube Answer

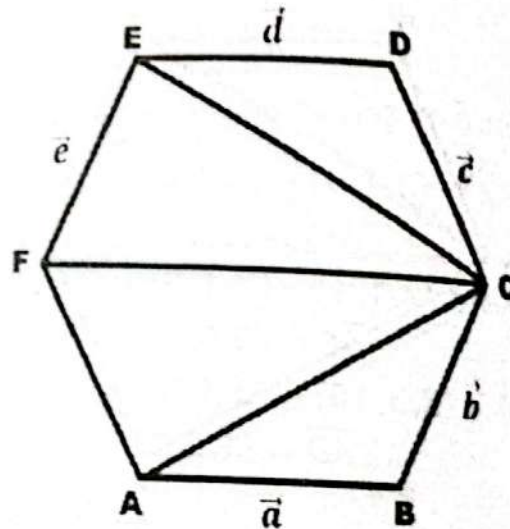
SOLUTION: Prove Hexagon Law of vector addition.

PROOF: Hexagon law of vector addition

Hexagon law of vector addition states that, if five vectors are represented by five consecutive sides of a pentagon in same order then the resultant is the sixth side in opposite order

Let $\vec{a}$, $\vec{b}$, $\vec{c}$, $\vec{d}$ and $\vec{e}$ be the five vectors represented by the sides $\overline{AB}$, $\overline{BC}$, $\overline{CD}$, $\overline{DE}$ and $\overline{EF}$ respectively of hexagon $ABCDEF$ as shown in figure

By Hexagon law of vectors, we have to prove that $\overline{AF} = \vec{a} + \vec{b} + \vec{c} + \vec{d} + \vec{e}$



In $\triangle ABC$, by triangle law of addition

$$\overline{AC} = \vec{a} + \vec{b} \quad \text{--- (i)}$$

In $\triangle CDE$, by triangle law of addition

$$\overline{CE} = \vec{c} + \vec{d}$$

In $\triangle CEF$, by triangle law of addition

$$\overline{CF} = \overline{CE} + \vec{e}$$

But $\overline{CE} = \vec{c} + \vec{d}$

$$\overline{CF} = \vec{c} + \vec{d} + \vec{e}$$

In $\triangle FCA$, by triangle law of addition

$$\overline{AF} = \overline{AC} + \overline{CF}$$

But $\overline{AC} = \vec{a} + \vec{b}$ and $\overline{CF} = \vec{c} + \vec{d} + \vec{e}$

$$\overline{AF} = \vec{a} + \vec{b} + \vec{c} + \vec{d} + \vec{e}$$

Hence Proved

8. Solve the Linear programming by graphical method
 $0 \leq x \leq 12$ and $y \geq 0$ Maximize $Z(x, y) = 30x + 36y$ subject to:
 $6x + 5y \leq 20$

SOLUTION:

CONSTRAINT 1:

$$4x + 2y \leq 12$$

Consider it as an equation $4x + 2y = 12$

Intercept put $y = 0$

$$4x + 2(0) = 12 \Rightarrow 4x = 12 \Rightarrow x = \frac{12}{4}$$

$$x = 3$$

Intersection point of line with x -axis is $(3, 0)$

Intercept put $x = 0$

$$4(0) + 2y = 12 \Rightarrow 2y = 12 \Rightarrow y = \frac{12}{2}$$

$$y = 6$$

Intersection point of line with y -axis is $(0, 6)$

put $x = 0$ and put $y = 0$ in $4x + 2y \leq 12$

$$4(0) + 2(0) \leq 12$$

$$0 \leq 12 \text{ (true)}$$

UNIT 2:

$$6x + 5y \leq 20$$

Consider it as an equation $6x + 5y = 20$

Intercept put $y = 0$

$$6x + 5(0) = 20 \Rightarrow 6x = 20 \Rightarrow x = \frac{20}{6}$$

$$x = \frac{10}{3}$$

Intersection point of line with x -axis is $(\frac{10}{3}, 0)$

Intercept put $x = 0$

$$6(0) + 5y = 20 \Rightarrow 5y = 20 \Rightarrow y = \frac{20}{5}$$

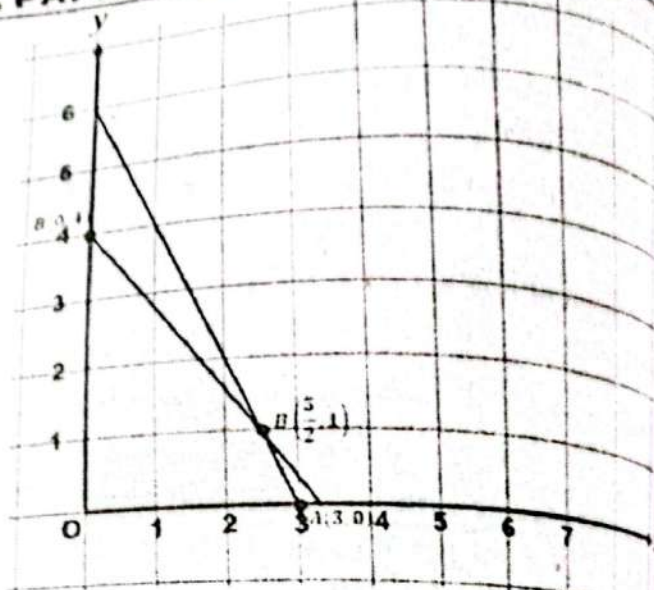
$$y = 4$$

Intersection point of line with y -axis is $(0, 4)$

put $x = 0$ and put $y = 0$ in $6x + 5y \leq 20$

$$6(0) + 5(0) \leq 20$$

$$0 \leq 20 \text{ (true)}$$



The intersection of two linear inequalities is the required feasible region OABC which is the shaded area in the graph.

The solution points or corner points of the region OABC are $O(0,0)$, $A(3,0)$, $B\left(\frac{5}{2}, 1\right)$, and $C(0,4)$ in the graph

OBJECTIVE FUNCTION

SOLUTION POINTS OR CORNER POINTS	OBJECTIVE FUNCTION $f(x, y) = 30x + 36y$
$O(0,0)$	$f(x, y) = 30(0) + 36(0) = 0$
$A(3,0)$	$f(x, y) = 30(3) + 36(0) = 90$
$B\left(\frac{5}{2}, 1\right)$	$f(x, y) = 30\left(\frac{5}{2}\right) + 36(1)$ $= 75 + 36 = 111$
$C(0,4)$	$f(x, y) = 30(0) + 36(4)$ $= 0 + 144 = 144$

Hence

Maximum at $(0,4)$; Maximum value = 144 Answer

9. Prove fundamental law $\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$

SOLUTION:

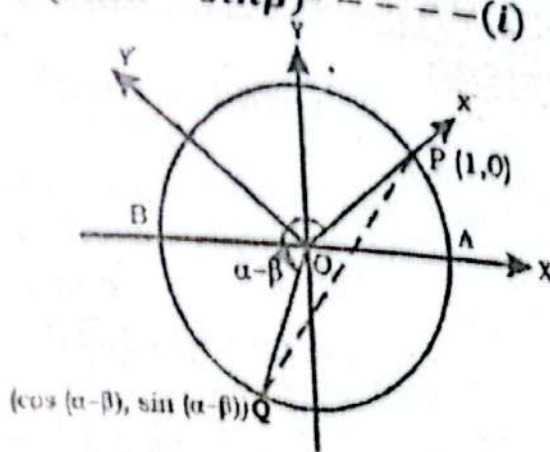
PROOF:

Consider a unit circle with centre at $O(0,0)$ as shown in the figure. Let $P(\cos\beta, \sin\beta)$ and $Q(\cos\alpha, \sin\alpha)$ be any two points on the circle.

Then by using distance formula

$PQ = \sqrt{(\cos\alpha - \cos\beta)^2 + (\sin\alpha - \sin\beta)^2} \dots\dots (i)$

Now rotate the axes so that the positive direction of the x -axis passes through the point P . Then with respect to this coordinate system. The coordinates of P and Q respectively become $(1, 0)$ and $(\cos(\alpha - \beta), \sin(\alpha - \beta))$ as shown in figure



$$PQ = \sqrt{\{(\cos\alpha - \cos\beta) - 1\}^2 + \{(\sin\alpha - \sin\beta) - 0\}^2}$$

$$PQ = \sqrt{\{\cos(\alpha - \beta) - 1\}^2 + \{\sin(\alpha - \beta)\}^2} \dots\dots (ii)$$

By comparing equation (i) and (ii), we get

$$\sqrt{(\cos\alpha - \cos\beta)^2 + (\sin\alpha - \sin\beta)^2} = \sqrt{\{(\cos\alpha - \cos\beta) - 1\}^2 + \sin^2(\alpha - \beta)}$$

Squaring on both sides

$$(\cos\alpha - \cos\beta)^2 + (\sin\alpha - \sin\beta)^2 = \{\cos(\alpha - \beta) - 1\}^2 + \{\sin(\alpha - \beta)\}^2$$

$$\begin{aligned} & (\cos\alpha)^2 - 2(\cos\alpha)(\cos\beta) + (\cos\beta)^2 + \{(\sin\alpha)^2 - 2(\sin\alpha)(\sin\beta) + (\sin\beta)^2\} \\ & = \{\cos(\alpha - \beta)\}^2 - 2\{\cos(\alpha - \beta)\}(1) + (1)^2 + \sin^2(\alpha - \beta) \end{aligned}$$

$$\begin{aligned} & \cos^2\alpha - 2\cos\alpha\cos\beta + \cos^2\beta + \sin^2\alpha - 2\sin\alpha\sin\beta + \sin^2\beta \\ & = \cos^2(\alpha - \beta) - 2\cos(\alpha - \beta) + 1 + \sin^2(\alpha - \beta) \\ & (\cos^2\alpha + \sin^2\alpha) + (\cos^2\beta + \sin^2\beta) - 2\cos\alpha\cos\beta - 2\sin\alpha\sin\beta \\ & = \{\cos^2(\alpha - \beta) + \sin^2(\alpha - \beta)\} - 2\cos(\alpha - \beta) + 1 \end{aligned}$$

$$\because \cos^2\theta + \sin^2\theta = 1$$

$$(1) + (1) - 2(\cos\alpha\cos\beta - \sin\alpha\sin\beta) = (1) - 2\cos(\alpha - \beta) + 1$$

$$2 - 2(\cos\alpha\cos\beta - \sin\alpha\sin\beta) = 2 - 2\cos(\alpha - \beta)$$

$$-2(\cos\alpha\cos\beta - \sin\alpha\sin\beta) = -2\cos(\alpha - \beta)$$

$$\cos(\alpha - \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$$

Hence Proved

OR--Prove that:

$$(i) \cos 4\theta = 8\cos^4\theta - 8\cos^2\theta + 1$$

$$(ii) \frac{\sin 6\theta + \sin 4\theta}{\cos 6\theta + \cos 4\theta} = \tan 5\theta$$

$$(i) \cos 4\theta = 8\cos^4\theta - 8\cos^2\theta + 1$$

SOLUTION:

$$\cos 4\theta = 8\cos^4\theta - 8\cos^2\theta + 1$$

Taking L.H.S.:

$$\begin{aligned} &= \cos 4\theta \\ &= \cos(2\theta + 2\theta) \\ &\because \cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta \\ &= \cos 2\theta \cos 2\theta - \sin 2\theta \sin 2\theta \\ &= \cos^2 2\theta - \sin^2 2\theta \\ &= (\cos 2\theta)^2 - (\sin 2\theta)^2 \\ &\because \cos 2\theta = 2\cos^2\theta - 1 \\ &\because \sin 2\theta = 2\sin\theta \cos\theta \\ &= (2\cos^2\theta - 1)^2 - (2\sin\theta \cos\theta)^2 \\ &= 4\cos^4\theta - 4\cos^2\theta + 1 - 4\sin^2\theta \cos^2\theta \\ &\because \sin^2\theta = 1 - \cos^2\theta \\ &= 4\cos^4\theta - 4\cos^2\theta + 1 - 4(1 - \cos^2\theta)\cos^2\theta \\ &= 4\cos^4\theta - 4\cos^2\theta + 1 - 4\cos^2\theta + 4\cos^4\theta \\ &= 8\cos^4\theta - 8\cos^2\theta + 1 \end{aligned}$$

Hence Proved L.H.S = R.H.S.

(ii) $\frac{\sin 6\theta + \sin 4\theta}{\cos 6\theta + \cos 4\theta} = \tan 5\theta$

SOLUTION:

Taking L.H.S.:

$$\frac{\sin 6\theta + \sin 4\theta}{\cos 6\theta + \cos 4\theta} = \tan 5\theta$$

$$\begin{aligned} &= \frac{\sin 6\theta + \sin 4\theta}{\cos 6\theta + \cos 4\theta} \\ &= \frac{\sin 6\theta + \sin 4\theta}{\cos 6\theta + \cos 4\theta} \end{aligned}$$

$$\because \sin u + \sin v = 2\sin \frac{u+v}{2} \cos \frac{u-v}{2}$$

$$\because \cos u + \cos v = 2\cos \frac{u+v}{2} \cos \frac{u-v}{2}$$

$$\begin{aligned} &= \frac{\left(2\sin \frac{6\theta + 4\theta}{2} \cos \frac{6\theta - 4\theta}{2}\right)}{\left(2\cos \frac{6\theta + 4\theta}{2} \cos \frac{6\theta - 4\theta}{2}\right)} \\ &= \frac{2\sin \frac{10\theta}{2} \cos \frac{2\theta}{2}}{2\cos \frac{10\theta}{2} \cos \frac{2\theta}{2}} \end{aligned}$$

$$\frac{2\sin 5\theta \cos \theta}{2\cos 5\theta \cos \theta} = \frac{\sin 5\theta}{\cos 5\theta}$$

$$\therefore \frac{\sin \theta}{\cos \theta} = \tan \theta$$

$$= \tan 5\theta$$

Hence Proved L.H.S. = R.H.S.

Find the general solution set of the trigonometric equation $\csc \theta = \sqrt{3} + \cot \theta$ and also verify the solution.

SOLUTION:

$$\csc \theta = \sqrt{3} + \cot \theta$$

$$\therefore \csc \theta = \frac{1}{\sin \theta} \text{ and } \cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$\frac{1}{\sin \theta} = \sqrt{3} + \frac{\cos \theta}{\sin \theta}$$

$$\frac{1}{\sin \theta} = \frac{\sqrt{3} \sin \theta + \cos \theta}{\sin \theta}$$

$$\sqrt{3} \sin \theta + \cos \theta = 1$$

$$\sqrt{3} \sin \theta = 1 - \cos \theta$$

Squaring both sides

$$(\sqrt{3} \sin \theta)^2 = (1 - \cos \theta)^2$$

$$3 \sin^2 \theta = 1 - 2 \cos \theta + \cos^2 \theta$$

$$\therefore \sin^2 \theta = 1 - \cos^2 \theta$$

$$3(1 - \cos^2 \theta) = 1 + \cos^2 \theta - 2 \cos \theta$$

$$3 - 3 \cos^2 \theta = 1 + \cos^2 \theta - 2 \cos \theta$$

$$3 - 1 = \cos^2 \theta - 2 \cos \theta + 3 \cos^2 \theta$$

$$4 \cos^2 \theta - 2 \cos \theta = 2$$

Divided by "2"

$$2 \cos^2 \theta - \cos \theta = 1$$

$$2 \cos^2 \theta - \cos \theta - 1 = 0$$

$$2 \cos^2 \theta - \cos \theta - \cos \theta + \cos \theta - 1 = 0$$

$$2 \cos^2 \theta - 2 \cos \theta + \cos \theta - 1 = 0$$

$$2 \cos \theta (\cos \theta - 1) + 1(\cos \theta - 1) = 0$$

$$(\cos \theta - 1)(2 \cos \theta + 1) = 0$$

Either

$$\cos \theta - 1 = 0$$

$$\cos \theta = 1$$

We know that the

$$\cos 0 = 1.$$

So, the reference angle is

Or

$$2 \cos \theta + 1 = 0$$

$$2 \cos \theta = -1$$

$$\cos \theta = -\frac{1}{2}$$

$$\cos \theta = -\frac{1}{2}$$

We know that the

$$x = 0, 2\pi$$

$$\cos \frac{\pi}{3} = \frac{1}{2}, \quad \cos \frac{2\pi}{3} = -\frac{1}{2}$$

So, the reference angle is $\frac{\pi}{3}$

$$x = \frac{\pi}{3}, \frac{2\pi}{3}$$

FOR $\cos x = 1$

$$x = 0, \quad x = 2\pi$$

FOR $\cos x = -\frac{1}{2}$

Since $\cos x$ is negative in 2nd and 3rd quadrants therefore

FOR 2ND QUADRANT

FOR $\cos x = \frac{\pi}{3}$

$$\begin{aligned} \therefore x &= \frac{\pi}{2} + x \\ x &= \frac{\pi}{2} + \left(\frac{\pi}{3}\right) \\ x &= \frac{3\pi + 2\pi}{6} \\ x &= \frac{5\pi}{6} \end{aligned}$$

Also

$$\begin{aligned} \therefore x &= \pi - x \\ x &= \pi - \left(\frac{\pi}{3}\right) \\ x &= \frac{3\pi - \pi}{3} \\ x &= \frac{2\pi}{3} \end{aligned}$$

FOR $\cos x = \frac{2\pi}{3}$

$$\begin{aligned} \therefore x &= \frac{\pi}{2} + x \\ x &= \frac{\pi}{2} + \left(\frac{2\pi}{3}\right) \\ x &= \frac{3\pi + 4\pi}{6} \\ x &= \frac{7\pi}{6} \end{aligned}$$

Also

$$\therefore x = \pi - x$$

FOR 3RD QUADRANT

FOR $\cos x = \frac{\pi}{3}$

$$\begin{aligned} \therefore x &= \pi + x \\ x &= \pi + \left(\frac{\pi}{3}\right) \\ x &= \frac{3\pi + \pi}{3} \\ x &= \frac{4\pi}{3} \end{aligned}$$

Also

$$\begin{aligned} \therefore x &= \frac{3\pi}{2} - x \\ x &= \frac{3\pi}{2} - \left(\frac{\pi}{3}\right) \\ x &= \frac{9\pi - 2\pi}{6} \\ x &= \frac{7\pi}{6} \end{aligned}$$

FOR $\cos x = \frac{2\pi}{3}$

$$\begin{aligned} \therefore x &= \pi + x \\ x &= \pi + \left(\frac{2\pi}{3}\right) \\ x &= \frac{3\pi + 2\pi}{3} \\ x &= \frac{5\pi}{3} \end{aligned}$$

Also

$$\therefore x = \frac{3\pi}{2} - x$$

$$x = \frac{3\pi}{2} - \left(\frac{2\pi}{3}\right)$$

$$x = \frac{9\pi - 4\pi}{6}$$

$$x = \frac{5\pi}{6}$$

$$\operatorname{cosec} 0 = \sqrt{3} + \cot 0$$

$$\frac{1}{0} = \sqrt{3} - \frac{1}{0}$$

$$\infty = \sqrt{3} - \infty$$

this value because it is undefined.

$$\operatorname{cosec} 2\pi = \sqrt{3} + \cot 2\pi$$

$$\frac{1}{0} = \sqrt{3} - \frac{1}{0}$$

$$\infty = \sqrt{3} - \infty$$

$$\infty = \infty$$

this value because it is undefined.

$$\operatorname{cosec} \frac{5\pi}{3} = \sqrt{3} + \cot \frac{5\pi}{3}$$

$$-\frac{2}{\sqrt{3}} = \sqrt{3} - \frac{1}{\sqrt{3}}$$

$$-\frac{2}{\sqrt{3}} = \frac{3-1}{\sqrt{3}}$$

$$-\frac{2}{\sqrt{3}} = \frac{2}{\sqrt{3}} \text{ (not True)}$$

$$\operatorname{cosec} \frac{2\pi}{3} = \sqrt{3} + \cot \frac{2\pi}{3}$$

$$\frac{2}{\sqrt{3}} = \sqrt{3} - \frac{1}{\sqrt{3}}$$

$$\frac{2}{\sqrt{3}} = \frac{3-1}{\sqrt{3}}$$

$$\frac{2}{\sqrt{3}} = \frac{2}{\sqrt{3}} \text{ (True)}$$

$$\text{FOR } \theta = \frac{4\pi}{3}$$

$$\operatorname{cosec} \frac{4\pi}{3} = \sqrt{3} + \cot \frac{4\pi}{3}$$

$$-\frac{2}{\sqrt{3}} = \sqrt{3} + \frac{1}{\sqrt{3}}$$

$$-\frac{2}{\sqrt{3}} = \frac{3+1}{\sqrt{3}}$$

$$-\frac{2}{\sqrt{3}} = \frac{4}{\sqrt{3}} \text{ (not True)}$$

$$\text{FOR } \theta = \frac{7\pi}{6}$$

$$\operatorname{cosec} \frac{7\pi}{6} = \sqrt{3} + \cot \frac{7\pi}{6}$$

$$-2 = \sqrt{3} + \sqrt{3}$$

$$-2 = 2\sqrt{3} \text{ (not True)}$$

$$\text{FOR } \theta = \frac{\pi}{3}$$

$$\operatorname{cosec} \frac{\pi}{3} = \sqrt{3} + \cot \frac{\pi}{3}$$

$$\frac{2}{\sqrt{3}} = \sqrt{3} - \frac{1}{\sqrt{3}}$$

$$\frac{2}{\sqrt{3}} = \frac{3-1}{\sqrt{3}}$$

$$\frac{2}{\sqrt{3}} = \frac{2}{\sqrt{3}} \text{ (True)}$$

$$\text{FOR } \theta = \frac{5\pi}{3}$$

$$\operatorname{cosec} \frac{5\pi}{3} = \sqrt{3} + \cot \frac{5\pi}{3}$$

$$-\frac{2}{\sqrt{3}} = \sqrt{3} - \frac{1}{\sqrt{3}}$$

$$-\frac{2}{\sqrt{3}} = \frac{3-1}{\sqrt{3}}$$

$$-\frac{2}{\sqrt{3}} = \frac{2}{\sqrt{3}} \text{ (not True)}$$

$$\text{Solution set} = \{n\pi\} \left\{ \frac{2\pi}{3} + 2n\pi \right\} n \in \mathbb{Z}$$

MATHEMATICS 2024

(Science General Group)

(100 Marks)

TIME: 3 Hours

SECTION 'A' (Multiple Choice Questions)(40)

NOTE: i) This section consists of 20 part questions and all are to be answered. Each part question carries one mark.
Do not copy the part questions in your answer script. Write only the answer in full against the proper number of the question and its part.
All notations are used in their usual meanings. The use of Scientific Calculator is allowed.

Choose the correct answer for each from the given options:

1) If $Z = 3i + 4$ then $Z + \bar{Z} =$:

- * -6 * 8✓ * -8

2) The product of two conjugate complex numbers is:
a real number ✓

- * an imaginary number
* undefined

3) If A is the singular matrix, then AA^{-1} is:

- * $A^{-1}A$ * I * 0 * undefined✓

4) A square matrix A is called idempotent matrix if:

- * $A^2 = A$ ✓ * $A^3 = A$ * $A^2 = I$ * $A^2 = 0$

5) $\begin{vmatrix} 1 & 0 \\ 5 & 6 \end{vmatrix} - \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix} =$:

- * -2✓ * 4 * 8 * 10

6) If $|\vec{a} \times \vec{b}| = |\vec{a} \cdot \vec{b}|$, then the angle between $\vec{a}$ and $\vec{b} =$:

- * 0 * $\frac{\pi}{4}$ ✓ * $\frac{\pi}{2}$ * π

7) This position vector of the point $(1, 0, 2)$ is:

- * $\hat{i} + \hat{j} + 2\hat{k}$ * $\hat{i} + 2\hat{j}$ * $2\hat{i} + 2\hat{k}$ * $\hat{i} + 2\hat{k}$ ✓

8) The A.M's between 4 and 10 is:

- * 4 * 5 * 6 * 7✓

9) A sequence is a function whose domain is a set of:

- * integers * rational numbers
* real numbers * natural numbers✓

10) $\sum_{n=1}^{50} n =$:

- * 1274 * 1275✓ * 1280 * 1285

11) Probability of getting 7 in throwing a dice is:

- * $\frac{1}{6}$ ✓ * 1 * -1 * undefined

12) The sum of the expansion of $(a + b)^{2n}$ is:

- * n * $n + 1$ ✓ * $2n + 1$ * $2n$

xiii) If $n = 0$, then $(n + 1)! = :$

- * 0 * n * 1 ✓ * ∞

xiv) If $f(x, y) = 5x + 4y$ is an objective function and the corner points of the feasible region are $(5, 4)$, $(0, 0)$, $(0, 6)$ and $(4, 0)$ then the function is maximum at:

- * $(4, 0)$ * $(0, 6)$ * $(0, 0)$ * $(5, 4)$ ✓

xv) The region of solution in L.P. problem is called:

- * Infeasible region * Unbounded region
* Infinite region * feasible region ✓

xvi) $1 + \cos \theta = :$

- * $2\sin^2 \frac{\theta}{2}$ * $\sin^2 \frac{\theta}{2}$ * $2\cos^2 \frac{\theta}{2}$ ✓ * $\cos^2 \frac{\theta}{2}$

xvii) $c^2 \frac{\sin \beta \sin \alpha}{\sin \gamma} = :$

- * Δ * Δ^2 * $\frac{1}{2} \Delta$ * 2Δ ✓

xviii) $\sin \frac{\alpha}{2}$ is equal to:

- * $\pm \sqrt{\frac{1+\sin \alpha}{2}}$ * $\pm \sqrt{\frac{1-\cos \alpha}{2}}$ ✓ * $\pm \sqrt{\frac{1+\cos \alpha}{2}}$ * $\pm \sqrt{\frac{1-\sin \alpha}{2}}$

xix) Period of $\cos x$ is:

- * $\frac{\pi}{2}$ * π * 2π ✓ * 3π

xx) $\cos \left(\sin^{-1} \frac{1}{\sqrt{2}} \right) = :$

- * $\frac{1}{\sqrt{2}}$ ✓ * $-\frac{1}{\sqrt{2}}$ * $\frac{\sqrt{3}}{2}$ * $-\frac{1}{2}$

SECTION 'B' (Short-Answer Questions) (40)

NOTE: Attempt any TEN part questions in all. All questions carry equal marks. (i.e. 4 marks of each part).

2. i) Find the real and imaginary parts of the $\left(\frac{1-i}{2-i} \right)^{-1}$

SOLUTION:

$$\begin{aligned} \left(\frac{1-i}{2-i} \right)^{-1} &= \frac{(1-i)^{-1}}{(2-i)^{-1}} \\ &= \frac{2-i}{1-i} \\ &= \frac{2-i}{1-i} \times \frac{1+i}{1+i} \\ &= \frac{(2-i)(1+i)}{(1-i)(1+i)} \end{aligned}$$

QUESTIONS XI (S)

$$= \frac{2(1+i) - i(1+i)}{(1)^2 - (i)^2}$$

$$= \frac{2+2i-i-i^2}{1-i^2}$$

$$\because i^2 = -1$$

$$= \frac{2+3i-(-1)}{1-(-1)}$$

$$= \frac{3+3i}{2}$$

$$= \frac{3}{2} + \frac{3}{2}i$$

$$\text{Re}(z) = \frac{3}{2}; \quad \text{Imaginary Part} = \text{Im}(z) = \frac{3}{2}i$$

Let $A = \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix}$ is an involuntary matrix:

$$A = \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix}$$

$$A.A = \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix} \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} (-5)(-5) + (-8)(3) + (0)(1) & (-5)(-8) + (-8)(5) + (0)(2) & (-5)(0) + (-8)(0) + (0)(-1) \\ (3)(-5) + (5)(3) + (0)(1) & (3)(-8) + (5)(5) + (0)(2) & (3)(0) + (5)(0) + (0)(-1) \\ (1)(-5) + (2)(3) + (-1)(1) & (1)(-8) + (2)(5) + (-1)(2) & (1)(0) + (2)(0) + (-1)(-1) \end{bmatrix}$$

$$= \begin{bmatrix} 25-24+0 & 40-40+0 & 0+0+0 \\ -15+15+0 & -24+25+0 & 0+0+0 \\ -5+6-1 & -8+10-2 & 0+0+1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$$

$A^2 = I$, So A involuntary matrix

expanding show that $\begin{bmatrix} x+1 & x+3 & x+5 \\ x+4 & x+6 & x+8 \\ x+7 & x+9 & x+11 \end{bmatrix} = 0$

$$\begin{vmatrix} x+1 & x+3 & x+5 \\ x+4 & x+6 & x+8 \\ x+7 & x+9 & x+11 \end{vmatrix} = 0$$

$$= \begin{vmatrix} x+1 & x+3 & x+5 \\ x+4 & x+6 & x+8 \\ x+7 & x+9 & x+11 \end{vmatrix}$$

Subtracting R_1 from R_2 and R_3 from R_2

$$= \begin{vmatrix} x+1 & x+3 & x+5 \\ x+4-x-1 & x+6-x-3 & x+8-x-5 \\ x+7-x-4 & x+9-x-6 & x+11-x-8 \end{vmatrix}$$

$$= \begin{vmatrix} x+1 & x+3 & x+5 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{vmatrix}$$

According to property 3, "If two rows of a matrix are identical, the determinant is zero" in this case R_2 and R_3 are identical, the

$$= 0$$

L.H.S = R.H.S Hence proved

iv) If α, β, γ are the direction angles of a vector then show that $\sin^2 \beta + \sin^2 \gamma = 2$.

SOLUTION:

$$\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 2$$

Taking L.H.S.:

$$= \sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma$$

$$\because \sin^2 \theta = 1 - \cos^2 \theta$$

$$= (1 - \cos^2 \alpha) + (1 - \cos^2 \beta) + (1 - \cos^2 \gamma)$$

$$= 1 - \cos^2 \alpha + 1 - \cos^2 \beta + 1 - \cos^2 \gamma$$

$$= 3 - (\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma)$$

$$\because \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$= 3 - (1)$$

$$= 2$$

$$L.H.S = R.H.S$$

Hence Shown

v) Find three consecutive numbers in G.P whose sum is 39 and product is 729.

SOLUTION:

Let the three numbers in G.P. are : a, ar and ar^2

As,

$$\text{Sum of three numbers} = a + ar + ar^2 = 39$$

$$a(1 + r + r^2) = 39 \quad \dots (i)$$

$$\text{product of three numbers} = a \times ar \times ar^2 = 729$$

$$a^3 r^3 = 729$$

Taking cube root on both sides

$$\sqrt[3]{a^3 r^3} = \sqrt[3]{729}$$

$$ar = 9$$

$$a = \frac{9}{r} \quad \dots \dots (ii)$$

value of 'r' in equation (i), we get

$$\frac{9}{r}(1 + r + r^2) = 39$$

$$9 + 9r + 9r^2 = 39r$$

$$9r^2 + 9r - 39r + 9 = 0$$

$$9r^2 - 30r + 9 = 0$$

$$9r^2 - 27r - 3r + 9 = 0$$

$$9r(r - 3) - 3(r - 3) = 0$$

$$(r - 3)(9r - 3) = 0$$

Or

$$9r - 3 = 0$$

$$9r = 3$$

$$3$$

$$r = \frac{3}{9}$$

$$1$$

$$r = \frac{1}{3}$$

$$r - 3 = 0$$

$$r = 3$$

value of 'r' in equation (ii), we get

$$9$$

$$a = \frac{9}{3}$$

$$a = 3$$

$$a = \frac{9}{1}$$

$$3$$

$$a = 9 \times 3 = 27$$

the numbers are

$$\text{First number} = a = 3$$

$$\text{Second number} = ar$$

$$= 3 \times 3 = 9$$

$$\text{Third number} = ar^2 = 3 \times 3^2$$

$$= 3 \times 9 = 27$$

$$\text{First number} = a = 27$$

$$\text{Second number} = ar$$

$$= 27 \times \frac{1}{3} = 9$$

$$\text{Third number} = ar^2$$

$$= 27 \times \left(\frac{1}{3}\right)^2 = 27 \times \frac{1}{9} = 3$$

Required three consecutive numbers in G.P. are

3, 9, 27 and 27, 9, 3 Answer

Series $1 + 2.5 + 3.5^2 + 4.5^3 + \dots$ to 30 terms.

$$1 + 2.5 + 3.5^2 + 4.5^3 + \dots + 30.5^{29} \dots \dots (i)$$

Multiplying both sides by '5'

$$5 + 2.5^2 + 3.5^3 + 4.5^4 + \dots + 30.5^{30} \dots \dots (ii)$$

Subtracting equation (ii) from equation (i)

$$S = 1 + 2.5 + 3.5^2 + 4.5^3 + \dots + 30.5^{29}$$

$$S = 1.5 + 2.5^2 + 3.5^3 + 4.5^4 + \dots + 30.5^{30}$$

$$-4S = 1 + 1.5 + 1.5^2 + 1.5^3 + 1.5^4 + 1.5^5 + \dots + 1.5^{29}$$

$$-4S = 1 + 5(1 + 1.5 + 1.5^2 + 1.5^3 + 1.5^4 + \dots + 1.5^{29})$$

$$\therefore S_n = a + ar + ar^2 + ar^3 + ar^4 + \dots + ar^{n-1}$$

$$-4S = 1 + 5(S_n) - 30.5^{30}$$

$$\therefore S_n = \frac{a(r^n - 1)}{r - 1}$$

$$-4S = 1 + 5 \left\{ \frac{a(r^n - 1)}{r - 1} \right\} - 30.5^{30}$$

$$a = 1, \quad r = 5, \quad n = 30$$

$$-4S = 1 + 5 \left\{ \frac{1(5^{30} - 1)}{5 - 1} \right\} - 30.5^{30}$$

$$-4S = 1 + \frac{5(5^{30} - 1)}{4} - 30.5^{30}$$

$$-4S = 1 + \frac{5 \times 5^{30} - 5}{4} - 30.5^{30}$$

$$-4S = \frac{4 + 5^{30} - 5 - 120.5^{30}}{4}$$

$$-4S = \frac{-1 + 5^{30} - 120.5^{30}}{4}$$

$$-4S = \frac{-1 + 5^{30}(1 - 120)}{4}$$

$$-4S = \frac{-1 - 119(5^{30})}{4}$$

$$-4S = - \left\{ \frac{1 + 119(5^{30})}{4} \right\}$$

$$S = \frac{1 + 119(5^{30})}{16}$$

$$S = \frac{1}{16} + \frac{119}{16}(5^{30}) \text{ Answer}$$

vii) How many distinct permutations of letters of 'ESSENTIAL' are possible?

SOLUTION:

Total Number of Letter in ESSENTIAL = 9

Total Number of Letter E = 2

Total Number of Letter S = 2

- Total Number of Letter N = 1
- Total Number of Letter T = 1
- Total Number of Letter I = 1
- Total Number of Letter A = 1
- Total Number of Letter L = 1

$$\therefore P = \binom{n}{r, s, t} = \frac{n!}{r! s! t!}$$

Permutations when 'S, s' together

In this case, we consider both 'S' as a single unit because both will be placed in the same place

$$\begin{aligned} \text{Permutations} &= \frac{(9-1)!}{2! \times 1! \times 1! \times 1! \times 1! \times 1! \times 1!} \\ &= \frac{8!}{2!} \\ &= \frac{40320}{2} \end{aligned}$$

Permutations = 20160 Answer

viii) Prove the preposition by mathematical induction for every positive integer n.

$$1 + 4 + 7 + \dots + (3n-2) = \frac{n(3n-1)}{2}$$

SOLUTION:

$$1 + 4 + 7 + \dots + (3n-2) = \frac{n(3n-1)}{2}$$

Condition I:

For n = 1

$$1 + 4 + 7 + \dots + \{3(1)-2\} = \frac{(1)\{3(1)-1\}}{2}$$

$$1 + 4 + 7 + \dots + (3-2) = \frac{(1)(3-1)}{2}$$

$$1 + 4 + 7 + \dots + 1 = \frac{2}{2}$$

$$1 + 4 + 7 + \dots + 1 = 1$$

So, P(n) is true for n = 1

Condition II:

Assuming the result to be true for n = k, we get the hypothesis

$$1 + 4 + 7 + \dots + (3k-2) = \frac{k(3k-1)}{2}$$

Adding (3k + 1) On both Sides

$$\left| \begin{array}{l} \text{R.W.} \\ \text{last term} = (3k-2) \end{array} \right.$$

$$\begin{aligned}
 & 1 + 4 + 7 + \dots + (3k - 2) + (3k + 1) \\
 &= \frac{k(3k - 1)}{2} + (3k + 1) \\
 &= \frac{(3k^2 - k) + 2(3k + 1)}{2} \\
 &= \frac{3k^2 - k + 6k + 2}{2} \\
 &= \frac{3k^2 + 5k + 2}{2} \\
 &= \frac{3k^2 + 3k + 2k + 2}{2} \\
 &= \frac{3k(k + 1) + 2(k + 1)}{2} \\
 &= \frac{(k + 1)(3k + 2)}{2} \\
 &= \frac{(k + 1)(3k + 2 + 1 - 1)}{2} \\
 &= \frac{(k + 1)(3k + 3 - 1)}{2} \\
 &= \frac{(k + 1)\{(k + 1) - 1\}}{2}
 \end{aligned}$$

$$\begin{aligned}
 & \text{Put } n = k + 1 \\
 &= \{3(k + 1) - 2\} \\
 &= (3k + 3 - 2) \\
 &= (3k + 1)
 \end{aligned}$$

i.e. $p(n)$ is true for $n = k + 1$

Thus, by the principle of mathematical induction $P(n)$ is true for all natural numbers n . Answer

ix) Find the middle term in the expansion of $\left(\frac{a}{b} - \frac{b}{a}\right)^{18}$.

ANSWER: (See Question No. 2(ix), 2024 P.E.)

---OR--- Use binomial theorem to find the value of $(19)^4$.

SOLUTION:

$$(19)^4 = (20 - 1)^4$$

$$\begin{aligned}
 \therefore (a + b)^n &= a^n + na^{n-1}b + \frac{n(n-1)}{2!}a^{n-2}b^2 + \dots \\
 &\quad + \frac{n(n-1)\dots(n-r+2)}{r!}a^{n-r}b^r + \dots
 \end{aligned}$$

$$a = 20, \quad b = -1, \quad n = 4, \quad r = n - 1 = 3$$

$$\begin{aligned}
 (20-1)^4 &= (20)^4 + 4(20)^{4-1}(-1)^1 + \frac{4(4-1)}{2!}(20)^{4-2}(-1)^2 \\
 &\quad + \frac{4(4-1)(4-2)}{3!}(20)^{4-3}(-1)^3 \\
 &\quad + \frac{4(4-1)(4-2)(4-3)}{4!}(20)^{4-4}(-1)^4
 \end{aligned}$$

$$\begin{aligned}
 &= 160000 + 4(8000)(-1) + \frac{4(3)}{2}(400)(1) + \frac{4(3)(2)}{6}(20)(-1) \\
 &\quad + \frac{4(3)(2)(1)}{24}(1)(1)
 \end{aligned}$$

$$\begin{aligned}
 &= 160000 - 32000 + 2400 - 80 + 1 \\
 &= 130321 \text{ Answer}
 \end{aligned}$$

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by $f(x) = -5x + 1$, find the value of $f^{-1}(0)$.

SOLUTION:

$$f(x) = -5x + 1$$

$$f(x) = -5x + 1 = y$$

$$-5x = y - 1 \Rightarrow x = \frac{y-1}{-5}$$

$$x = \frac{1-y}{5}$$

replace x by $f^{-1}(y)$, we get

$$f^{-1}(y) = \frac{1-y}{5} \text{ --- (i)}$$

in this case $f^{-1}(y) = f^{-1}(0)$,

$$f^{-1}(0) = \frac{1-0}{5}$$

$$f^{-1}(0) = \frac{1}{5}$$

find $f^{-1}(x)$, replace y by x . i.e.

$$f^{-1}(x) = \frac{1-x}{5}$$

$$f^{-1}(0) = \frac{1}{5}, \quad f^{-1}(x) = \frac{1}{5} - \frac{x}{5} \text{ Answer}$$

The sides of a parallelogram are 25cm and 35cm long and one of its angles is 36° . Find the length of the diagonal(s).

SOLUTION:

By the properties of the parallelogram sum of all angles of the parallelogram is equal to 360° , and the opposite angles are equal, i.e.

$$\angle A + \angle B + \angle C + \angle D = 360^\circ$$

And

$$\angle A = \angle C, \angle B = \angle D$$

$$\therefore \angle B = \angle D = 36^\circ \text{ and } \angle A = \angle C = \theta$$

Therefore

$$\theta + 36^\circ + \theta + 36^\circ = 360^\circ$$

$$2\theta + 72^\circ = 360^\circ \Rightarrow 2\theta = 360^\circ - 72^\circ$$

$$2\theta = 288^\circ \Rightarrow \theta = \frac{288^\circ}{2}$$

$$\theta = 144^\circ$$

Now

For the lengths of diagonal $|\overline{AC}|$

Applying Law of on triangle ABC

$$\therefore b^2 = a^2 + c^2 - 2ac \cos \beta$$

$$|\overline{AC}|^2 = |\overline{AB}|^2 + |\overline{BC}|^2 - 2|\overline{AB}||\overline{BC}| \cos \theta$$

$$|\overline{AB}| = 35 \text{ cm}, \quad |\overline{BC}| = 25 \text{ cm}, \quad \theta = 144^\circ$$

$$|\overline{AC}|^2 = (35)^2 + (25)^2 - 2(35)(25) \cos 144^\circ$$

$$= 1225 + 625 - 1750(-0.8090)$$

$$= 1850 + 1415.7797$$

$$|\overline{AC}|^2 = 3265.77$$

$$|\overline{AC}| = 57.140 \text{ cm}$$

For the lengths of diagonal $|\overline{BD}|$

Applying Law of on triangle ABD

$$\therefore a^2 = b^2 + c^2 - 2bc \cos \alpha$$

$$|\overline{BD}|^2 = |\overline{AB}|^2 + |\overline{AD}|^2 - 2|\overline{AB}||\overline{AD}| \cos \theta$$

$$|\overline{AB}| = 35 \text{ cm}, \quad |\overline{AD}| = 25 \text{ cm}, \quad \theta = 36^\circ$$

$$|\overline{BD}|^2 = (35)^2 + (25)^2 - 2(35)(25) \cos 36^\circ$$

$$= 1225 + 625 - 1750(0.8090)$$

$$= 1850 - 1415.77$$

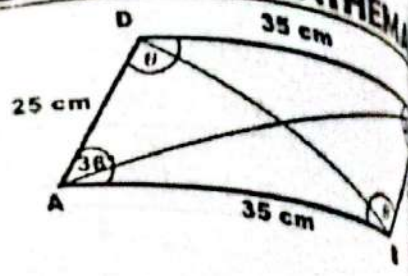
$$|\overline{BD}|^2 = 434.220$$

$$|\overline{BD}| = 20.837 \text{ cm}$$

Hence the lengths of its diagonals are 57.140 cm and 20.837 cm

xii) Prove that: $r = 4R \sin \frac{\alpha}{2} \sin \frac{\beta}{2} \sin \frac{\gamma}{2}$

SOLUTION:



$$r = 4R \sin \frac{\alpha}{2} \sin \frac{\beta}{2} \sin \frac{\gamma}{2}$$

R.H.S.:

$$= 4R \sin \frac{\alpha}{2} \sin \frac{\beta}{2} \sin \frac{\gamma}{2}$$

$$= 4R \left(\sqrt{\frac{(s-b)(s-c)}{bc}} \right) \left(\sqrt{\frac{(s-a)(s-c)}{ac}} \right) \left(\sqrt{\frac{(s-a)(s-b)}{ac}} \right)$$

$$= \frac{abc}{4R} \left(\sqrt{\frac{(s-b)(s-c) \times (s-a)(s-c) \times (s-a)(s-b)}{bc \times ac \times ac}} \right)$$

$$= \frac{abc}{4R} \left(\sqrt{\frac{(s-a)^2 (s-b)^2 (s-c)^2}{a^2 b^2 c^2} \times \frac{s^2}{s^2}} \right)$$

$$= \frac{abc}{4R} \left(\sqrt{\frac{s^2 (s-a)^2 (s-b)^2 (s-c)^2}{a^2 b^2 c^2} \times \frac{1}{s^2}} \right)$$

$$\because \sqrt{s^2 (s-a)^2 (s-b)^2 (s-c)^2} = \Delta^2$$

$$= \frac{abc}{4R} \left(\frac{\Delta^2}{abc \times s} \right)$$

$$= \frac{\Delta}{s}$$

$$= r$$

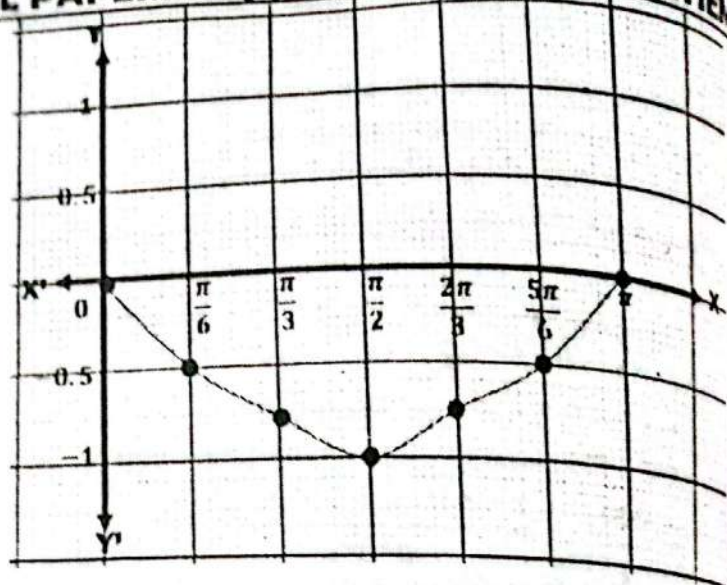
L.H.S = R.H.S

Hence Proved

Draw the graph of $y = -\sin \theta$; where $0 \leq \theta \leq \pi$

$$y = -\sin x$$

0	$\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{5\pi}{6}$	π
0	-0.5	-0.87	-1	-0.87	-0.5	0



---OR---Show that $\cos^{-1}(y) + \cos^{-1}(-y) = \pi$.

SOLUTION:

$$\cos^{-1}y + \cos^{-1}(-y) = \pi$$

Taking L.H.S:

$$\begin{aligned} &= \cos^{-1}y + \cos^{-1}(-y) \\ \because \cos^{-1}A + \cos^{-1}B &= \cos^{-1}(AB - \sqrt{(1-A^2)}\sqrt{1-B^2}) \\ &= \cos^{-1}\{(y)(-y) - \sqrt{1-y^2}\sqrt{1-(-y)^2}\} \\ &= \cos^{-1}\{-y^2 - \sqrt{(1-y^2)}\sqrt{1-y^2}\} \\ &= \cos^{-1}\{-y^2 - \sqrt{(1-y^2)^2}\} \\ &= \cos^{-1}\{-y^2 - (1-y^2)\} \\ &= \cos^{-1}(-y^2 - 1 + y^2) \\ &= \cos^{-1}(-1) \\ &= \pi \end{aligned}$$

L.H.S. = R.H.S. Hence Proved

xiv) If $\alpha + \beta + \gamma = 180^\circ$, then show that $\cos\left(\frac{\beta+\gamma}{2}\right) = \sin\frac{\alpha}{2}$

SOLUTION:

Given that

$$\begin{aligned} \alpha + \beta + \gamma &= 180^\circ \\ \alpha &= 180^\circ - (\beta + \gamma) \end{aligned}$$

Whole equation divided by '2'

$$\frac{\alpha}{2} = \frac{180^\circ}{2} - \frac{(\beta + \gamma)}{2}$$

$$\frac{\alpha}{2} = 90^\circ - \left(\frac{\beta + \gamma}{2}\right)$$

Taking sin on both sides

$$\sin \frac{\alpha}{2} = \sin \left\{ 90^\circ - \left(\frac{\beta + \gamma}{2}\right) \right\}$$

$$\because \sin(90^\circ - \theta) = \cos \theta$$

$$\sin \frac{\alpha}{2} = \cos \left(\frac{\beta + \gamma}{2}\right)$$

Find the equation of the graph of the function of the type $y = ax^2 + bx + c$ which crosses the x -axis at the points $(-4, 0)$ and $(3, 0)$ and passes through the point $(2, -4)$.

SOLUTION:

Equation of the curve which passes through x -axis at the point $(p, 0)$ and $(q, 0)$ has the form

$$f(x) = a(x - p)(x - q) \text{ --- (i)}$$

the curve which passes through the points $(-4, 0)$, $(3, 0)$ and

$$p = -4 \text{ and } q = 3$$

substituting these values in equation (i), we get

$$f(x) = a\{x - (-4)\}\{x - (3)\}$$

$$f(x) = a(x + 4)(x - 3) \text{ --- (ii)}$$

point $(2, -4)$ lies on the curve, so it must satisfy equation (ii), so

$$-4 = a(2 + 4)(2 - 3)$$

$$-4 = a(6)(-1)$$

$$-4 = -6a$$

$$a = \frac{4}{6}$$

$$a = \frac{2}{3}$$

Therefore equation (ii) becomes

$$f(x) = \frac{2}{3}(x + 4)(x - 3)$$

$$f(x) = \frac{2}{3}\{x(x - 3) + 4(x - 3)\}$$

$$f(x) = \frac{2}{3}(x^2 - 3x + 4x - 12)$$

$$f(x) = \frac{2}{3}(x^2 + x - 12) \text{ Answer}$$

SECTION "C" (Detailed Answer Questions)(40)

NOTE: Attempt any **FIVE** question from this section.
All questions carry equal marks.

3. Use Gauss, Jordan method to solve the system of linear equations
 $2x + 2y - z = 4$, $x - 2y + z = 2$ $x + y = 0$

SOLUTION:

$$2x + 2y - z = 4$$

$$x - 2y + z = 2$$

$$x + y = 0$$

Transform the system of the linear equation into matrix form

$$\text{Where } A = \begin{bmatrix} 2 & 2 & -1 \\ 1 & -2 & 1 \\ 1 & 1 & 0 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 4 \\ 2 \\ 0 \end{bmatrix}$$

Here

$$\text{Augmented matrix} = A_B = \begin{bmatrix} 2 & 2 & -1 & 4 \\ 1 & -2 & 1 & 2 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

$$A_B = \begin{bmatrix} 2 & 2 & -1 & 4 \\ 1 & -2 & 1 & 2 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \times \frac{1}{2} & 2 \times \frac{1}{2} & -1 \times \frac{1}{2} & 4 \times \frac{1}{2} \\ 1 & -2 & 1 & 2 \\ 1 & 1 & 0 & 0 \end{bmatrix} \text{ by } \left(\frac{1}{2}\right)R_1$$

$$= \begin{bmatrix} 1 & 1 & -\frac{1}{2} & 2 \\ 1 & -2 & 1 & 2 \\ 1 & 1 & 0 & 0 \end{bmatrix} \text{ by } (-1)R_1 + R_2 \text{ and } (-1)R_1 + R_3$$

$$= \begin{bmatrix} 1 & 1 & -\frac{1}{2} & 2 \\ 0 & -3 \times -\frac{1}{3} & \frac{3}{2} \times -\frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{2} & -2 \end{bmatrix} \text{ by } \left(-\frac{1}{3}\right)R_2$$

$$\begin{aligned}
 &= \begin{bmatrix} 1 & 1-1 & -\frac{1}{2} + \frac{1}{2} & 2 \\ 0 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} & -2 \end{bmatrix} \text{ by } (-1)R_2 + R_1 \\
 &= \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} & -2 \end{bmatrix} \text{ by } (2)R_3 \\
 &= \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & -\frac{1}{2} + \frac{1}{2} & 0-2 \\ 0 & 0 & 1 & -4 \end{bmatrix} \text{ by } \left(\frac{1}{2}\right)R_3 + R_2 \\
 &= \begin{bmatrix} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & -4 \end{bmatrix}
 \end{aligned}$$

by detaching the last column back to its original position from above matrix. We get

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \\ -4 \end{bmatrix}$$

OR

$x = -2, \quad y = -2, \quad z = -4$ Answer

If the p^{th} term of a H.P is q , the q^{th} term is p , then the $(p+q)^{\text{th}}$ term

SOLUTION:

Let
 $a = \frac{1}{p}$
 $d = \frac{1}{q} - \frac{1}{p}$
 $a_p = \frac{1}{p}$

$$\begin{aligned}
 &\therefore a_n = a + (n-1)d \\
 &a_p = a + (p-1)d \quad \left| \begin{array}{l} \text{FOR } a_q \\ a_q = a + (q-1)d \end{array} \right.
 \end{aligned}$$

$$\frac{1}{q} = a + dp - d \dots (i)$$

$$\frac{1}{p} = a + dq - d \dots (ii)$$

On subtracting equation (ii) from equation (i), we get

$$\frac{1}{q} = a + dp - d$$

$$\frac{1}{p} = a + dq - d$$

$$\begin{array}{r} - \quad - \quad - \quad + \\ \hline \end{array}$$

$$\frac{1}{q} - \frac{1}{p} = dp - dq$$

$$\frac{p - q}{pq} = d(p - q)$$

$$\frac{p - q}{pq(p - q)} = d$$

$$d = \frac{1}{pq} \dots (iii)$$

Putting the value of 'd' in equation (i), we get

$$\frac{1}{q} = a + \left(\frac{1}{pq}\right)p - \left(\frac{1}{pq}\right)$$

$$\frac{1}{q} = a + \frac{1}{q} - \frac{1}{pq}$$

$$\frac{1}{q} - \frac{1}{q} + \frac{1}{pq} = a$$

$$a = \frac{1}{pq}$$

For a_{p+q}

$$a_{p+q} = a + (p + q - 1)d$$

$$a_{p+q} = \frac{1}{pq} + (p + q - 1)\left(\frac{1}{pq}\right)$$

$$a_{p+q} = \frac{1}{pq} + \frac{p + q - 1}{pq}$$

$$a_{p+q} = \frac{1 + p + q - 1}{pq}$$

$$a_{p+q} = \frac{p + q}{pq}$$

For H.P.

$$\therefore T_n = \frac{1}{a_n}$$

$$T_{p+q} = \frac{1}{p+q}$$

$$T_{p+q} = \frac{\frac{pq}{pq}}{p+q}$$

Hence Proved

king, the queen and the jack of clubs are removed from a deck of playing cards and the shuffled. A card is drawn from the remaining

Find the probability of getting:

(ii) A Queen

(iii) A Club

SOLUTION:

No. of playing cards = 52

cards after removing king, queen and jack = $O(S) = 52 - 3 =$

Heart = $O(A) = 13$

queens card remaining = $O(B) = 3$

clubs card = $O(C) = 10$

$$\therefore \text{Probability} = \frac{\text{Successful outcomes}}{\text{Possible outcomes}}$$

$$\therefore P(A) = \frac{O(A)}{O(S)} = \frac{13}{49}$$

$$P(A) = \frac{13}{49}$$

$$P(B) = \frac{O(B)}{O(S)} = \frac{3}{49}$$

$$P(B) = \frac{3}{49}$$

$$P(C) = \frac{O(C)}{O(S)} = \frac{10}{49}$$

$$P(C) = \frac{10}{49}$$

Binomial Theorem to show that $4x^2 - 2x - 1 = 0$ If

$$\frac{1}{x} = \frac{2}{5} + \frac{1 \cdot 3}{2!} \left(\frac{2}{5}\right)^2 + \frac{1 \cdot 3 \cdot 5}{3!} \left(\frac{2}{5}\right)^3 + \dots$$

SOLUTION:

$$\frac{1}{x} = \frac{2}{5} + \frac{1 \cdot 3}{2!} \left(\frac{2}{5}\right)^2 + \frac{1 \cdot 3 \cdot 5}{3!} \left(\frac{2}{5}\right)^3 + \dots$$

1 to both the sides of the given series,

$$1 + \frac{1}{x} = 1 + \frac{2}{5} + \frac{1 \cdot 3}{2!} \left(\frac{2}{5}\right)^2 + \frac{1 \cdot 3 \cdot 5}{3!} \left(\frac{2}{5}\right)^3 + \dots$$

Let the series on the right hand side of (i) be identified with

$$\therefore (1+x)^n = 1 + nx + \frac{n(n-1)}{2!} x^2 + \frac{n(n-1)(n-2)}{3!} x^3 + \dots$$

By comparing above expression with equation (i)

$$1 + \frac{1}{x} = (1+x)^n \quad \dots \dots (ii)$$

And

$$nx = \frac{2}{5} \quad \dots \dots (iii)$$

Squaring both sides

$$(nx)^2 = \left(\frac{2}{5}\right)^2$$

$$n^2 x^2 = \frac{4}{25} \quad \dots \dots (iv)$$

Also

$$\frac{n(n-1)}{2!} x^2 = \frac{1 \cdot 3}{2!} \left(\frac{2}{5}\right)^2$$

$$n(n-1)x^2 = 3 \left(\frac{4}{25}\right)$$

$$n(n-1)x^2 = \frac{12}{25} \quad \dots \dots (v)$$

Equation (v) divided by (iv)

$$\frac{n(n-1)x^2}{n^2 x^2} = \frac{\frac{12}{25}}{\frac{4}{25}}$$

$$\frac{(n-1)}{n} = 3$$

$$\frac{(n-1)}{n} = 3$$

$$n-1 = 3n$$

$$3n - n = -1$$

$$2n = -1$$

$$n = -\frac{1}{2}$$

Putting the value of "n" in equation (iii) we get

$$\left(-\frac{1}{2}\right)x = \frac{2}{5}$$

$$x = -\frac{4}{5}$$

substituting the value of n and x in right hand sides of equation (II) have

$$1 + x = \left(1 - \frac{4}{5}\right)^{-\frac{1}{2}}$$

$$1 + \frac{1}{x} = \left(\frac{5-4}{5}\right)^{-\frac{1}{2}}$$

$$1 + \frac{1}{x} = \left(\frac{1}{5}\right)^{-\frac{1}{2}}$$

squaring both sides

$$\left(\frac{x+1}{x}\right)^2 = \left\{\left(\frac{1}{5}\right)^{-\frac{1}{2}}\right\}^2$$

$$\frac{x^2 + 2x + 1}{x^2} = \left(\frac{1}{5}\right)^{-1}$$

$$\frac{x^2 + 2x + 1}{x^2} = 5$$

$$x^2 + 2x + 1 = 5x^2$$

$$5x^2 - x^2 - 2x - 1 = 0$$

$$4x^2 - 2x - 1 = 0$$

$$4x^2 - 2x - 1 = 0 \text{ Hence Proved Answer}$$

The points $A(1, 1)$, $B(5, 4)$, $C(8, 9)$ and $D(0, 3)$ represent the position vectors. Show that the $ABCD$ is a Trapezium.

SOLUTION:

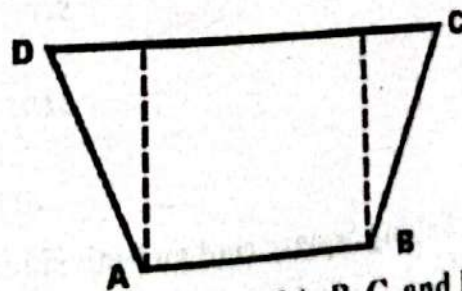
Let

$$A = (1, 1)$$

$$B = (5, 4)$$

$$C = (8, 9)$$

$$D = (0, 3)$$



Since the given points represent the position vectors of A , B , C , and D of the trapezium. So we have to show that two opposite vectors (sides) of the quadrilateral are parallel.

$$\begin{aligned} \overrightarrow{AB} &= B - A = (5, 4) - (1, 1) \\ &= (5 - 1, 4 - 1) \\ &= (4, 3) \end{aligned}$$

$$\begin{aligned}\overrightarrow{DC} &= C - D = (8, 9) - (0, 3) \\ &= (8 - 0, 9 - 3) \\ &= (8, 6) \\ &= 2(4, 3)\end{aligned}$$

$$\begin{aligned}\overrightarrow{BC} &= C - B = (8, 9) - (5, 4) \\ &= (8 - 5, 9 - 4) \\ &= (3, 5)\end{aligned}$$

$$\begin{aligned}\overrightarrow{AD} &= D - A = (0, 3) - (1, 1) \\ &= (0 - 1, 3 - 1) \\ &= (-1, 2)\end{aligned}$$

Since only one pair of opposite vectors is parallel, i.e.

$$|\overrightarrow{AB}| \parallel \overrightarrow{CD}$$

Hence Shown ABCD is a trapezium Answer

8. Solve the linear programming problem when $x \geq 0$ and $y \geq 0$
Maximize $Z(x, y) = 30x + 36y$, subject to: $4x + 2y \leq 12$; $6x + 20y \leq 20$

ANSWER: (See Question No. 8, 2024 P.E.)

9. If $\cos \theta = \frac{3}{5}$, where $0 < \theta < \frac{\pi}{2}$, then find the value of $\tan 3\theta$.

SOLUTION:

$$\because \sin^2 \theta + \cos^2 \theta = 1$$

$$\sin^2 \theta = 1 - \cos^2 \theta$$

$$\text{But } \cos \theta = \frac{3}{5}$$

$$\sin^2 \theta = 1 - \left(\frac{3}{5}\right)^2$$

$$\sin^2 \theta = 1 - \frac{9}{25}$$

$$\sin^2 \theta = \frac{25 - 9}{25}$$

$$\sin^2 \theta = \frac{16}{25}$$

Taking Square root on both sides

$$\sin \theta = \pm \frac{4}{5}$$

Since θ is in 1st Quadrant, therefore

$$\sin \theta = \frac{4}{5}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\tan \theta = \frac{4}{3}$$

$$\Rightarrow \tan 3\theta$$

$$\therefore \tan 3\theta = \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta}$$

$$\begin{aligned} \tan 3\theta &= \frac{3\tan\theta - \tan^3\theta}{1 - 3\tan^2\theta} \\ &= \frac{3\tan\theta - (\tan\theta)^3}{1 - 3(\tan\theta)^2} \end{aligned}$$

$$\begin{aligned} \text{But } \tan \theta &= \frac{4}{3} \\ &= \frac{3\left(\frac{4}{3}\right) - \left(\frac{4}{3}\right)^3}{1 - 3\left(\frac{4}{3}\right)^2} \\ &= \frac{4 - \frac{64}{27}}{1 - 3\left(\frac{16}{9}\right)} \\ &= \frac{4 - \frac{64}{27}}{1 - \frac{16}{3}} \\ &= \frac{\frac{108 - 64}{27}}{\frac{3 - 16}{3}} \\ &= \frac{44}{27} \times \frac{3}{-13} \end{aligned}$$

$$\tan 3\theta = \frac{44}{117} \text{ Answer}$$

10. Find the general solution of the trigonometric equation $4\cos^2 x - 8\sin x + 1 = 0$. Also verify the solution.

SOLUTION:

$$4\cos^2 x - 8\sin x + 1 = 0$$

$$\therefore \cos^2 \theta = 1 - \sin^2 \theta$$

$$4(1 - \sin^2 x) - 8\sin x + 1 = 0$$

$$4 - 4\sin^2 x - 8\sin x + 1 = 0$$

$$4\sin^2 x + 8\sin x - 5 = 0$$

$$4\sin^2 x + 8\sin x + 2\sin x - 2\sin x - 5 = 0$$

$$4\sin^2 x + 10\sin x - 2\sin x - 5 = 0$$

$$2\sin x(2\sin x + 5) - 1(2\sin x + 5) = 0$$

$$(2\sin x + 5)(2\sin x - 1) = 0$$

Either

$$2\sin x + 5 = 0$$

$$2\sin x = -5$$

$$\sin x = -\frac{5}{2}$$

It is not possible, because range of $\sin x$ is $-1 \leq \sin x \leq 1$

Or

$$2\sin x - 1 = 0$$

$$2\sin x = 1$$

$$\sin x = \frac{1}{2}$$

We know that the

$$\sin \frac{\pi}{6} = \frac{1}{2}$$

So, the reference angle is

$$x = \frac{\pi}{6}$$

Since $\sin \theta$ is positive in 1st and 2nd quadrants

FOR 1ST QUADRANT

$$x = \frac{\pi}{6}$$

FOR 2ND QUADRANT

$$\therefore x = \frac{\pi}{2} + x$$

$$x = \frac{\pi}{2} + \left(\frac{\pi}{6}\right)$$

$$x = \frac{3\pi + \pi}{2}$$

$$x = \frac{4\pi}{2}$$

$$x = \frac{12\pi}{12}$$

$$x = \frac{\pi}{3}$$

Also

$$\therefore x = \pi - x$$

$$x = \pi - \left(\frac{\pi}{6}\right)$$

$$x = \frac{6\pi - \pi}{6}$$

$$x = \frac{5\pi}{6}$$

$$x = \frac{5\pi}{6}$$

VERIFICATION:

FOR $x = \frac{\pi}{6}$

$$4 \cos^2 \left(\frac{\pi}{6}\right) - 8 \sin \left(\frac{\pi}{6}\right) + 1 = 0$$

$$4 \left(\cos \frac{\pi}{6} \right)^2 - 8 \sin \left(\frac{\pi}{6} \right) + 1 = 0$$

$$4 \left(\frac{\sqrt{3}}{2} \right)^2 - 8 \left(\frac{1}{2} \right) + 1 = 0$$

$$4 \left(\frac{3}{4} \right) - 4 + 1 = 0$$

$$3 - 4 + 1 = 0$$

$$0 = 0 \text{ (True)}$$

 $\frac{\pi}{3}$

$$4 \cos^2 \left(\frac{\pi}{3} \right) - 8 \sin \left(\frac{\pi}{3} \right) + 1 = 0$$

$$4 \left(\cos \frac{\pi}{3} \right)^2 - 8 \sin \left(\frac{\pi}{3} \right) + 1 = 0$$

$$4 \left(\frac{1}{2} \right)^2 - 8 \left(\frac{\sqrt{3}}{2} \right) + 1 = 0$$

$$4 \left(\frac{1}{4} \right) - 4(\sqrt{3}) + 1 = 0$$

$$1 - 4\sqrt{3} + 1 = 0$$

$$2 - 4\sqrt{3} = 0 \text{ (Not True)}$$

 $\frac{5\pi}{6}$

$$4 \cos^2 \left(\frac{5\pi}{6} \right) - 8 \sin \left(\frac{5\pi}{6} \right) + 1 = 0$$

$$4 \left(\cos \frac{5\pi}{6} \right)^2 - 8 \sin \left(\frac{5\pi}{6} \right) + 1 = 0$$

$$4 \left(-\frac{\sqrt{3}}{2} \right)^2 - 8 \left(\frac{1}{2} \right) + 1 = 0$$

$$4 \left(\frac{3}{4} \right) - 4 + 1 = 0$$

$$3 - 4 + 1 = 0$$

$$0 = 0 \text{ (True)}$$

$$\text{Solution set} = \left\{ \frac{\pi}{6} + 2n\pi \right\} \cup \left\{ \frac{5\pi}{6} + 2n\pi \right\}, n \in \mathbb{Z}$$

TIME: 3 Hours

SECTION 'A' (Multiple Choice Questions)

(100 Marks)

1. Choose the correct answer for each from the given options:

1) If $z = 4 + 3i$ then $z + \bar{z}$:

* 6i

* 10

* 8 ✓

* 9i

2) $(1 + i)^2 =$ (where $i = \sqrt{-1}$):* $-2i$ * $2i$ ✓

* 2

* -2

3) If $A = [-7]$, then $|A|$

* 7

* -7 ✓

* 1

* 0

4) If $|\vec{a} \times \vec{b}| = |\vec{a} \cdot \vec{b}|$ then the angle between $\vec{a}$ and $\vec{b}$ is:

* 0

* $\frac{\pi}{2}$ * $\frac{\pi}{4}$ ✓* π 5) $\sum_{n=3}^{20} n^0$:

* 1 ✓

* 18

* 19

* 20

6) The factorial form of 12.11.10 is:

* $\frac{12!}{9!}$ ✓

* 12!

* $\binom{12}{9}$ * $(12!)(9!)$

7) No term of a G.P. is:

* 0 ✓

* 1

* Negative

* ∞ 8) $\sin \frac{\alpha}{2} =$:* $\sqrt{\frac{(s-b)(s-c)}{bc}}$ ✓* $\sqrt{\frac{(s-c)(s-a)}{ca}}$ * $\sqrt{\frac{(s-a)(s-b)}{ba}}$ * $\sqrt{s(s-a)(s-b)(s-c)}$ 9) Solution space of the linear inequality $2x + 3y \leq 6$ includes:

* All points above the line

* All points on and below the line

* All points below the line

* All points on and above the line

10) A circle, which passes through all vertices of a triangle is:

* tri-circle

* e-circle

* incircle

* circum circle

11) If K is a scalar and A, B, C are the square matrices of the same order, then $(KABC)^t =$:* $K A^t B^t C^t$ * $K C^t B^t A^t$ ✓* $K (BA)^t$ 12) If $\frac{a-b}{b-c} = \frac{a}{b}$, then a, b and c are in:

* A.P

* G.P

* H.P ✓

$a_n = 5 - 3n + 2n^2$, then $a_{2n} =$:

* $5 - 6n + 4n^2$

* $5 - 6n + 8n^2$ ✓

When a coin is tossed, the probability of getting the tail is:

* $\frac{1}{3}$

* $\frac{1}{4}$

* $\frac{1}{6}$

$x + x^2 + x^3 + \dots =$:

* $(1 - x)^{-2}$

* $(1 + x)^{-1}$ ✓

Graph of a linear function is:

* straight line ✓

* triangle

parabola

If f and g are equal, where $f(x) = 7x - 4$ and $g(x) = x$, then $x =$

* $\frac{2}{3}$ ✓

* 1

* $\frac{4}{3}$

$\cos(2\pi + \theta) =$:

* $-\sin \theta$

* $\cos \theta$ ✓

* $-\cos \theta$

If $y = \sin x$, then its range is:

* $y \leq 1$ ✓

* $\mathbb{R}$

* $0 \leq y \leq \pi$

* $-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$

Solution of $\sqrt{2} \sin \theta - 1 = 0$ in the interval $[\frac{\pi}{2}, \pi]$ is:

* $\frac{5\pi}{4}$

* $\frac{3\pi}{4}$ ✓

* $\frac{7\pi}{4}$

SECTION 'B' (Short-Answer Questions) (40)

Attempt any Ten part questions in all. All questions equal marks. (i.e. 4 marks of each part).

Solve the quadratic equation: $z^2 - 4z + 5 = 0$ by completing the square where z is a complex number.

Solution:

$$z^2 - 4z + 5 = 0$$

$$(z)^2 - 2(z)(2) = -5$$

$$(z)^2 - 2(z)(2) + (2)^2 = -5 + (2)^2$$

$$(z - 2)^2 = -5 + 4$$

$$(z - 2)^2 = -1$$

Take square root of both sides

$$z - 2 = \sqrt{-1}$$

$$z - 2 = i$$

$$z = 2 + i \text{ Answer}$$

ii) Find the real numbers x, y, z such that matrix A is Hermitian

where $A = \begin{bmatrix} 3 & x+2i & yi \\ 3-2i & 0 & 1+zi \\ yi & i-xi & -1 \end{bmatrix}$

SOLUTION:

$$A = \begin{bmatrix} 3 & x+2i & yi \\ 3-2i & 0 & 1+zi \\ yi & i-xi & -1 \end{bmatrix}$$

$$\bar{A} = \begin{bmatrix} 3 & x+2i & yi \\ 3-2i & 0 & 1+zi \\ yi & i-xi & -1 \end{bmatrix}$$

$$\bar{A} = \begin{bmatrix} 3 & x-2i & -yi \\ 3+2i & 0 & 1-zi \\ -yi & 1+xi & -1 \end{bmatrix}$$

Taking transpose on both sides

$$(\bar{A})^t = \begin{bmatrix} 3 & x-2i & -yi \\ 3+2i & 0 & 1-zi \\ -yi & 1+xi & -1 \end{bmatrix}$$

$$(\bar{A})^t = \begin{bmatrix} 3 & 3+2i & -yi \\ x-2i & 0 & 1+xi \\ -yi & 1-zi & -1 \end{bmatrix}$$

Since the given matrix A is Hermitian therefore

$$A = (\bar{A})^t$$

$$\begin{bmatrix} 3 & x+2i & yi \\ 3-2i & 0 & 1+zi \\ yi & i-xi & -1 \end{bmatrix} = \begin{bmatrix} 3 & 3+2i & -yi \\ x-2i & 0 & 1+xi \\ -yi & 1-zi & -1 \end{bmatrix}$$

By equating elements of both matrices, we get

$\begin{aligned} x+2i &= 3+2i \\ x+2i &= 3+2i \\ x &= 3 \end{aligned}$	$\begin{aligned} yi &= -yi \\ y &= -y \\ y+y &= 0 \\ 2y &= 0 \\ y &= 0 \end{aligned}$	$\begin{aligned} 1+zi &= -1-zi \\ 1+zi &= -1-zi \\ zi &= -1-zi \\ z &= -1-zi \\ \text{But } x &= 3 \end{aligned}$
--	---	---

$x = 3, \quad y = 0, \quad z = 3$ Answer

iii) Without expanding determinants, prove that

$$\begin{vmatrix} \alpha & \alpha^2 & \alpha^3 \\ \beta & \beta^2 & \beta^3 \\ \gamma & \gamma^2 & \gamma^3 \end{vmatrix}$$

$$\begin{vmatrix} \alpha & \beta\gamma & \alpha\beta\gamma \\ \beta & \gamma\alpha & \alpha\beta\gamma \\ \gamma & \alpha\beta & \alpha\beta\gamma \end{vmatrix} = \begin{vmatrix} \alpha & \alpha^2 & \alpha^3 \\ \beta & \beta^2 & \beta^3 \\ \gamma & \gamma^2 & \gamma^2 \end{vmatrix}$$

Proved L.H.S.:

$$= \begin{vmatrix} \alpha & \beta\gamma & \alpha\beta\gamma \\ \beta & \gamma\alpha & \alpha\beta\gamma \\ \gamma & \alpha\beta & \alpha\beta\gamma \end{vmatrix}$$

Taking Common $\alpha\beta\gamma$ from C_3

$$= \alpha\beta\gamma \begin{vmatrix} \alpha & \beta\gamma & 1 \\ \beta & \gamma\alpha & 1 \\ \gamma & \alpha\beta & 1 \end{vmatrix}$$

Now Multiply α to R_1 , β to R_2 and γ to R_3

$$= \begin{vmatrix} \alpha^2 & \alpha\beta\gamma & \alpha \\ \beta^2 & \alpha\beta\gamma & \beta \\ \gamma^2 & \alpha\beta\gamma & \gamma \end{vmatrix}$$

Again Taking Common $\alpha\beta\gamma$ from C_2

$$= \alpha\beta\gamma \begin{vmatrix} \alpha^2 & 1 & \alpha \\ \beta^2 & 1 & \beta \\ \gamma^2 & 1 & \gamma \end{vmatrix}$$

Now Multiply α to R_1 , β to R_2 and γ to R_3

$$= \begin{vmatrix} \alpha^3 & \alpha & \alpha^2 \\ \beta^3 & \beta & \beta^2 \\ \gamma^3 & \gamma & \gamma^2 \end{vmatrix}$$

Interchange C_1 and C_2

$$= - \begin{vmatrix} \alpha & \alpha^3 & \alpha^2 \\ \beta & \beta^3 & \beta^2 \\ \gamma & \gamma^3 & \gamma^2 \end{vmatrix}$$

Interchange again C_2 and C_3

$$= \begin{vmatrix} \alpha & \alpha^2 & \alpha^3 \\ \beta & \beta^2 & \beta^3 \\ \gamma & \gamma^2 & \gamma^3 \end{vmatrix}$$

Proved L.H.S. = R.H.S

Q. 2. Find λ if the vectors $\hat{i} + \hat{j} + 2\hat{k}$, $\lambda\hat{i} - \hat{j} + \hat{k}$ and $3\hat{i} - 2\hat{j} - \hat{k}$ are coplanar.

SOLUTION:

$\vec{a} = \hat{i} + \hat{j} + 2\hat{k}$, $\vec{b} = \lambda\hat{i} - \hat{j} + \hat{k}$, $\vec{c} = 3\hat{i} - 2\hat{j} - \hat{k}$
 Since vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$ are coplanar, therefore

$$\begin{vmatrix} a & b & c \end{vmatrix} = 0$$

$$\begin{vmatrix} 1 & 1 & 2 \\ \lambda & -1 & 1 \\ 3 & -2 & -1 \end{vmatrix} = 0$$

$$1 \begin{vmatrix} -1 & 1 \\ -2 & -1 \end{vmatrix} - 1 \begin{vmatrix} \lambda & 1 \\ 3 & -1 \end{vmatrix} + 2 \begin{vmatrix} \lambda & -1 \\ 3 & -2 \end{vmatrix} = 0$$

$$1((-1)(-1) - (1)(-2)) - 1((\lambda)(-1) - (1)(3)) + 2((\lambda)(-2) - (-1)(3)) = 0$$

$$1(1 + 2) - 1(-\lambda - 3) + 2(-2\lambda + 3) = 0$$

$$1(3) - 1(-\lambda - 3) + 2(-2\lambda + 3) = 0$$

$$3 + \lambda + 3 - 4\lambda + 6 = 0$$

$$-3\lambda + 12 = 0$$

$$12 = 3\lambda$$

$$\lambda = \frac{12}{3}$$

$$\lambda = 4 \text{ Answer}$$

v) Find n so that $\frac{x^{n+5} + y^{n+5}}{x^{n+4} + y^{n+4}}$ may be A.M between x and y .

SOLUTION:

$$H.M. = \frac{x^{n+5} + y^{n+5}}{x^{n+4} + y^{n+4}} \quad \text{--- (i)}$$

$$H.M. = \frac{2ab}{a+b}$$

Also Harmonic mean between x and y is

$$H.M. = \frac{2xy}{x+y} \quad \text{--- (ii)}$$

On comparing expression (i) and (ii), we get

$$\frac{x^{n+5} + y^{n+5}}{x^{n+4} + y^{n+4}} = \frac{2xy}{x+y}$$

$$(x+y)(x^{n+5} + y^{n+5}) = 2xy(x^{n+4} + y^{n+4})$$

$$x(x^{n+5} + y^{n+5}) + y(x^{n+5} + y^{n+5}) = 2x \cdot x^{n+4} + 2xy \cdot y^{n+4}$$

$$x \cdot x^{n+5} + x \cdot y^{n+5} + x^{n+5} y + y \cdot y^{n+5} = 2x \cdot x^{n+4} + 2xy \cdot y^{n+4}$$

$$x \cdot x^{n+5} + x \cdot y^{n+5} + x^{n+5} y + y \cdot y^{n+5} = 2x^{n+5} + 2xy \cdot y^{n+4}$$

$$x \cdot x^{n+5} + y \cdot y^{n+5} + x \cdot y^{n+5} + x^{n+5} y = 2x^{n+5} + 2xy \cdot y^{n+4}$$

$$x \cdot x^{n+5} + y \cdot y^{n+5} + x \cdot y^{n+5} + x^{n+5} y - 2x^{n+5} - 2xy \cdot y^{n+4} = 0$$

$$x \cdot x^{n+5} + y \cdot y^{n+5} - x^{n+5} y - x \cdot y^{n+5} = 0$$

$$x \cdot x^{n+5} - x^{n+5} y - x \cdot y^{n+5} + y \cdot y^{n+5} = 0$$

$$x^{n+5}(x - y) - y^{n+5}(x - y) = 0$$

$$(x - y)(x^{n+5} - y^{n+5}) = 0$$

$$x - y = 0$$

$$x = y$$

$$x^{n+5} - y^{n+5} = 0$$

$$x^{n+5} = y^{n+5}$$

$$\frac{x^{n+5}}{y^{n+5}} = 1$$

$$\left(\frac{x}{y}\right)^{n+5} = \left(\frac{x}{y}\right)^0$$

By equating powers of both sides, we get,

$$n + 5 = 0$$

$$n = -5 \text{ Answer}$$

Find the sum of the series $3 + 6 + 15 + 42 + 123 + \dots$ to n

SOLUTION:

$$3 + 6 + 15 + 42 + 123$$

+ ... to n terms

Here, differences of consecutive terms are in G.P.

$3, 9, 27, \dots$ is in G.P.

Since the differences of consecutive terms are in G.P. so we use method of differences

$$S_n = 3 + 6 + 15 + 42 + 123 + \dots \text{ to } n \text{ term}$$

$$S_n = 3 + 6 + 15 + 42 + \dots + a_n$$

$$0 = 3 + (3 + 9 + 27 + 81 + \dots \text{ to } (n-1)) - a_n$$

$$a_n = 3 + (3 + 9 + 27 + 81 + \dots \text{ to } (n-1))$$

Therefore $(3 + 9 + 27 + 81 + \dots \text{ to } (n-1)) = S_n$

$$a_n = 3 + S_n$$

$$\therefore S_n = \frac{a(r^{n+1} - 1)}{r - 1}$$

$$\therefore a_n = 3 + \frac{a(r^{n+1} - 1)}{r - 1}$$

In case $a = 3, r = \frac{9}{3} = 3$ and number of terms = n

$$a_n = 3 + \frac{3(3^{n-1} - 1)}{3 - 1}$$

$$a_n = 3 + \frac{3(3^{n-1} - 1)}{2}$$

$$a_n = \frac{6 + 3(3^{n-1} - 1)}{2}$$

$$a_n = \frac{3}{2}(2 + 3^{n-1} - 1)$$

$$a_n = \frac{3}{2}\left(1 + \frac{3^n}{3}\right)$$

$$a_n = \frac{3}{2}\left(\frac{3 + 3^n}{3}\right)$$

$$a_n = \frac{1}{2}(3^n + 3)$$

For sum of the series,

$$S_n = \frac{1}{2} \left[\sum_{1}^n 3^n + 3 \sum_{1}^n n^0 \right]$$

$$S_n = \frac{1}{2} [(3^1 + 3^2 + 3^3 + \dots + 3^n) + 3n]$$

By using Geometrical Progression for $3^1 + 3^2 + 3^3 + \dots + 3^n$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$S_n = \frac{3(3^n - 1)}{3 - 1}$$

$$S_n = \frac{1}{2} \left[\frac{3(3^n - 1)}{2} + 3n \right]$$

$$S_n = \left[\frac{3(3^n - 1) + 6n}{4} \right]$$

$$S_n = \frac{6n + 3(3^n - 1)}{4} \quad \text{Answer}$$

vii) In how many ways can two English books, three chemistry books and four physics books be arranged on a shelf so that all the books of the same subject are together?

SOLUTION:

The Two English books can be arranged in 2P_2 , i.e. 2! Ways among themselves

The Three Chemistry books can be arranged in 3P_3 , i.e. 3! Ways among themselves

Physics books can be arranged in 4P_4 , i.e. 4! Ways among

groupings of the books on the three subjects can be

in 3P_3 , i.e. 3! Ways.

by fundamental principle,

$$\therefore \text{total number of arrangements} = {}^2P_2 \cdot {}^3P_3 \cdot {}^4P_4 \cdot {}^3P_3$$

$$\therefore {}^nP_n = n!$$

$$= 2! \times 3! \times 4! \times 3!$$

$$= (2 \times 1) \times (3 \times 2 \times 1) \times (4 \times 3 \times 2 \times 1) \times (3 \times 2 \times 1)$$

$$= 2 \times 6 \times 24 \times 6$$

$$\therefore \text{total number of arrangements} = 1728 \text{ Ways Answer}$$

Prove the proposition by mathematical induction for every positive

$$n: 2 + 4 + 6 + \dots + 2n = n(n + 1)$$

PROOF:

$$2 + 4 + 6 + \dots + 2n = n(n + 1)$$

Step I:

$$\text{For } n = 1$$

$$2 + 4 + 6 + \dots + 2(1) = (1)(1 + 1)$$

$$2 + 4 + 6 + \dots + 2 = (1)(2)$$

$$2 + 4 + 6 + \dots + 2 = 2$$

$$\text{So, } P(n) \text{ is true for } n = 1$$

Step II:

Assuming the result to be true for $n = k$, we get the hypothesis

$$2 + 4 + 6 + \dots + 2k = k(k + 1)$$

Adding $2(k + 1)$ On both Sides

$$2 + 4 + 6 + \dots + 2k + 2(k + 1)$$

$$= k(k + 1) + 2(k + 1)$$

$$= (k + 1)(k + 2)$$

$$= (k + 1)(k + 1 + 1)$$

$$= (k + 1)\{(k + 1) + 1\}$$

$$\text{i.e. } P(n) \text{ is true for } n = k + 1$$

By the principle of mathematical induction $P(n)$ is true for all numbers n . Answer

OR) Write in the simplified form the term independent of x in the

$$\text{Expansion of } \left(2x + \frac{1}{x^2}\right)^9$$

SOLUTION: (See Question No. 2[(ix) OR], 2024 P.E.)

Find the measure of the largest angle in $\triangle ABC$ with $a = 10\text{cm}$, $b = 20\text{cm}$ and $c = 26\text{cm}$.

SOLUTION:

$$a = 10\text{cm}, b = 20\text{cm and } c = 26\text{cm}$$

We know that the largest angle is opposite to the largest sides of triangle, in this case, the largest side is 'c', therefore

Applying Law of cosine

$$c^2 = a^2 + b^2 - 2ab \cos \gamma$$

$$2ab \cos \gamma = a^2 + b^2 - c^2$$

$$\cos \gamma = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\cos \gamma = \frac{(10)^2 + (20)^2 - (26)^2}{2(10)(20)}$$

$$\cos \gamma = \frac{100 + 400 - 676}{200}$$

$$\cos \gamma = \frac{400 - 676}{200}$$

$$\cos \gamma = \frac{-276}{200}$$

$$\cos \gamma = -0.44$$

$$\gamma = \cos^{-1}(-0.44)$$

$$\gamma = 116^\circ \text{ Answer}$$

The largest angle in ΔABC is γ Answer

x) Solve the following linear programming problem by graphical method when $x \leq 0, y \geq 0$ maximize the objective function $10x + 11y$, subject to constraints $2x + 3y \leq 8; 6x + 3y \leq 8$

SOLUTION:

CONSTRAINT 1:

$$2x + 3y \leq 8$$

We consider it as an equation $2x + 3y = 8$

For x - intercept put $y = 0$

$$2x + 3(0) = 8 \Rightarrow 2x = 8 \Rightarrow x = \frac{8}{2}$$

$$x = 4$$

The intersection point of line with x - axis is $(4, 0)$

For y - intercept put $x = 0$

$$2(0) + 3y = 8 \Rightarrow 3y = 8$$

$$y = \frac{8}{3}$$

The intersection point of line with y - axis is $(0, \frac{8}{3})$

For origin put $x = 0$ and put $y = 0$ in $2x + 3y \leq 8$

$$2(0) + 3(0) \leq 8$$

$$0 \leq 8 \text{ (true)}$$

CONSTRAINT 2:

$$6x + 3y \leq 10$$

consider it as an equation $6x + 3y = 10$
 x-intercept put $y = 0$

$$6x + 3(0) = 10 \Rightarrow 6x = 10 \Rightarrow x = \frac{10}{6}$$

$$x = \frac{5}{3}$$

intersection point of line with x-axis is $(\frac{5}{3}, 0)$

y-intercept put $x = 0$

$$6(0) + 3y = 10 \Rightarrow 3y = 10$$

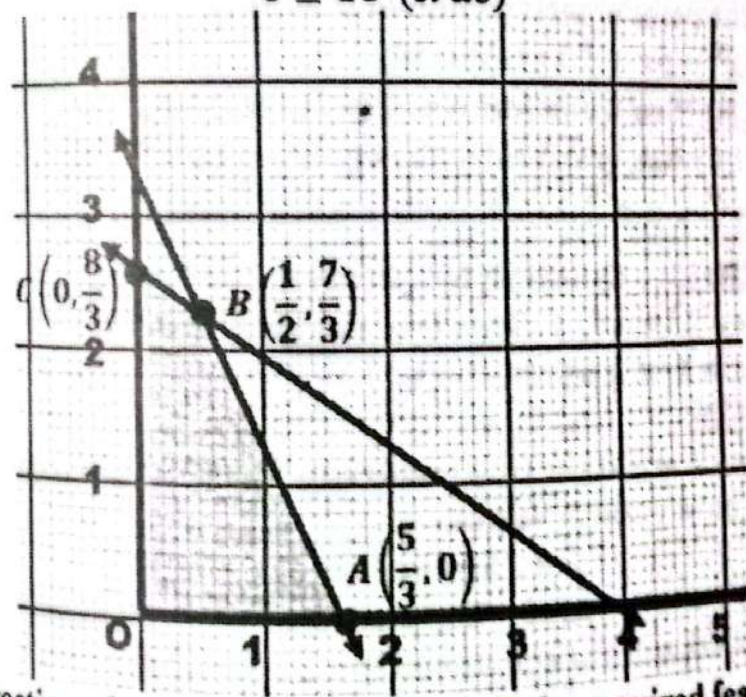
$$y = \frac{10}{3}$$

intersection point of line with y-axis is $(0, \frac{10}{3})$

origin put $x = 0$ and put $y = 0$ in $6x + 3y \leq 10$

$$2(0) + 3(0) \leq 10$$

$$0 \leq 10 \text{ (true)}$$



intersection of two linear inequalities is the required feasible region OABC which is the shaded area in the graph.

The solution points or corner points of the region OABC are $(0, 0)$, $A(\frac{5}{3}, 0)$, $B(\frac{1}{2}, \frac{7}{3})$, and $C(0, \frac{8}{3})$ in the graph

OBJECTIVE FUNCTION	
SOLUTION POINTS OR CORNER POINTS	OBJECTIVE FUNCTION $f(x, y) = 10x + 11y$

$O(0, 0)$	$f(x, y) = 10(0) + 11(0) = 0 + 0 = 0$
$A\left(\frac{5}{3}, 0\right)$	$f(x, y) = 10\left(\frac{5}{3}\right) + 11(0) = \frac{50}{3} + 0$ $= \frac{50}{3} = 16.66$
$B\left(\frac{1}{2}, \frac{7}{3}\right)$	$f(x, y) = 10\left(\frac{1}{2}\right) + 11\left(\frac{7}{3}\right) = 5 + \frac{77}{3}$ $= \frac{92}{3} = 30.66$
$C\left(0, \frac{8}{3}\right)$	$f(x, y) = 10(0) + 11\left(\frac{8}{3}\right) = 0 + \frac{88}{3}$ $= \frac{88}{3} = 29.33$

Hence

Maximum at $\left(\frac{1}{2}, \frac{7}{3}\right)$; Maximum value = 30.66 Answer

xi) Find the point of intersection of the function $f(x) = x + 4$ and $g(x) = x^2 - 6x + 10$ graphically.

ANSWER: (See Question No. 2(xv), 2024 P.E.)

xi) (OR) By using graph solve the equation $\cos \theta - \theta = 0$ for the interval $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$.

ANSWER: (See Question No. 2(xiv), 2024 P.E.)

xii) Express $15\sin \theta + 8\cos \theta$ in the form of $r \sin(\theta + \phi)$ where θ and ϕ are in first quadrant.

SOLUTION:

Since θ and ϕ are in the first quadrant, therefore, all trigonometric functions have positive values.

$$\text{Let } 15 = r \cos \phi \text{ --- (i)}$$

$$\text{and } 8 = r \sin \phi \text{ --- (ii)}$$

Squaring both sides of equation (i) and (ii), and then adding.

We get

$$(15)^2 + (8)^2 = (r \cos \phi)^2 + (r \sin \phi)^2$$

$$225 + 64 = r^2 \cos^2 \phi + r^2 \sin^2 \phi$$

$$289 = r^2 (\cos^2 \phi + \sin^2 \phi) \quad \because \cos^2 \phi + \sin^2 \phi = 1$$

$$289 = r^2 (1)$$

$$r^2 = 289$$

$$r = 17$$

So,

$$15 = (17)\cos\phi \quad \text{and} \quad 8 = (17)\sin\phi$$

$$15 = 17\cos\phi \quad \text{and} \quad 8 = 17\sin\phi$$

$$15\sin\theta + 8\cos\theta = (17\cos\phi)\sin\theta + (17\sin\phi)\cos\theta$$

$$15\sin\theta + 8\cos\theta = 17(\sin\theta\cos\phi + \cos\theta\sin\phi)$$

$$\therefore \sin(\theta + \phi) = \sin\theta\cos\phi + \cos\theta\sin\phi$$

$$15\sin\theta + 8\cos\theta = 17\sin(\theta + \phi)$$

$$\text{Where } \cos\phi = \frac{15}{17} \quad \text{and} \quad \sin\phi = \frac{8}{17} \quad \text{Answer}$$

$$\text{Prove that } r_1r_2 + r_2r_3 + r_3r_1 = s^2.$$

PROOF:

$$r_1r_2 + r_2r_3 + r_3r_1 = s^2$$

L.H.S.:

$$= r_1r_2 + r_2r_3 + r_3r_1$$

$$= \left(\frac{\Delta}{s-a} \right) \left(\frac{\Delta}{s-b} \right) + \left(\frac{\Delta}{s-b} \right) \left(\frac{\Delta}{s-c} \right) + \left(\frac{\Delta}{s-c} \right) \left(\frac{\Delta}{s-a} \right)$$

$$= \frac{\Delta^2}{(s-a)(s-b)} + \frac{\Delta^2}{(s-b)(s-c)} + \frac{\Delta^2}{(s-c)(s-a)}$$

$$= \frac{\Delta^2 (s-c) + \Delta^2 (s-a) + \Delta^2 (s-b)}{(s-a)(s-b)(s-c)}$$

$$= \frac{\Delta^2 \{(s-c) + (s-a) + (s-b)\}}{(s-a)(s-b)(s-c)}$$

$$= \frac{\Delta^2 (s-c + s-a + s-b)}{(s-a)(s-b)(s-c)}$$

$$= \frac{\Delta^2 (3s - (a+b+c))}{(s-a)(s-b)(s-c)}$$

$$\therefore s = \frac{a+b+c}{2} \Rightarrow 2s = a+b+c$$

$$= \frac{\Delta^2 (3s - 2s)}{(s-a)(s-b)(s-c)}$$

$$= \frac{\Delta^2 s}{(s-a)(s-b)(s-c)} \times \frac{s}{s}$$

$$= \frac{\Delta^2 s^2}{s(s-a)(s-b)(s-c)}$$

$$\begin{aligned} & \therefore \frac{s(s-a)(s-b)(s-c)}{s^2} \\ & = \frac{s^2}{s^2} \\ & = s^2 \end{aligned}$$

L.H.S = R.H.S

Hence Proved

xiv) Show that: $\tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{3}{5}\right) - \tan^{-1}\left(\frac{8}{19}\right) = \frac{\pi}{4}$

SOLUTION:

$$\tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{3}{5}\right) - \tan^{-1}\left(\frac{8}{19}\right) = \frac{\pi}{4}$$

Taking L.H.S.:

$$\begin{aligned} & = \tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{3}{5}\right) - \tan^{-1}\left(\frac{8}{19}\right) \\ & = \left\{ \tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{3}{5}\right) \right\} - \tan^{-1}\left(\frac{8}{19}\right) \end{aligned}$$

$$\therefore \tan^{-1}A + \tan^{-1}B = \tan^{-1}\left(\frac{A+B}{1-AB}\right)$$

$$= \tan^{-1}\left\{ \frac{\frac{3}{4} + \frac{3}{5}}{1 - \left(\frac{3}{4}\right)\left(\frac{3}{5}\right)} \right\} - \tan^{-1}\left(\frac{8}{19}\right)$$

$$= \tan^{-1}\left(\frac{\frac{15+12}{20}}{1 - \frac{9}{20}} \right) - \tan^{-1}\left(\frac{8}{19}\right)$$

$$= \tan^{-1}\left(\frac{\frac{27}{20}}{\frac{20-9}{20}} \right) - \tan^{-1}\left(\frac{8}{19}\right)$$

$$= \tan^{-1}\left(\frac{27}{20} \times \frac{20}{11} \right) - \tan^{-1}\left(\frac{8}{19}\right)$$

$$= \tan^{-1}\left(\frac{27}{11}\right) - \tan^{-1}\left(\frac{8}{19}\right)$$

$$\therefore \tan^{-1}A - \tan^{-1}B = \tan^{-1}\left(\frac{A-B}{1+AB}\right)$$

$$= \tan^{-1}\left(\frac{\frac{27}{11} - \frac{8}{19}}{1 + \left(\frac{27}{11}\right)\left(\frac{8}{19}\right)} \right)$$

$$= \tan^{-1} \left(\frac{\frac{513 - 88}{209}}{1 + \frac{216}{209}} \right)$$

$$= \tan^{-1} \left(\frac{\frac{425}{209}}{\frac{209 + 216}{209}} \right)$$

$$= \tan^{-1} \left(\frac{425}{209} \times \frac{209}{425} \right)$$

$$= \tan^{-1} \left(\frac{425}{209} \times \frac{209}{425} \right)$$

$$= \tan^{-1}(1)$$

$$= \frac{\pi}{4}$$

Hence Proved

L.H.S. = R.H.S.

SECTION "C" (Detailed Answer Questions)(40)

Attempt any FIVE question from this section.

Questions carry equal marks.

Solve the following non-homogenous system of linear equations using Jordan - Method.

$$x - y + 4z = 4$$

$$2x + 2y - z = 2$$

$$3x - 2y + 3z = -3$$

SOLUTION:

$$4z = 4$$

$$2x + 2y - z = 2$$

$$3x - 2y + 3z = -3$$

Convert the system of the linear equation into matrix form of $AX = B$

$$\begin{bmatrix} 1 & -1 & 4 \\ 2 & 2 & -1 \\ 3 & -2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ -3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -1 & 4 \\ 2 & 2 & -1 \\ 3 & -2 & 3 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } B = \begin{bmatrix} 4 \\ 2 \\ -3 \end{bmatrix}$$

$$\text{Augmented matrix} = A_B = \begin{bmatrix} 1 & -1 & 4 & 4 \\ 2 & 2 & -1 & 2 \\ 3 & -2 & 3 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -1 & 4 & 4 \\ 2-2 & 2+2 & -1-8 & 2-8 \\ 3-3 & -2+3 & 3-12 & -3-12 \end{bmatrix}$$

by $R_2 - (2)R_1$ and $R_3 - (3)R_1$

$$\begin{aligned}
 A_B &= \left[\begin{array}{ccc|c} 1 & -1 & 4 & 4 \\ 0 & \frac{4}{4} & -\frac{9}{4} & -\frac{6}{4} \\ 0 & 1 & -9 & -15 \end{array} \right] \text{By } \left(\frac{1}{4}\right)R_2 \\
 &= \left[\begin{array}{ccc|c} 1 & -1+1 & 4-\frac{9}{4} & 4-\frac{3}{2} \\ 0 & 1 & -\frac{9}{4} & -\frac{3}{2} \\ 0 & 1-1 & -9+\frac{9}{4} & -15+\frac{3}{2} \end{array} \right] \text{By } R_1 + R_2 \text{ and } R_3 \\
 &= \left[\begin{array}{ccc|c} 1 & 0 & \frac{7}{4} & \frac{5}{2} \\ 0 & 1 & -\frac{9}{4} & -\frac{3}{2} \\ 0 & 0 & \frac{-27}{4} \times -\frac{4}{27} & \frac{-27}{2} \times -\frac{4}{27} \end{array} \right] \text{By } \left(-\frac{4}{27}\right)R_3 \\
 &= \left[\begin{array}{ccc|c} 1 & 0 & \frac{7}{4} - \frac{7}{4} & \frac{5}{2} - \frac{7}{2} \\ 0 & 1 & -\frac{9}{4} + \frac{9}{4} & -\frac{3}{2} + \frac{9}{2} \\ 0 & 0 & 1 & 2 \end{array} \right] \text{By } R_1 - \left(\frac{7}{4}\right)R_3 \text{ and } R_2 + \\
 &= \left[\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 2 \end{array} \right]
 \end{aligned}$$

Find X by detaching the last column back to its original position in the above matrix. We get

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \\ 2 \end{bmatrix}$$

OR

$$x = -1, \quad y = 3, \quad z = 2 \quad \text{Answer}$$

4. A room has 3 lamps from a collection of 10 light bulbs of which 6 are no good, a person selects 3 at random and puts them in the sockets. What is the probability that he will light?

SOLUTION:

No. of lamps = 3

Total No. of light bulbs = 10

No. of bulbs which are not good = 6

No. of bulbs which are good = 4

there is following four probability combinations

Good bulbs	Not Good bulbs	Not Good bulbs
Good bulbs	Good bulbs	Not Good bulbs
Good bulbs	Good bulbs	Good bulbs

$$\text{No. of combinations} = ({}^4C_1 \times {}^6C_2) + ({}^4C_2 \times {}^6C_1) + ({}^4C_3 \times {}^6C_0)$$

$$= (1 \times 20) + (4 \times 15) + (6 \times 6) + (4 \times 1)$$

$$\text{No. of combinations} = 60 + 36 + 4 = 100$$

$$\text{Successful outcomes} = 100$$

$$\text{Possible outcomes} = {}^{10}C_3 = 120$$

$$\therefore \text{Probability} = \frac{\text{Successful outcomes}}{\text{Possible outcomes}}$$

$$\text{Probability} = \frac{100}{120}$$

$$\text{Probability} = \frac{5}{6}$$

probability that he will have light is $\frac{5}{6}$ Answer

$$\frac{1}{x} = \frac{2}{5} + \frac{1 \cdot 3}{2!} \left(\frac{2}{5}\right)^2 + \frac{1 \cdot 3 \cdot 5}{3!} \left(\frac{2}{5}\right)^3 + \dots \text{ then show that } 4x^2 - 1 = 0.$$

SOLUTION:

$$\frac{1}{x} = \frac{2}{5} + \frac{1 \cdot 3}{2!} \left(\frac{2}{5}\right)^2 + \frac{1 \cdot 3 \cdot 5}{3!} \left(\frac{2}{5}\right)^3 + \dots$$

Multiplying both sides of the given series, we have

$$1 + \frac{1}{x} = 1 + \frac{2}{5} + \frac{1 \cdot 3}{2!} \left(\frac{2}{5}\right)^2 + \frac{1 \cdot 3 \cdot 5}{3!} \left(\frac{2}{5}\right)^3 + \dots \text{ --- (i)}$$

The series on the right hand side of (i) be identified with

$$\therefore (1+x)^n = 1 + nx + \frac{n(n-1)}{2!} x^2 + \dots$$

Comparing above expression with equation (i)

$$1 + \frac{1}{x} = (1+x)^n \text{ --- (ii)}$$

Comparing both sides

$$nx = \frac{2}{5} \text{ --- (iii)}$$

$$(nx)^2 = \left(\frac{2}{5}\right)^2$$

$$n^2 x^2 = \frac{4}{25} \dots\dots (iv)$$

Also

$$\frac{n(n-1)}{2!} x^2 = \frac{1 \cdot 3}{2!} \left(\frac{2}{5}\right)^2$$

$$n(n-1)x^2 = 3\left(\frac{4}{25}\right)$$

$$n(n-1)x^2 = \frac{12}{25} \dots\dots (v)$$

Equation (v) divided by (iv)

$$\frac{n(n-1)x^2}{n^2 x^2} = \frac{\frac{12}{25}}{\frac{4}{25}}$$

$$\frac{(n-1)}{n} = 3$$

$$\frac{(n-1)}{n} = 3$$

$$n-1 = 3n$$

$$3n - n = -1$$

$$2n = -1$$

$$n = -\frac{1}{2}$$

Putting the value of "n" in equation (iii) we get

$$\left(-\frac{1}{2}\right)x = \frac{2}{5}$$

$$x = -\frac{4}{5}$$

By substituting the value of n and x in right hand sides of equation (ii) we have

$$1+x = \left(1-\frac{4}{5}\right)^{-\frac{1}{2}}$$

$$1+\frac{1}{x} = \left(\frac{5-4}{5}\right)^{-\frac{1}{2}}$$

$$1+\frac{1}{x} = \left(\frac{1}{5}\right)^{-\frac{1}{2}}$$

Squaring both sides

$$\left(\frac{x+1}{x}\right)^2 = \left\{\left(\frac{1}{5}\right)^{-\frac{1}{2}}\right\}^2$$

$$\frac{x^2 + 2x + 1}{x^2} = \left(\frac{1}{5}\right)^{-1}$$

$$\frac{x^2 + 2x + 1}{x^2} = 5$$

$$x^2 + 2x + 1 = 5x^2$$

$$5x^2 - x^2 - 2x - 1 = 0$$

$$4x^2 - 2x - 1 = 0$$

$4x^2 - 2x - 1 = 0$ Hence Proved Answer

$f: \mathbb{R} \rightarrow \mathbb{R}$ is the function defined by $f(x) = 5x + 7$. Find $f^{-1}(x)$

SOLUTION:

$$f(x) = 5x + 7$$

$$f(x) = 5x + 7 = y$$

$$5x + 7 = y$$

$$5x = y - 7$$

$$x = \frac{y - 7}{5}$$

Replacing x by $f^{-1}(y)$, we get

$$f^{-1}(y) = \frac{y - 7}{5}$$

For $f^{-1}(x)$, replace y by x , i.e.

$$f^{-1}(x) = \frac{x - 7}{5}$$

Verification

$$\begin{aligned} f^{-1}[f(x)] &= f^{-1}(5x + 7) = f^{-1}(y) \\ &= \frac{y - 7}{5} = \frac{5x + 7 - 7}{5} = \frac{5x}{5} \end{aligned}$$

$$f^{-1}[f(x)] = x$$

Hence Verified

Verify that $\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ = \frac{1}{16}$ by using trigonometric formulae:

SOLUTION:

$$\begin{aligned} &\Rightarrow \cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ \\ &= \cos 60^\circ (\cos 20^\circ \cos 40^\circ \cos 80^\circ) \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{2} \{ \cos(30^\circ - 10^\circ) \cos(30^\circ + 10^\circ) \cos(90^\circ - 10^\circ) \} \\
&\quad \because \cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta \\
&\quad \because \cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta \\
&= \frac{1}{2} [\{ \cos 30^\circ \cos 10^\circ + \sin 30^\circ \sin 10^\circ \} \{ \cos 30^\circ \cos 10^\circ \\
&\quad - \sin 30^\circ \sin 10^\circ \} \{ \cos 90^\circ \cos 10^\circ + \sin 90^\circ \sin 10^\circ \}] \\
&= \frac{1}{2} \left[\left\{ \left(\frac{\sqrt{3}}{2} \right) \cos 10^\circ + \left(\frac{1}{2} \right) \sin 10^\circ \right\} \left\{ \left(\frac{\sqrt{3}}{2} \right) \cos 10^\circ \right. \right. \\
&\quad \left. \left. - \left(\frac{1}{2} \right) \sin 10^\circ \right\} \{ (0) \cos 10^\circ + (1) \sin 10^\circ \} \right] \\
&= \frac{1}{2} \left[\left(\frac{\sqrt{3}}{2} \cos 10^\circ + \frac{1}{2} \sin 10^\circ \right) \left(\frac{\sqrt{3}}{2} \cos 10^\circ - \frac{1}{2} \sin 10^\circ \right) (\sin 10^\circ) \right] \\
&= \frac{1}{2} \left[\left(\frac{\sqrt{3}}{2} \cos 10^\circ \right)^2 - \left(\frac{1}{2} \sin 10^\circ \right)^2 \right] (\sin 10^\circ) \\
&= \frac{1}{2} \left(\frac{3}{4} \cos^2 10^\circ - \frac{1}{4} \sin^2 10^\circ \right) (\sin 10^\circ) \\
&= \frac{1}{2} \left(\frac{3 \cos^2 10^\circ - \sin^2 10^\circ}{4} \right) (\sin 10^\circ) \\
&= \frac{1}{8} (3 \cos^2 10^\circ - \sin^2 10^\circ) (\sin 10^\circ) \\
&= \frac{1}{8} \{ 3(1 - \sin^2 10^\circ) - \sin^2 10^\circ \} (\sin 10^\circ) \\
&= \frac{1}{8} (3 - 3 \sin^2 10^\circ - \sin^2 10^\circ) (\sin 10^\circ) \\
&= \frac{1}{8} (3 - 4 \sin^2 10^\circ) (\sin 10^\circ) \\
&= \frac{1}{8} (3 \sin 10^\circ - 4 \sin^3 10^\circ) \\
&\quad \because 3 \sin \theta - 4 \sin^3 \theta = \sin 3\theta \\
&= \frac{1}{8} \{ \sin 3(10^\circ) \} \\
&= \frac{1}{8} (\sin 30^\circ) \\
&= \frac{1}{8} \left(\frac{1}{2} \right) \\
&= \frac{1}{16} \text{ Answer}
\end{aligned}$$

General solution of the trigonometric equation $\sin 4\theta - \cos 3\theta = 0$ and verify the solution.

$$\sin 4\theta - \sin 2\theta - \cos 3\theta = 0$$

$$\sin 4\theta - \sin 2\theta = \cos 3\theta$$

$$\therefore \sin u - \sin v = 2 \cos \frac{u+v}{2} \sin \frac{u-v}{2}$$

$$2 \cos \frac{4\theta + 2\theta}{2} \sin \frac{4\theta - 2\theta}{2} = \cos 3\theta$$

$$2 \cos \frac{6\theta}{2} \sin \frac{2\theta}{2} = \cos 3\theta$$

$$2 \cos 3\theta \sin \theta = \cos 3\theta$$

$$2 \cos 3\theta \sin \theta - \cos 3\theta = 0$$

$$\cos 3\theta (2 \sin \theta - 1) = 0$$

Or

$$\cos 3\theta = 0$$

$$3\theta = \cos^{-1}(0)$$

$$3\theta = \frac{\pi}{2}$$

$$\theta = \frac{\pi}{6}$$

$$2 \sin \theta - 1 = 0$$

$$2 \sin \theta = 1$$

$$\sin \theta = \frac{1}{2}$$

$$\theta = \sin^{-1}\left(\frac{1}{2}\right)$$

$$\theta = \frac{\pi}{6}$$

Since $\sin \theta$ is positive in 1st and 2nd quadrants

FOR 2ND QUADRANTS

$$\therefore x = \pi - x$$

$$x = \pi - \frac{\pi}{6}$$

$$x = \frac{6\pi - \pi}{6}$$

$$x = \frac{5\pi}{6}$$

Also

$$\therefore x = \frac{\pi}{2} + x$$

$$x = \frac{\pi}{2} + \frac{\pi}{6}$$

$$x = \frac{3\pi + \pi}{6} = \frac{4\pi}{6}$$

$$x = \frac{2\pi}{3}$$

VERIFICATION:

FOR $\theta = \frac{\pi}{6}$

$$\sin 4\left(\frac{\pi}{6}\right) - \sin 2\left(\frac{\pi}{6}\right) = \cos 3\left(\frac{\pi}{6}\right)$$

$$\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} = 0$$

$$0 = 0 \text{ (True)}$$

FOR $\theta = \frac{5\pi}{6}$

$$\sin 4\left(\frac{5\pi}{6}\right) - \sin 2\left(\frac{5\pi}{6}\right) = \cos 3\left(\frac{5\pi}{6}\right)$$

$$-\frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} = 0$$

$$0 = 0 \text{ (True)}$$

FOR $\theta = \frac{2\pi}{3}$

$$\sin 4\left(\frac{2\pi}{3}\right) - \sin 2\left(\frac{2\pi}{3}\right) = \cos 3\left(\frac{2\pi}{3}\right)$$

$$\frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} = 1$$

$$\frac{2\sqrt{3}}{2} = 1$$

$$\sqrt{3} = 1 \text{ (Not True)}$$

Hence,

$$\text{Solution set} = \left\{\frac{\pi}{6} + 2n\pi\right\} \cup \left\{\frac{5\pi}{6} + 2n\pi\right\} n \in \mathbb{Z}$$

9. A, B, C are the points $\vec{a}, \vec{b}$ and $2\vec{a} - \vec{b}$ respectively. D divides $\overline{AC}$ in 2 : 3 and E divides $\overline{BD}$ in 4 : 1. Find the position vector of E .

SOLUTION:

$$A = \vec{a}$$

$$B = \vec{b}$$

$$C = 2\vec{a} - \vec{b}$$

Since D divides the $\overline{AC}$ in 2 : 3 internally (i.e. $m + n = 2 + 3$), then

$$\therefore \vec{r} = \frac{m\vec{b} + n\vec{a}}{m + n}$$

$$\therefore \vec{d} = \frac{(2)(2\vec{a} - \vec{b}) + (3)\vec{a}}{2 + 3}$$

$$\vec{d} = \frac{4\vec{a} - 2\vec{b} + 3\vec{a}}{5}$$

$$\vec{d} = \frac{7\vec{a} - 2\vec{b}}{5}$$

$$\text{Point D} = \frac{7\vec{a} - 2\vec{b}}{5}$$

Point D divides the $\vec{BD}$ in 4:1 internally (i.e. $m + n = 4:1$), therefore

$$\therefore \vec{r} = \frac{m\vec{b} + n\vec{d}}{m + n}$$

$$(4) \left(\frac{7\vec{a} - 2\vec{b}}{5} \right) + (1)\vec{b}$$

$$\therefore \vec{e} = \frac{4 + 1}{5}$$

$$\frac{28\vec{a} - 8\vec{b}}{5} + \vec{b}$$

$$= \frac{28\vec{a} - 8\vec{b} + 5\vec{b}}{5} \times \frac{1}{5}$$

$$\vec{e} = \frac{28\vec{a} - 3\vec{b}}{25} \text{ Answer}$$

NEW MODEL QUESTION PAPER

Annual Examination 2023

MATHEMATICS

CLASS XI (SCIENCE)

Time: 3 Hours

Total Marks: 100

SECTION 'A' (COMPULSORY)

MULTIPLE CHOICE QUESTIONS (M.C.Qs)

(20 Marks)

Choose the correct answer for each of the given options:
The part of $\frac{2+i}{1}$ is equal to:

* 2

* -1

* $\frac{1}{2}$

* -1

* i

* -i

iii) Let k be a scalar and A, B, C be the square matrix of the same order, then $(kABC)^t =$
 * $kA^tB^tC^t$ * $kC^tB^tA^t$ ✓ * $kABC$ * $kCBA$

iv) If A is idempotent matrix, then:
 * $A^2 = 1$ * $A^2 = 0$ * $A^2 = A$ ✓ * $A^2 = A^t$

v) The distance of the point $(-3, 4, 5)$ from the origin is:
 * 50 * $5\sqrt{2}$ * 6 * $2\sqrt{5}$ ✓

vi) A sequence is a function whose domain is set of:
 * integers * rational numbers
 * natural numbers ✓ * real numbers

vii) Let r be the common ratio of a geometric series, if $|r| > 1$, the infinite geometric series is:
 * convergent * divergent ✓ * undefined * harmonic series

viii) $\sum_{n=40}^{46} 100n^0 =$
 * 100 * 46 * 600 * 4600

ix) The product of number 16.17.18.19 is also equal to:
 * ${}^{20}P_4$ * ${}^{20}C_4$ * ${}^{19}P_4$ ✓ * ${}^{19}P_4$

x) Two teams A and B are playing a match, the probability that A wins is:

* $\frac{1}{3}$ ✓ * $\frac{2}{3}$ * $\frac{1}{2}$ * 1

xi) $1 + 2x + 3x^2 + 4x^3 + \dots =$
 * $(1+x)^{-1}$ * $(1-x)^{-1}$ * $(1+x)^{-2}$ * $(1-x)^{-2}$ ✓

xii) Domain of the function $f(x) = \frac{2x-3}{4x-5}$ is the set of all real numbers except

* $\frac{3}{2}$ * 3 * $\frac{5}{4}$ * 5

xiii) The function $\sin(\theta)$ is an/a:
 * even ✓ * odd * linear * neither even nor odd

xiv) Solution space of linear inequality $2x + 3y \leq 6, \forall x, y \in \mathbb{R}$ includes all points

* above the line * below the line
 * on and above the line * on and below the line ✓

xv) The feasible solution which maximizes or minimizes the objective function is called the:

* optimal solution ✓ * corner solution
 * initial solution * complex solution

xvi) A group of angles are allied angle of basic angle θ , if the difference of any two of them gives the integral multiple of θ .

* 180° or π radians* 45° or $\frac{\pi}{4}$ radians

A circle, which passes through all vertices of a triangle is called

*incircle

*tri-circle

*e-circle

In any equilateral triangle ABC, with usual notations, $r:R:r_1 =$

*intransitive✓

*auxiliary

*modal

* 3:2:1

* 1:3:2

*1:1:1

$$\sec\left[\sin^{-1}\left(-\frac{1}{2}\right)\right] =$$

$$*\frac{\sqrt{3}}{2}$$

$$*\frac{1}{2}$$

$$*-\frac{2}{\sqrt{3}}$$

The solution of $\sqrt{2}\sin\theta - 1 = 0$ in the interval $\left[\frac{\pi}{2}, \pi\right]$

$$*\frac{3\pi}{4}$$

$$*\frac{5\pi}{4}$$

$$*\frac{7\pi}{4}$$

SECTION 'B' (SHORT-ANSWER QUESTIONS)(Marks: $10 \times 5 = 50$)

NOTE: Attempt any TEN part questions from this section, selecting at least THREE part questions from each question. You may choose the tenth part question from any one sub-section. All questions carry equation marks.

COMPLEX NUMBER, MATRIX AND DETERMINANT.**VECTORS AND FUNCTION & GRAPHS**

1. i). Solve the quadratic equation $z^2 - 6z = -13$ by completing the squares, where z is a complex number.

SOLUTION:

$$z^2 - 6z = -13$$

$$(z)^2 - 2(z)(3) + (3)^2 = -13 + (3)^2$$

$$(z - 3)^2 = -13 + 9$$

$$(z - 3)^2 = -4$$

Taking square root of both sides

$$z - 3 = \pm\sqrt{-4}$$

$$z - 3 = \pm 2i$$

$$z = 3 \pm 2i \text{ Answer}$$

Find the period of the following period matrix. $\begin{vmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{vmatrix}$

SOLUTION:

$$\text{Let } A = \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}$$

$$\begin{aligned} \text{Now } A^2 &= \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix} \cdot \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix} \\ &= \begin{bmatrix} (1)(1) + (-2)(-3) + (-6)(2) & (1)(-2) + (-2)(2) + (-6)(0) & (1)(-6) + (-2)(9) + (-6)(-3) \\ (-3)(1) + (2)(-3) + (9)(2) & (-3)(-2) + (2)(2) + (9)(0) & (-3)(-6) + (2)(9) + (9)(-3) \\ (2)(1) + (0)(-3) + (-3)(2) & (2)(-2) + (0)(2) + (-3)(0) & (2)(-6) + (0)(9) + (-3)(-3) \end{bmatrix} \\ &= \begin{bmatrix} 1 + 6 - 12 & -2 - 4 + 0 & -6 - 18 + 18 \\ -3 - 6 + 18 & 6 + 4 + 0 & 18 + 18 - 27 \\ 2 + 0 - 6 & -4 + 0 + 0 & -12 + 0 + 9 \end{bmatrix} \\ &= \begin{bmatrix} -5 & -6 & -6 \\ 9 & 10 & 9 \\ -4 & -4 & -3 \end{bmatrix} \end{aligned}$$

Again we multiply by A

$$\begin{aligned} A^2 \cdot A &= \begin{bmatrix} -5 & -6 & -6 \\ 9 & 10 & 9 \\ -4 & -4 & -3 \end{bmatrix} \cdot \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix} \\ &= \begin{bmatrix} (-5)(1) + (-6)(-3) + (-6)(2) & (-5)(-2) + (-6)(2) + (-6)(0) & (-5)(-6) + (-6)(9) + (-6)(-3) \\ (9)(1) + (10)(-3) + (9)(2) & (9)(-2) + (10)(2) + (9)(0) & (9)(-6) + (10)(9) + (9)(-3) \\ (-4)(1) + (-4)(-3) + (-3)(2) & (-4)(-2) + (-4)(2) + (-3)(0) & (-4)(-6) + (-4)(9) + (-3)(-3) \end{bmatrix} \\ &= \begin{bmatrix} -5 + 18 - 12 & 10 - 12 + 0 & 30 - 54 + 18 \\ 9 - 30 + 18 & -18 + 20 + 0 & -54 + 90 - 27 \\ -4 + 12 - 6 & 8 - 8 + 0 & 24 - 36 + 9 \end{bmatrix} \\ &= \begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix} \end{aligned}$$

The period of matrix is $A^{k+1} = A$

$$A^3 = A^{2+1} = A$$

Hence the period of given matrix is 2.

iii). Without expanding determinants, prove that.

$$\begin{vmatrix} \alpha & \beta\gamma & \alpha\beta\gamma \\ \beta & \gamma\alpha & \alpha\beta\gamma \\ \gamma & \alpha\beta & \alpha\beta\gamma \end{vmatrix} = \begin{vmatrix} \alpha & \alpha^2 & \alpha^3 \\ \beta & \beta^2 & \beta^3 \\ \gamma & \gamma^2 & \gamma^3 \end{vmatrix}$$

ANSWER: (See Question No. 2(iii), 2023)

iv). A force of 22N is applied to the end of 0.15 meter wrench at an angle of 75 degrees with the axis of rotation. Calculate the magnitude of the moment $\vec{M}_O$ produced by applied force.

SOLUTION:

$$\text{Force} = \vec{F} = 22N$$

$$r = 0.15m$$

$$\theta = 75^\circ$$

$$\text{Magnitude of the moment} = \tau = ?$$

$$\therefore \vec{M}_o = \vec{r} \times \vec{F}$$

$$M_o = |\vec{r}| |\vec{F}| \sin \theta$$

$$\tau = (0.15)(22)(\sin 75^\circ)$$

$$\tau = 3.3(0.9659)$$

$$\tau = 3.187 \text{ N.m Answer}$$

Find the point of intersection of the following function graphically.

$$f(x) = x + 2 \text{ and } g(x) = x^2 - 4x + 6$$

SOLUTION:

$$f(x) = x + 2 \text{ and } g(x) = x^2 - 4x + 6$$

$$y = x + 2$$

$$y = x^2 - 4x + 6$$

$$y = x + 2$$

$$\text{Put } x = 1$$

$$y = 1 + 2$$

$$y = 3$$

$$\text{Put } x = 2$$

$$y = 2 + 2$$

$$y = 4$$

$$\text{Put } x = 3$$

$$y = 3 + 2$$

$$y = 5$$

$$\text{Put } x = 4$$

$$y = 4 + 2$$

$$y = 6$$

1	2	3	4
3	4	5	6

$$y = x^2 - 4x + 6$$

$$\text{Put } x = 1$$

$$y = (1)^2 - 4(1) + 6$$

$$y = 1 - 4 + 6$$

$$y = 3$$

$$\text{Put } x = 2$$

$$y = (2)^2 - 4(2) + 6$$

$$y = 4 - 8 + 6$$

$$y = 2$$

$$\text{Put } x = 3$$

$$y = (3)^2 - 4(3) + 6$$

$$y = 9 - 12 + 6$$

$$y = 3$$

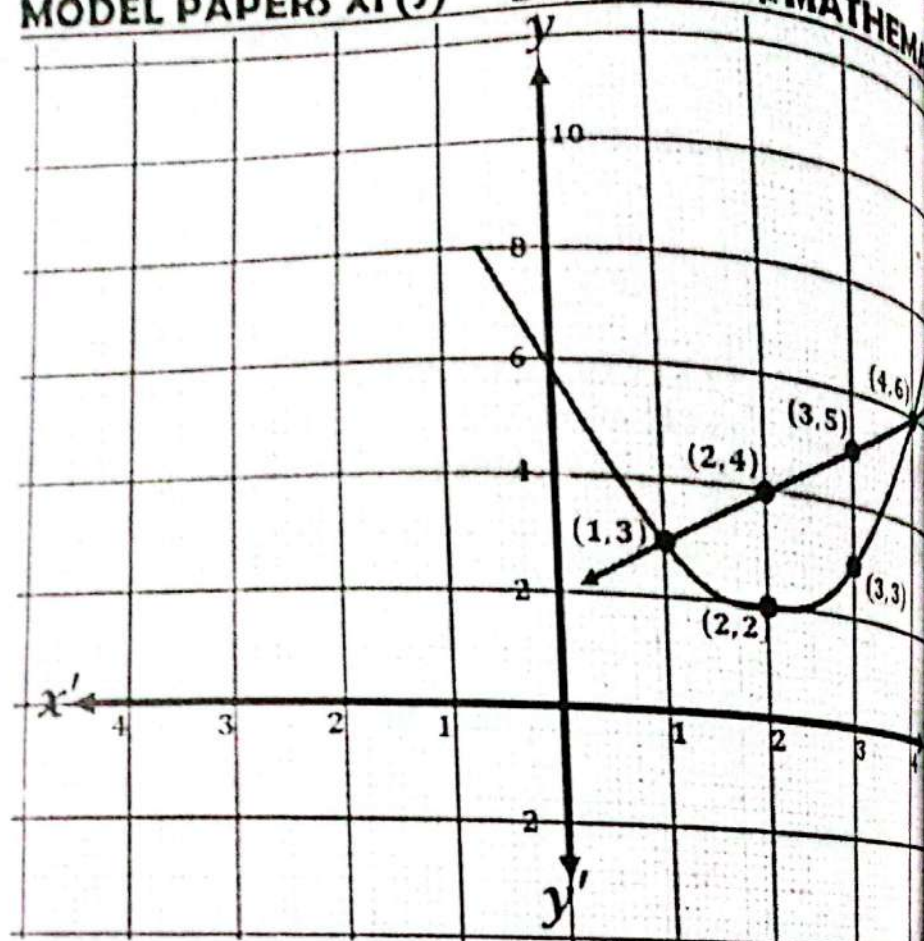
$$\text{Put } x = 4$$

$$y = (4)^2 - 4(4) + 6$$

$$y = 16 - 16 + 6$$

$$y = 6$$

x	1	2	3	4
y	3	2	3	6

**SEQUENCE & SERIES, PROBABILITY, MATHEMATICS****INDUCTION & BINOMIAL THEOREM**

Q. 3. i). If the p th term of an H. P. is q , the q th term is p ; prove that
 $(p + q)^{th}$ term is $\frac{pq}{(p+q)}$.

ANSWER: (See Question No. 4, 2024 G.S.)

ii). Find the sum of the following series. $3 + 6 + 21 + 96 + 471 + \dots$ to n terms

ANSWER: (See Question No. 2(vi), 2024 P.E.)

iii). How many words can be formed by 3 vowels and 4 consonants of 5 vowels and 7 consonants.

SOLUTION:

No. of vowels = 5

No. of consonants = 9

No. of letters in a word = 7

A word formed by 3 vowels out of 5 is $\binom{5}{3}$ and 4 consonants out of 9 is $\binom{9}{4}$. Also there is no limits of arrangements of letters therefore,

Product Principle, the required number of words are:

$$\begin{aligned} \text{Total No. of words} &= \binom{5}{3} \cdot \binom{7}{4} \cdot 5! = {}^5C_3 \times {}^7C_4 \times 5! \\ &= 10 \times 35 \times 120 \end{aligned}$$

$$\text{Total No. of words} = 42000 \text{ words Answer}$$

If the probability of solving a problem by two students Ahsan and Umar are $\frac{1}{2}$ and $\frac{1}{3}$ respectively then what is the probability of the problem to be solved.

SOLUTION:

Let A and B be the events of solving the problem by Ahsan and Umar respectively

$$\text{Probability of Ahsan} = P(A) = \frac{1}{2}$$

$$\text{Probability of Umar} = P(B) = \frac{1}{3}$$

$$\text{Probability of the problem to be solved} = P(A \cup B) = ?$$

Since both events are independent therefore,

$$\therefore P(A \cap B) = P(A) \cdot P(B)$$

$$P(A \cap B) = \frac{1}{3} \times \frac{1}{2}$$

$$P(A \cap B) = \frac{1}{6}$$

By addition theorem of probability, we have

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= \frac{1}{3} + \frac{1}{2} - \frac{1}{6}$$

$$= \frac{2 + 3 - 1}{6} = \frac{4}{6}$$

$$P(A \cup B) = \frac{2}{3}$$

Hence the probability of the problem to be solved is $\frac{2}{3}$ Answer

v). Prove the following statement by mathematical induction. $7^n - 4^n$ is divisible by 3.

SOLUTION:

Condition I:

$$7^n - 4^n$$

$$= 7^n - 4^n$$

$$\text{Put } n = 1$$

$$7^{(1)} - 4^{(1)} = 7 - 4$$

$$= 3$$

Since 3 is divisible by 3, the given statement is true for $n = 1$

Condition II:

Suppose the given statement is true for $n = k$
 $= 7^k - 4^k$

Assume that the above statement is divisible by 3

Condition III:

Put $n = k + 1$

$$7^{(k+1)} - 4^{(k+1)} = 7^1 \cdot 7^k - 4^1 \cdot 4^k \\ = 7 \cdot 7^k - 4 \cdot 4^k$$

Adding and subtracting $7 \cdot 4^k$

$$= 7 \cdot 7^k - 4 \cdot 4^k + 7 \cdot 4^k - 7 \cdot 4^k \\ = 7 \cdot 7^k - 7 \cdot 4^k + 7 \cdot 4^k - 4 \cdot 4^k \\ = 7 \cdot (7^k - 4^k) + 3 \cdot 4^k$$

By our hypothesis $(7^k - 4^k)$ is divisible by 3 and $3 \cdot 4^k$ is obviously divisible by 3. Therefore, the given expression is also divisible by 3
 $n = k + 1$

Thus, by the principle of mathematical induction it is true for all positive integer n .

LINEAR PROGRAMING AND TRIGONOMETRY

Q. 4. i). Solve the following LP programming problems by graphical method when $x \geq 0, y \geq 0$.

Maximize the objective function $z = (x, y) = 10x + 11y$,
 Subject to the constraints $2x + 3y \leq 8$; $6x + 3y \leq 10$.

ANSWER: (See Question No. 2(x), 2023)

ii). Express $4\sin\theta + 3\cos\theta$ in the form $r\sin(\theta + \phi)$, where θ and ϕ are in first quadrant.

SOLUTION:

Since θ and ϕ are in the first quadrant, therefore, all trigonometric functions have positive values.

$$\text{Let } 4 = r\cos\phi \text{ --- (i)}$$

$$\text{and } 3 = r\sin\phi \text{ --- (ii)}$$

Squaring both sides of equation (i) and (ii), and then adding.
 We get

$$(4)^2 + (3)^2 = (r\cos\phi)^2 + (r\sin\phi)^2$$

$$16 + 9 = r^2\cos^2\phi + r^2\sin^2\phi$$

$$25 = r^2(\cos^2\phi + \sin^2\phi)$$

$$\because \cos^2\phi + \sin^2\phi = 1$$

$$25 = r^2(1)$$

$$r^2 = 25$$

$$r = 5$$

$$4 = (5)\cos\phi \quad \text{and} \quad 3 = (5)\sin\phi$$

$$4 = 5\cos\phi \quad \text{and} \quad 3 = 5\sin\phi$$

$$4\sin\theta + 3\cos\theta = (5\cos\phi)\sin\theta + (5\sin\phi)\cos\theta$$

$$4\sin\theta + 3\cos\theta = 5(\sin\theta\cos\phi + \cos\theta\sin\phi)$$

$$\because \sin(\theta + \phi) = \sin\theta\cos\phi + \cos\theta\sin\phi$$

$$4\sin\theta + 3\cos\theta = 5\sin(\theta + \phi)$$

$$\text{Where } \cos\phi = \frac{4}{5} \quad \text{and} \quad \sin\phi = \frac{3}{5} \quad \text{Answer}$$

Prove that:

$$\sin 3\theta = 3\sin\theta - 4\sin^3\theta$$

$$\frac{\sin 6\theta + \sin 4\theta}{\cos 6\theta + \cos 4\theta} = \tan 5\theta$$

$$\frac{\sin 6\theta + \sin 4\theta}{\cos 6\theta + \cos 4\theta} = \tan 5\theta$$

$$\sin 3\theta = 3\sin\theta - 4\sin^3\theta$$

SOLUTION:

ing L.H.S.:

$$\sin 3\theta = 3\sin\theta - 4\sin^3\theta$$

$$= \sin 3\theta$$

$$= \sin(2\theta + \theta)$$

$$\because \sin(\alpha + \beta) = \sin\alpha \cos\beta + \cos\alpha \sin\beta$$

$$= \sin 2\theta \cos\theta + \cos 2\theta \sin\theta$$

$$\because \sin 2\theta = 2\sin\theta \cos\theta$$

$$\because \cos 2\theta = 2\cos^2\theta - 1$$

$$= (2\sin\theta \cos\theta) \cos\theta + (2\cos^2\theta - 1) \sin\theta$$

$$= 2\sin\theta \cos^2\theta + 2\cos^2\theta \sin\theta - \sin\theta$$

$$\because \cos^2\theta = 1 - \sin^2\theta$$

$$= 2\sin\theta(1 - \sin^2\theta) + 2(1 - \sin^2\theta)\sin\theta - \sin\theta$$

$$= 2\sin\theta - 2\sin^3\theta + 2\sin\theta - 2\sin^3\theta - \sin\theta$$

$$= 3\sin\theta - 4\sin^3\theta$$

Hence Proved L.H.S. = R.H.S.

$$\frac{\sin 6\theta + \sin 4\theta}{\cos 6\theta + \cos 4\theta} = \tan 5\theta$$

$$\frac{\sin 6\theta + \sin 4\theta}{\cos 6\theta + \cos 4\theta} = \tan 5\theta$$

SOLUTION:

ing L.H.S.:

$$\frac{\sin 6\theta + \sin 4\theta}{\cos 6\theta + \cos 4\theta} = \tan 5\theta$$

$$= \frac{\sin 6\theta + \sin 4\theta}{\cos 6\theta + \cos 4\theta}$$

$$= \frac{\sin 6\theta + \sin 4\theta}{\cos 6\theta + \cos 4\theta}$$

$$\therefore \sin u + \sin v = 2 \sin \frac{u+v}{2} \cos \frac{u-v}{2}$$

$$\therefore \cos u + \cos v = 2 \cos \frac{u+v}{2} \cos \frac{u-v}{2}$$

$$= \frac{\left(2 \sin \frac{6\theta + 4\theta}{2} \cos \frac{6\theta - 4\theta}{2}\right)}{\left(2 \cos \frac{6\theta + 4\theta}{2} \cos \frac{6\theta - 4\theta}{2}\right)}$$

$$= \frac{2 \sin \frac{10\theta}{2} \cos \frac{2\theta}{2}}{2 \cos \frac{10\theta}{2} \cos \frac{2\theta}{2}}$$

$$= \frac{2 \sin 5\theta \cos \theta}{2 \cos 5\theta \cos \theta} = \frac{\sin 5\theta}{\cos 5\theta}$$

$$= \tan 5\theta$$

Hence Proved L.H.S. = R.H.S.

iv). The area of triangle is 3.346 square unit. If $\beta = 20.9^\circ$, $\gamma = 117.2^\circ$. Find α and angle α .

SOLUTION:

$$\Delta = 3.346 \text{ square unit, } \beta = 20.9^\circ, \gamma = 117.2^\circ$$

$$\therefore \alpha + \beta + \gamma = 180^\circ$$

$$\alpha = 180^\circ - (\beta + \gamma)$$

$$\alpha = 180^\circ - (20.9^\circ + 117.2^\circ) = 180^\circ - 138.1^\circ$$

$$\alpha = 41.9^\circ$$

$$\therefore \Delta = \frac{1}{2} a^2 \frac{\sin \beta \sin \gamma}{\sin \alpha}$$

$$3.346 = \frac{1}{2} a^2 \frac{\sin 20.9^\circ \sin 117.2^\circ}{\sin 41.9^\circ}$$

$$2(3.346) = a^2 \frac{(0.3567)(0.8894)}{(0.6678)}$$

$$2(3.346) = a^2 (0.4751)$$

$$\frac{6.692}{0.4751} = a^2$$

$$a^2 = 14.083$$

Solve the equation $\cos \theta - \theta = 0$, graphically for the interval $-\frac{\pi}{2} \leq$

ANSWER: (See Question No. 2(xiv), 2024 P.E.)

SECTION 'C' (DESCRIPTIVE-ANSWER QUESTIONS)

(Marks: $15 \times 2 = 30$)

TE: Attempt any TWO questions. All questions equal marks.

Q1. Solve the non-homogeneous system of linear equations using Jordan method.

$$2y - z = 4 \quad x - 2y + z = 2 \quad x + y = 0 \quad \text{(Marks 08)}$$

ANSWER: (See Question No. 3, 2024 G.S.)

Q2. Find the volume of the tetrahedron whose vertices are $A(1, 8)$, $B(3, 2, 9)$, $C(2, 1, 4)$, and $D(3, 3, 10)$. (Marks 07)

ANSWER: (See Question No. 7, 2024 P.E.)

Q3. If $x = \frac{1}{3} + \frac{1.3}{3.6} + \frac{1.3.6}{3.6.9} + \frac{1.3.6.9}{3.6.9.12} + \dots$, prove that $x^2 + 2x -$ (Marks 08)

SOLUTION:

$$x = \frac{1}{3} + \frac{1.3}{3.6} + \frac{1.3.6}{3.6.9} + \frac{1.3.6.9}{3.6.9.12} + \dots$$

1 to both the sides of the given series,

$$1 + x = 1 + \frac{1}{3} + \frac{1.3}{3.6} + \frac{1.3.6}{3.6.9} + \frac{1.3.6.9}{3.6.9.12} + \dots \quad \text{--- (i)}$$

series on the right hand side of (i) be identified with

$$\therefore (1 + x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots$$

Comparing above expression with equation (i)

$$1 + x = (1 + x)^n \quad \text{--- (ii)}$$

on both sides

$$nx = \frac{1}{3} \quad \text{--- (iii)}$$

$$(nx)^2 = \left(\frac{1}{3}\right)^2$$

$$n^2 x^2 = \frac{1}{9} \quad \text{--- (iv)}$$

Also

$$\frac{n(n-1)}{2!} x^2 = \frac{1.3}{3.6}$$

$$\frac{n(n-1)}{2 \times 1} x^2 = \frac{1}{6}$$

$$n(n-1)x^2 = \frac{2}{6}$$

$$n(n-1)x^2 = \frac{1}{3} \quad \dots \dots (v)$$

Equation (v) divided by (iv)

$$\frac{n(n-1)x^2}{n^2 x^2} = \frac{\frac{1}{3}}{\frac{1}{9}}$$

$$\frac{(n-1)}{n} = \frac{1}{3} \times \frac{9}{1}$$

$$\frac{(n-1)}{n} = 3$$

$$n-1 = 3n$$

$$3n - n = -1$$

$$2n = -1$$

$$n = -\frac{1}{2}$$

Putting the value of "n" in equation (iii) we get

$$\left(-\frac{1}{2}\right)x = \frac{1}{3}$$

$$x = -\frac{2}{3}$$

By substituting the value of n and x in right hand sides of equation (i) we have

$$1+x = \left(1 - \frac{2}{3}\right)^{-\frac{1}{2}}$$

$$1+x = \left(\frac{3-2}{3}\right)^{-\frac{1}{2}}$$

$$1+x = \left(\frac{1}{3}\right)^{-\frac{1}{2}}$$

Squaring both sides

$$(1+x) = \left\{ \left(\frac{1}{3} \right)^{-\frac{1}{2}} \right\}^2$$

$$1+2x+x^2 = \left(\frac{1}{3} \right)^{-1}$$

$$x^2 + 2x + 1 = 3$$

$$x^2 + 2x - 2 = 0$$

$5y^2 + 20y - 16 = 0$ Hence Proved Answer

The starting salary of a peon was Rs. 8000/= and after each year his salary was increased by 15%. What total amount of he got for the first twelve years? (Marks 07)

Salary of peon = $a = 8000$

Increase in salary = $r = 1 + 15\% = 1 + 0.15 = 1.15$

Number of years = $n = 12$

Amount of salary $S_n = ?$

$$\therefore S_n = \frac{a(r^n - 1)}{r - 1}$$

$$S_n = \frac{8000\{(1.15)^{12} - 1\}}{1.15 - 1}$$

$$S_n = \frac{8000(5.350250 - 1)}{0.15}$$

$$S_n = \frac{8000(4.350250)}{0.15} = \frac{34802.00084}{0.15}$$

$$S_n = 232013.339$$

Amount of salary of the peon will be Rs. 232013 (approximately) Find and verify the general solution of the following metric equation. (Marks 08)

ON:

$$3\tan^2 x + 2\sqrt{3}\tan x = -1$$

$$3\tan^2 x + 2\sqrt{3}\tan x = -1$$

$$3\tan^2 x + 2\sqrt{3}\tan x + 1 = 0$$

$$(\sqrt{3}\tan x)^2 + 2(\sqrt{3}\tan x)(1) + (1)^2 = 0$$

$$(\sqrt{3}\tan x + 1)^2 = 0$$

Take square root on both sides, we get

$$\sqrt{3}\tan x + 1 = 0$$

$$\sqrt{3}\tan x = -1$$

$$\tan x = -\frac{1}{\sqrt{3}}$$

$$x = \tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)$$

$$x = -\frac{\pi}{6}$$

Since $\tan x$ is negative in 2nd and 4th quadrants therefore

FOR 2ND QUADRANT

$$\therefore x = \frac{\pi}{2} + x$$

$$x = \frac{\pi}{2} + \left(-\frac{\pi}{6}\right)$$

$$x = \frac{\pi}{2} - \frac{\pi}{6}$$

$$x = \frac{6\pi - \pi}{6}$$

$$x = \frac{5\pi}{6}$$

Also

$$\therefore x = \pi + x$$

$$x = \pi + \left(-\frac{\pi}{6}\right)$$

$$x = \frac{6\pi - \pi}{6}$$

$$x = \frac{11\pi}{6}$$

VERIFICATION:

FOR $x = -\frac{\pi}{6}$

$$3 \left\{ \tan\left(-\frac{\pi}{6}\right) \right\}^2 + 2\sqrt{3} \tan\left(-\frac{\pi}{6}\right) = -1$$

$$3 \left(\frac{1}{\sqrt{3}} \right)^2 + 2\sqrt{3} \left(\frac{1}{\sqrt{3}} \right) = -1$$

$$3 \left(\frac{1}{3} \right) + 2 = -1$$

$$3 = -1 \quad (\text{True})$$

FOR $x = \frac{5\pi}{6}$

$$3 \left\{ \tan\left(\frac{5\pi}{6}\right) \right\}^2 + 2\sqrt{3} \tan\left(\frac{5\pi}{6}\right) = -1$$

FOR 4TH QUADRANT

$$\therefore x = \frac{3\pi}{2} + \theta$$

$$x = \frac{3\pi}{2} + \left(-\frac{\pi}{6}\right)$$

$$x = \frac{3\pi}{2} - \frac{\pi}{6}$$

$$x = \frac{9\pi - \pi}{6} = \frac{8\pi}{6}$$

$$x = \frac{4\pi}{3}$$

Also

$$\therefore x = 2\pi - \theta$$

$$x = 2\pi - \left(-\frac{\pi}{6}\right)$$

$$x = \frac{12\pi + \pi}{6}$$

$$x = \frac{13\pi}{6}$$

$$3\left(-\frac{1}{\sqrt{3}}\right)^2 + 2\sqrt{3}\left(-\frac{1}{\sqrt{3}}\right) = -1$$

$$3\left(\frac{1}{3}\right) - 2 = -1$$

$$-1 = -1 \text{ (True)}$$

FOR $x = \frac{11\pi}{6}$

$$3\left\{\tan\left(\frac{11\pi}{6}\right)\right\}^2 + 2\sqrt{3}\tan\left(\frac{11\pi}{6}\right) = -1$$

$$3\left(-\frac{1}{\sqrt{3}}\right)^2 + 2\sqrt{3}\left(-\frac{1}{\sqrt{3}}\right) = -1$$

$$3\left(\frac{1}{3}\right) - 2 = -1$$

$$-1 = -1 \text{ (True)}$$

FOR $x = \frac{4\pi}{3}$

$$3\left\{\tan\left(\frac{4\pi}{3}\right)\right\}^2 + 2\sqrt{3}\tan\left(\frac{5\pi}{6}\right) = -1$$

$$3(\sqrt{3})^2 + 2\sqrt{3}(\sqrt{3}) = -1$$

$$3(3) + 2(3) = -1$$

$$9 + 6 = -1$$

$$15 = -1 \text{ (Not True)}$$

FOR $x = \frac{13\pi}{6}$

$$3\left\{\tan\left(\frac{13\pi}{6}\right)\right\}^2 + 2\sqrt{3}\tan\left(\frac{13\pi}{6}\right) = -1$$

$$3\left(\frac{1}{\sqrt{3}}\right)^2 + 2\sqrt{3}\left(\frac{1}{\sqrt{3}}\right) = -1$$

$$3\left(\frac{1}{3}\right) + 2 = -1$$

$$3 = -1 \text{ (Not True)}$$

Hence,

$$\text{Solution set} = \left\{\frac{5\pi}{6} + 2n\pi\right\} \cup \left\{\frac{11\pi}{6} + 2n\pi\right\} n \in \mathbb{Z} \text{ Answer}$$

Q. 7. b). The sides of a parallelogram are 25cm and 35cm long and one of its angles is 36° . Find the lengths of its diagonals. (Marks 07)

ANSWER: (See Question No. 2(xi), 2024 G.S.)

YOUSUIFS MODEL QUESTION PAPER 1

Annual Examination 2023

MATHEMATICS**CLASS XI (SCIENCE)**

Total Time: 3 Hours

Total Marks: 100

SECTION 'A' (COMPULSORY)**MULTIPLE CHOICE QUESTIONS (M.C.Qs)**

Time: 20 Minutes

(20 Marks)

- NOTE:** (i). This section consists of 20 part questions and all are to be answered. Each question carries one marks.
 (ii). Do not copy the part questions in your answer-book. Fill the circle of correct option.
 (iii). The use of calculator is allowed. All notations are used in their usual meanings.

1. Choose the correct answer for each of the given options:
- (i) If $z = 3i - 4$, then $z + \bar{z} =$ _____
 * 8 * $-3i$ * -8 ✓ * $3i - 8$
- (ii) If $A = [-7]$, then $[A]$ is equal to;
 * 7 * -7 ✓ * 0 * Not possible
- (iii) The volume of tetrahedron determined by vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ is:
 * $[\vec{a}, \vec{b}, \vec{c}]$ * $\frac{1}{3}[\vec{a}, \vec{b}, \vec{c}]$ * $\frac{1}{6}[\vec{a}, \vec{b}, \vec{c}]$ ✓ * None
- (iv) Arithmetic mean between c and d is:
 * $\frac{c+d}{2}$ ✓ * $\frac{c+d}{2cd}$ * $\frac{2cd}{c+d}$ * $\frac{2}{c+d}$
- (v) If $a_{n-2} = 3n - 11$ then $a_n =$ _____
 * $3n + 5$ * $3n - 5$ ✓ * $3n - 9$ * $3n - 13$
- (vi) Probability of getting 7 in throwing a dice is:
 * 0 ✓ * 1 * -1 * Not defined
- (vii) The number of terms in the expansion of $(a + b)^{2n}$ are:
 * n * $2n + 1$ ✓ * 2^n * 2^{n-1}
- (viii) An one to one function is also called _____ function:
 * Injective ✓ * Surjective * Bijective * Inverse
- (ix) Linear Programming (LP) is used to obtain _____ solution.
 * Feasible * Trivial * Optimal ✓ * Infeasible
- (x) $\cos(-\theta) =$ _____
 * $\sin\theta$ * $-\sin\theta$ * $\cos\theta$ ✓ * $-\cos\theta$
- (xi) to solve an oblique triangle, we use:

of sines
of tangents

* Laws of cosines

* All of these ✓

If $y = \cos x$, then range is:

$-1 \leq y \leq 1$

* R

* $y \geq -1$ or $y \geq 1$

* $y \leq -1$ or $y \geq 1$

The value of i^9 is equal to:

* $-i$

* i ✓

* -1

If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then $\text{adj. } A =$:

$\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ ✓

* $\begin{bmatrix} a & -b \\ -c & d \end{bmatrix}$

* $\begin{bmatrix} -a & c \\ b & -d \end{bmatrix}$

* $\begin{bmatrix} a & c \\ b & d \end{bmatrix}$

If first term is a and second term is b of H.P. then $d =$:

$\frac{a-b}{ab}$ ✓

* $\frac{ab}{a-b}$

* $\frac{a+b}{a-b}$

* $\frac{ab}{a+b}$

The G.M between $\sqrt{2}$ and $\frac{1}{\sqrt{2}}$ is:

± 2 * $\pm \sqrt{2}$

* ± 1 ✓

* $\pm \frac{1}{\sqrt{2}}$

A coin is tossed twice, the probability of getting both tail is:

* $\frac{3}{4}$

* $\frac{1}{4}$ ✓

* 1

The probability of getting a head in single toss of a coin is:

* 1

* $1/2$ ✓

* $-1/2$

The magnitude of vector $\vec{a} = 2\hat{i} + 2\hat{j} + \hat{k}$ is:

* $\sqrt{3}$

* 5

* $\sqrt{5}$

The middle term in the expansion $(1+2x)^6$ is:

* term

* 4^{th} term ✓

* 2^{nd} term

* 3^{rd} term

SECTION 'B' (SHORT-ANSWER QUESTIONS)

(Marks: $10 \times 5 = 50$)

NOTE: Attempt any TEN part questions from this section, selecting at least THREE part questions from each question. You may choose the tenth part question from any one subsection. All questions carry equation marks.

COMPLEX NUMBER, MATRIX AND DETERMINANT.

VECTORS AND FUNCTION & GRAPHS

(i). Solve the following simultaneous linear equations with complex coefficients for w and z .

$3z + (2+i)w = 11-i, (2-i)z - w = -1+i$

SOLUTION:

$3z + (2+i)w = 11-i$ --- (1)

$(2-i)z - w = -1+i$ --- (2)

In order to equate the coefficients of z , we multiply equation (1) by (2) and equation (2) by 3

$$3z(2-i) + (2-i)(2+i)w = (2-i)(11-i)$$

$$3(2-i)z - 3w = 3(-1+i)$$

For equation (1)

$$3z(2-i) + (2-i)(2+i)w = (2-i)(11-i)$$

$$3z(2-i) + \{(2)^2 - (i)^2\}w = 2(11-i) - i(11-i)$$

$$3z(2-i) + \{(4) - (-1)\}w = 22 - 2i - 11i + i^2$$

$$3z(2-i) + (4+1)w = 22 - 13i + (-1)$$

$$3z(2-i) + 5w = 21 - 13i \quad \dots (1')$$

For equation (2)

$$3(2-i)z - 3w = 3(-1+i)$$

$$3z(2-i) - 3w = -3 + 3i \quad \dots (2')$$

Subtracting equation (2') from (1')

$$3z(2-i) + 5w = 21 - 13i$$

$$3z(2-i) - 3w = -3 + 3i$$

$$\begin{array}{r} = \quad \quad \quad + \quad \quad = + \quad - \\ \hline 8w = 24 - 16i \\ 8w = 8(3 - 2i) \\ w = \frac{8(3 - 2i)}{8} \\ w = 3 - 2i \end{array}$$

Putting the value of "w" in equation (2)

$$(2-i)z - w = -1 + i$$

$$(2-i)z - (3 - 2i) = -1 + i$$

$$(2-i)z - 3 + 2i = -1 + i$$

$$(2-i)z = -1 + i + 3 - 2i$$

$$(2-i)z = 2 - i$$

$$z = \frac{(2-i)}{(2-i)}$$

$$z = 1$$

$$w = 3 - 2i, \quad z = 1 \quad \text{Answer}$$

Q. 2(ii). Find real number x, y, z such that matrix A is Hermitian matrix.

ANSWER: (See Question No. 2(ii), 2023)

Q. 2(iii). Without expanding determinants, prove that.

$$\begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix} = 0$$

$$\begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix} = 0$$

$$= \begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix}$$

$$= \begin{vmatrix} 1 & a & a+b+c \\ 1 & b & c+a+b \\ 1 & c & a+b+c \end{vmatrix}$$

$$= \begin{vmatrix} 1 & a & a+b+c \\ 1 & b & a+b+c \\ 1 & c & a+b+c \end{vmatrix}$$

common $(a+b+c)$ from C_3

$$= (a+b+c) \begin{vmatrix} 1 & a & 1 \\ 1 & b & 1 \\ 1 & c & 1 \end{vmatrix}$$

C_3 By using Property 3

$$= (a+b+c) \times 0$$

$$= 0$$

Hence Proved $L.H.S. = R.H.S$

1). Find a unit vector which is orthogonal to both the vectors. $\vec{a} = \hat{i} + 3\hat{k}$ and $\vec{b} = 3\hat{i} - 2\hat{j} + \hat{k}$

SOLUTION:

$$\vec{a} = \hat{i} + 3\hat{k}$$

$$\vec{b} = 3\hat{i} - 2\hat{j} + \hat{k}$$

Required unit vector is orthogonal to both the vectors $\vec{a}$ and $\vec{b}$,
Therefore

$$\hat{u} = \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|}$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & 3 \\ 3 & -2 & 1 \end{vmatrix}$$

$$= \hat{i} \begin{vmatrix} 0 & 3 \\ -2 & 1 \end{vmatrix} - \hat{j} \begin{vmatrix} 1 & 3 \\ 3 & 1 \end{vmatrix} + \hat{k} \begin{vmatrix} 1 & 0 \\ 3 & -2 \end{vmatrix}$$

$$= \hat{i} \{(-2)(1) - (3)(-2)\} - \hat{j} \{(1)(1) - (3)(3)\} + \hat{k} \{(1)(-2) - (0)(3)\}$$

$$= \hat{i}(-2 + 6) - \hat{j}(1 - 9) + \hat{k}(-2 + 0)$$

$$= \hat{i}(4) - \hat{j}(-8) + \hat{k}(-2)$$

$$\begin{aligned}\vec{a} \times \vec{b} &= 4\vec{i} + 8\vec{j} + 4\vec{k} \\ |\vec{a} \times \vec{b}| &= \sqrt{(4)^2 + (8)^2 + (4)^2} \\ |\vec{a} \times \vec{b}| &= \sqrt{16 + 64 + 16} \\ |\vec{a} \times \vec{b}| &= \sqrt{96} \\ |\vec{a} \times \vec{b}| &= 4\sqrt{6} \\ \hat{u} &= \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|} = \frac{4\vec{i} + 8\vec{j} + 4\vec{k}}{4\sqrt{6}} \\ \hat{u} &= \frac{4(\vec{i} + 2\vec{j} + \vec{k})}{4\sqrt{6}} \\ \hat{u} &= \frac{\vec{i} + 2\vec{j} + \vec{k}}{\sqrt{6}}\end{aligned}$$

Hence the unit vector which is orthogonal to both the vectors $\vec{a}$ and $\vec{b}$ is

$$\hat{u} = \frac{1}{\sqrt{6}}(\vec{i} + 2\vec{j} + \vec{k}) \text{ Answer}$$

Q. 2(v). The paths of two aeroplanes A and B in the plane are determined by the straight lines $2x - y = 6$ and $3x + y = 4$ respectively. Graphically find the point where the two paths cross each other.

SOLUTION:

Since

$2x - y = 6$ represents the path of aeroplane A while

$3x + y = 4$ represents the path of aeroplane B, then

$$2x - y = 6$$

$$y = 2x - 6$$

$$\text{Put } x = 1$$

$$y = 2(1) - 6 = 2 - 6$$

$$y = -4$$

$$\text{Put } x = 2$$

$$y = 2(2) - 6 = 4 - 6$$

$$y = -2$$

$$\text{Put } x = 3$$

$$y = 2(3) - 6 = 6 - 6$$

$$y = 0$$

x	1	2	3
y	-4	-2	0

$$3x + y = 4$$

$$y = 4 - 3x$$

$$\text{Put } x = 1$$

$$y = 4 - 3(1) = 4 - 3$$

$$y = 1$$

$$\text{Put } x = 2$$

$$y = 4 - 3(2) = 4 - 6$$

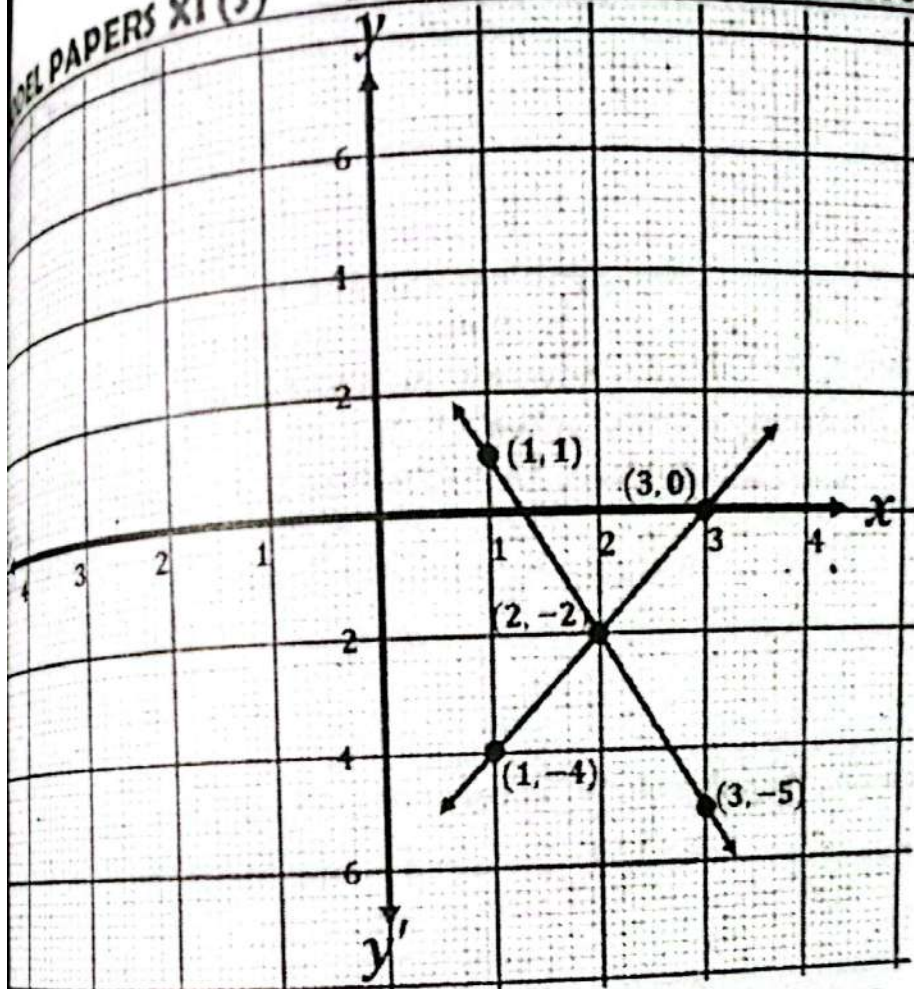
$$y = -2$$

$$\text{Put } x = 3$$

$$y = 4 - 3(3) = 4 - 9$$

$$y = -5$$

x	1	2
y	1	-2



SEQUENCE & SERIES, PROBABILITY, MATHEMATICAL

INDUCTION & BINOMIAL THEOREM

Q. If the 7th and 10th terms of an H.P are $\frac{1}{3}$ and $\frac{5}{21}$ respectively, find
the 3rd and 20th terms.

SOLUTION:

$$\frac{1}{3} \text{ and } T_{10} = \frac{5}{21}$$

?

terms in A.P are

$$3 \text{ and } a_{10} = \frac{21}{5}$$

term

term

$$\therefore a_n = a + (n - 1)d$$

$$a_7 = a + (7 - 1)d$$

$$3 = a + (6)d$$

$$a = 3 - 6d \text{ --- (1)}$$

$$a_{10} = a + (10 - 1)d$$

$$\frac{21}{5} = a + (9)d$$

$$\frac{21}{5} = a + 9d$$

$$21 = 5a + 45d$$

$$5a = 21 - 45d \quad \dots (2)$$

Putting the value of "a" in above equation we get

$$5(3 - 6d) = 21 - 45d$$

$$15 - 30d = 21 - 45d$$

$$45d - 30d = 21 - 15$$

$$15d = 6$$

$$d = \frac{6}{15}$$

$$d = \frac{2}{5}$$

Putting the value of "d" in equation (1) we get

$$a = 3 - 6\left(\frac{2}{5}\right)$$

$$a = \frac{15 - 12}{5}$$

$$a = \frac{3}{5}$$

14th and 20th terms of H.P. are

$$T_n = \frac{1}{a + (n - 1)d}$$

$$T_{14} = \frac{1}{\frac{3}{5} + (14 - 1)\left(\frac{2}{5}\right)}$$

$$= \frac{1}{\frac{3}{5} + (13)\left(\frac{2}{5}\right)}$$

$$= \frac{1}{\frac{3}{5} + \frac{26}{5}}$$

$$= \frac{1}{\frac{3 + 26}{5}}$$

$$T_{14} = \frac{5}{29}$$

$$T_{20} = \frac{1}{\frac{3}{5} + (20 - 1)\left(\frac{2}{5}\right)}$$

$$T_{20} = \frac{1}{\frac{3}{5} + (19)\left(\frac{2}{5}\right)}$$

$$T_{20} = \frac{1}{\frac{3}{5} + \frac{38}{5}}$$

$$T_{20} = \frac{1}{\frac{3 + 38}{5}}$$

$$T_{20} = \frac{5}{41}$$

$$T_{14} = \frac{5}{29} \text{ and } T_{20} = \frac{5}{41} \text{ Answer}$$

Find the sum of the following series. $3 + 6 + 15 + 42 + 123 + \dots$

Q. (See Question No. 2(vi), 2023)

There are 11 men and 9 women members of a club. How many committees of 8 members can be formed, having:

(i) At least five men

(ii) At most five women

Solution:

Men = 11

Women = 9

Committee members = 8

(i) At least five men

Men can be chosen in $\binom{11}{5}$ ways and the women in $\binom{9}{3}$ ways

By Product Principle, the required number of ways is:

$$\text{Total No. of ways} = \binom{11}{5} \cdot \binom{9}{3} = {}^{11}C_5 \times {}^9C_3$$

$$= 462 \times 84$$

$$= 38808 \text{ ways Answer}$$

(ii) At most five women

Women can be chosen in $\binom{9}{5}$ ways and the men in $\binom{11}{3}$ ways

By Sum Principle, the required number of ways is:

No. of ways =

$$= \binom{9}{5} \cdot \binom{11}{3} + \binom{9}{4} \cdot \binom{11}{4} + \binom{9}{3} \cdot \binom{11}{5} + \binom{9}{2} \cdot \binom{11}{6} \\ + \binom{9}{1} \cdot \binom{11}{7} + \binom{9}{0} \cdot \binom{11}{8}$$

$$= {}^{11}C_3 + {}^9C_4 \times {}^{11}C_4 + {}^9C_3 \times {}^{11}C_5 + {}^9C_2 \times {}^{11}C_6 + {}^9C_1 \times {}^{11}C_7 \\ + {}^9C_0 \times {}^{11}C_8$$

$$= 165 + 126 \times 330 + 84 \times 462 + 36 \times 462 + 9 \times 330 + 1 \times 165$$

$$= 20790 + 41580 + 38808 + 16632 + 2970 + 165$$

$$\text{Total No. of ways} = 120945 \text{ ways Answer}$$

(iii) At most five women.

Women can be chosen in $\binom{9}{5}$ ways and the men in $\binom{11}{3}$ ways

By Sum Principle, the required number of ways is:

Total No. of ways =

$$= \binom{9}{5} \cdot \binom{11}{3} + \binom{9}{6} \cdot \binom{11}{2} + \binom{9}{7} \cdot \binom{11}{1} + \binom{9}{8} \cdot \binom{11}{0}$$

$$= {}^9C_5 \times {}^{11}C_3 + {}^9C_6 \times {}^{11}C_2 + {}^9C_7 \times {}^{11}C_1 + {}^9C_8 \times {}^{11}C_0$$

$$= 126 \times 165 + 84 \times 55 + 36 \times 11 + 9 \times 1$$

$$= 20790 + 4620 + 396 + 9$$

Total No. of ways = 25815 ways Answer

Q. 3(iv). A bag contains 40 balls out of which 5 are green, 15 are red and the remaining are black. A ball is drawn out of the bag. Find the probability of getting:

- (i) The ball is green (ii) The ball is black
(iii) The ball is not green

SOLUTION:

Total No. of balls = $O(S) = 40$

No. of red balls = $O(R) = 15$

No. of green balls = $O(G) = 5$

No. of black balls = $O(B) = 40 - 5 - 15 = 20$

(i) The ball is green

$$\text{Probability} = \frac{\text{Successful outcomes}}{\text{Possible outcomes}}$$

In this case, the successful outcomes are 5 because the number of green balls is 5 while the possible outcomes are 40 because of the total balls are 40. i.e.

$$P(G) = \frac{O(G)}{O(S)} = \frac{5}{40}$$

$$P(G) = \frac{1}{8}$$

(ii) The ball is black

In this case, the successful outcomes are 20 because the number of black balls is 20 while the possible outcomes are 40 because of the total balls are 40. i.e.

$$P(B) = \frac{O(B)}{O(S)} = \frac{20}{40}$$

$$P(B) = \frac{1}{2}$$

(iii) The ball is not green

In this case, the successful outcomes are 35 because the sum of red and black balls is 35 while the possible outcomes are 40 because the total No. of balls are 40. i.e.

$$P(B \text{ or } R) = \frac{O(B \text{ or } R)}{O(S)} = \frac{35}{40}$$

$$P(B) = \frac{7}{8}$$

Prove by the mathematical induction for every positive integer

$$1.3 + 2.4 + 3.5 + \dots + n(n+2) = \frac{1}{6}n(n+1)(2n+7).$$

Proof:

$$1.3 + 2.4 + 3.5 + \dots + n(n+2) = \frac{1}{6}n(n+1)(2n+7).$$

Step I:

For $n = 1$

$$1.3 + 2.4 + 3.5 + \dots + 1(1+2) = \frac{1}{6}(1)(1+1)(2(1)+7).$$

$$1.3 + 2.4 + 3.5 + \dots + 1(3) = \frac{1}{6}(2)(2+7).$$

$$1.3 + 2.4 + 3.5 + \dots + 3 = \frac{1}{6}(2)(9).$$

$$1.3 + 2.4 + 3.5 + \dots + 3 = \frac{18}{6}$$

$$1.3 + 2.4 + 3.5 + \dots + 3 = 3$$

So, $P(n)$ is true for $n = 1$

Step II:

Assuming the result to be true for $n = k$, we get the hypothesis

$$1.3 + 2.4 + 3.5 + \dots + k(k+2) = \frac{1}{6}k(k+1)(2k+7).$$

Now, we have to prove that the result is true for $n = k+1$.
 On both Sides

$$\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots + \frac{1}{k(k+1)}$$

$$1.3 + 2.4 + 3.5 + \dots + k(k+2)$$

$$+ (k+1)(k+3)$$

$$= \frac{1}{6}k(k+1)(2k+7) + (k+1)(k+3)$$

$$= (k+1) \left\{ \frac{1}{6}k(2k+7) + (k+3) \right\}$$

$$= (k+1) \left\{ \frac{(2k^2 + 7k) + 6(k+3)}{6} \right\}$$

$$= (k+1) \left(\frac{2k^2 + 7k + 6k + 18}{6} \right)$$

R.W
 last term $n(n+2)$

$$\text{Put } n = k+1$$

$$= (k+1)(k+1+2)$$

$$= (k+1)(k+3)$$

$$\begin{aligned}
 &= \frac{(k+1)}{6} (2k^2 + 13k + 18) \\
 &= \frac{(k+1)}{6} (2k^2 + 4k + 9k + 18) \\
 &= \frac{(k+1)}{6} \{2k(k+2) + 9(k+2)\} \\
 &= \frac{(k+1)}{6} \{(k+2)(2k+9)\} \\
 &= \frac{(k+1)}{6} \{(k+1+1)(2k+2+7)\} \\
 &= \frac{1}{6} (k+1) \{(k+1)+1\} \{2(k+1)+7\} \\
 &\quad \text{i.e. } p(n) \text{ is true for } n = k+1
 \end{aligned}$$

Thus, by the principle of mathematical induction $P(n)$ is true for natural numbers n .

LINEAR PROGRAMING AND TRIGONOMET

Q. 4(i). Solve the following LP programming problems by graphical method when $x \geq 0, y \geq 0$. Minimize $Z = 200x + 500y$
Subject to the constraints: $x + 2y \geq 10$; $3x + 4y \leq 24$

SOLUTION:

CONSTRAINT 1:

$$x + 2y \geq 10$$

We consider it as an equation $x + 2y = 10$

For x -intercept put $y = 0$

$$x + 2(0) = 10$$

$$x = 10$$

The intersection point of line with x -axis is $(10, 0)$

For y -intercept put $x = 0$

$$(0) + 2y = 10 \Rightarrow y = \frac{10}{2}$$

$$y = 5$$

The intersection point of line with y -axis is $(0, 5)$

For origin put $x = 0$ and put $y = 0$ in $x + 2y \geq 10$

$$(0) + 2(0) \geq 10$$

$$0 \geq 10 \text{ (not true)}$$

CONSTRAINT 2:

$$3x + 4y \leq 24$$

We consider it as an equation $3x + 4y = 24$

For x -intercept put $y = 0$

$$3x + 4(0) = 24 \Rightarrow 3x = 24 \Rightarrow x = \frac{24}{3}$$

$$x = 8$$

Intersection point of line with x -axis is $(8, 0)$
 y -intercept put $x = 0$

$$3(0) + 4y = 24 \Rightarrow 4y = 24 \Rightarrow y = \frac{24}{4}$$

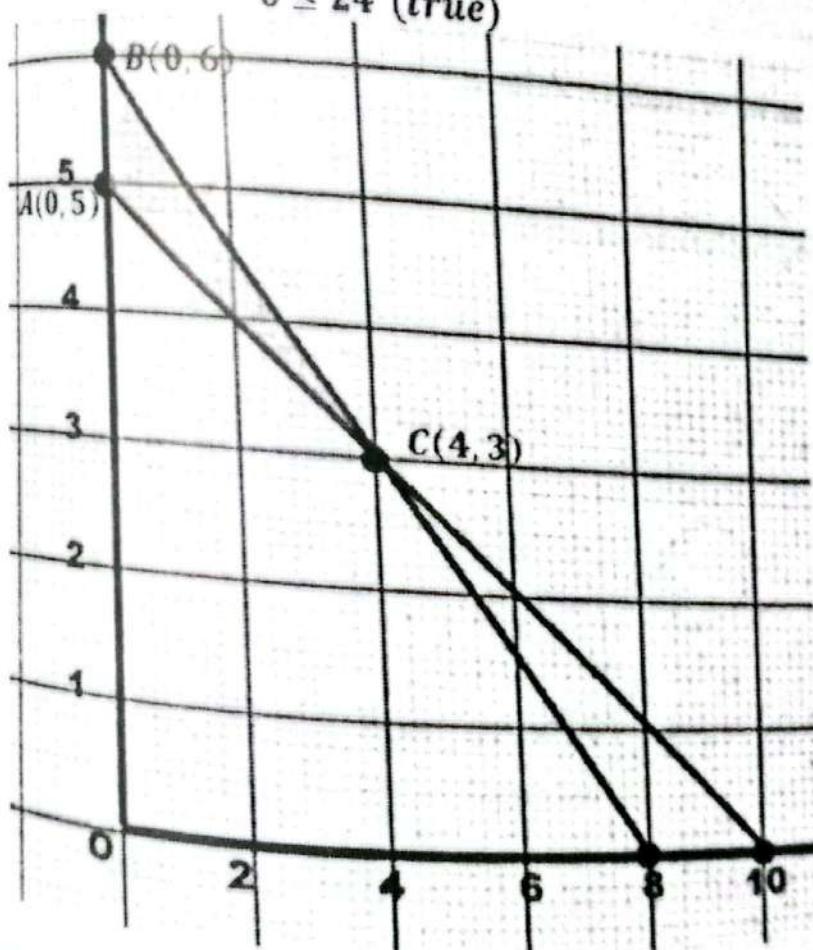
$$y = 6$$

Intersection point of line with y -axis is $(0, 6)$

Origin put $x = 0$ and put $y = 0$ in $3x + 4y \leq 24$

$$3(0) + 4(0) \leq 24$$

$$0 \leq 24 \text{ (true)}$$



Intersection of two linear inequalities is the required feasible region which is the shaded area in the graph.

Solution points or corner points of the region ABC are $A(0, 5)$, $B(0, 6)$, and $C(4, 3)$ in the graph

OBJECTIVE FUNCTION

SOLUTION POINTS
OR
CORNER POINTS

$$f(x, y) = 200x + 500y$$

$A(0, 5)$	$f(x, y) = 200(0) + 500(5)$ $= 0 + 2500 = 2500$
$B(0, 6)$	$f(x, y) = 200(0) + 500(6)$ $= 0 + 3000 = 3000$
$C(4, 3)$	$f(x, y) = 200(4) + 500(3)$ $= 800 + 1500 = 2300$

Hence Minimum at $(4, 3)$; Minimum value = 2300 Answer

Q. 4(ii). Express $\sin\theta + \cos\theta$ in the form $r\sin(\theta + \phi)$, where θ and ϕ are in first quadrant.

SOLUTION:

Since θ and ϕ are in the first quadrant, therefore, all trigonometrical functions have positive values.

$$\text{Let } 1 = r\cos\phi \text{ --- (i)}$$

$$\text{and } 1 = r\sin\phi \text{ --- (ii)}$$

Squaring both sides of equation (i) and (ii), and then adding.
We get

$$(1)^2 + (1)^2 = (r\cos\phi)^2 + (r\sin\phi)^2$$

$$1 + 1 = r^2\cos^2\phi + r^2\sin^2\phi$$

$$2 = r^2(\cos^2\phi + \sin^2\phi)$$

$$\because \cos^2\phi + \sin^2\phi = 1$$

$$2 = r^2(1)$$

$$r^2 = 2$$

$$r = \sqrt{2}$$

So,

$$1 = (\sqrt{2})\cos\phi \text{ and } 1 = (\sqrt{2})\sin\phi$$

$$1 = \sqrt{2}\cos\phi \text{ and } 1 = \sqrt{2}\sin\phi$$

Now

$$1.\sin\theta + 1.\cos\theta = (\sqrt{2}\cos\phi)\sin\theta + (\sqrt{2}\sin\phi)\cos\theta$$

$$\sin\theta + \cos\theta = \sqrt{2}(\sin\theta\cos\phi + \cos\theta\sin\phi)$$

$$\because \sin(\theta + \phi) = \sin\theta\cos\phi + \cos\theta\sin\phi$$

$$\sin\theta + \cos\theta = \sqrt{2}\sin(\theta + \phi)$$

$$\text{Where } \cos\phi = \frac{1}{\sqrt{2}} \text{ and } \sin\phi = \frac{1}{\sqrt{2}} \text{ Answer}$$

Q. 4(iii) Prove that:

(a). $\sin 4\theta = 4\sin\theta\cos\theta\cos 2\theta$

(b). $\frac{\sin\alpha + \sin\beta}{\sin\alpha - \sin\beta} = \tan\left(\frac{\alpha + \beta}{2}\right)\cot\left(\frac{\alpha - \beta}{2}\right)$

$$\sin 4\theta = 4 \sin \theta \cos \theta \cos 2\theta$$

$$\sin 4\theta = \sin (2\theta + 2\theta)$$

$$\begin{aligned} \therefore \sin(\alpha + \beta) &= \sin \alpha \cos \beta + \cos \alpha \sin \beta \\ &= \sin 2\theta \cos 2\theta + \cos 2\theta \sin 2\theta \\ &= 2 \sin 2\theta \cos 2\theta \end{aligned}$$

$$\begin{aligned} \therefore \sin 2\theta &= 2 \sin \theta \cos \theta \\ &= 2(2 \sin \theta \cos \theta) \cos 2\theta \\ &= 4 \sin \theta \cos \theta \cos 2\theta \end{aligned}$$

Hence Proved L.H.S. = R.H.S.

$$\tan\left(\frac{\alpha+\beta}{2}\right) \cot\left(\frac{\alpha-\beta}{2}\right)$$

$$\frac{\sin \alpha + \sin \beta}{\sin \alpha - \sin \beta} = \tan\left(\frac{\alpha + \beta}{2}\right) \cot\left(\frac{\alpha - \beta}{2}\right)$$

$$= \frac{\sin \alpha + \sin \beta}{\sin \alpha - \sin \beta}$$

$$\therefore \sin u + \sin v = 2 \sin \frac{u+v}{2} \cos \frac{u-v}{2}$$

$$\therefore \sin u - \sin v = 2 \cos \frac{u+v}{2} \sin \frac{u-v}{2}$$

$$= \frac{2 \sin\left(\frac{\alpha+\beta}{2}\right) \cos\left(\frac{\alpha-\beta}{2}\right)}{2 \cos\left(\frac{\alpha+\beta}{2}\right) \sin\left(\frac{\alpha-\beta}{2}\right)}$$

$$= \frac{\sin\left(\frac{\alpha+\beta}{2}\right) \cos\left(\frac{\alpha-\beta}{2}\right)}{\cos\left(\frac{\alpha+\beta}{2}\right) \sin\left(\frac{\alpha-\beta}{2}\right)}$$

$$= \frac{\sin\left(\frac{\alpha+\beta}{2}\right)}{\cos\left(\frac{\alpha+\beta}{2}\right)} \cdot \frac{\cos\left(\frac{\alpha-\beta}{2}\right)}{\sin\left(\frac{\alpha-\beta}{2}\right)}$$

$$= \tan\left(\frac{\alpha+\beta}{2}\right) \cot\left(\frac{\alpha-\beta}{2}\right)$$

$$= \tan\left(\frac{\alpha+\beta}{2}\right) \cot\left(\frac{\alpha-\beta}{2}\right)$$

$$\therefore \frac{\sin \theta}{\cos \theta} = \tan \theta \text{ and } \frac{\cos \theta}{\sin \theta} = \cot \theta$$

Hence Proved L.H.S. = R.H.S.

Two sides of a triangle are 4 and 5 units.
so that area of the triangle is 6 square units.
the three sides of the given triangle then,

$a = 4$ units
 $b = 5$ units
 $\Delta = 6$ square units
 $c = ?$

$$\therefore \Delta = \frac{1}{2} ab \sin \gamma$$

$$6 = \frac{1}{2} (4)(5) \sin \gamma$$

$$6 = 10 \sin \gamma$$

$$\sin \gamma = \frac{6}{10}$$

$$\sin \gamma = \frac{3}{5}$$

$$\gamma = \sin^{-1} \left(\frac{3}{5} \right)$$

$$\gamma = 36.8699^\circ$$

$$\therefore c^2 = a^2 + b^2 - 2ab \cos \gamma$$

$$c^2 = (4)^2 + (5)^2 - 2(4)(5) \cos(36.8699^\circ)$$

$$c^2 = 16 + 25 - 40 \left(\frac{4}{5} \right)$$

$$c^2 = 41 - 32$$

$$c^2 = 9$$

Taking square root on both sides

$$c = 3 \text{ Answer}$$

The third side of the triangle is 3 units

Q. 4(v). Find the solution set of the equation $\tan^{-2\theta} = 1$

for the interval $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$

SOLUTION:

$$\tan \theta - 2\theta = 0$$

$$\tan \theta = 2\theta$$

$$\text{Let } \tan \theta = y$$

$$y = \tan \theta \text{ --- (i)}$$

Put $\theta = -\frac{\pi}{2}$	Put $\theta =$
$\tan \left(-\frac{\pi}{2} \right) = y$	$\tan \left(-\frac{\pi}{2} \right) =$
$-\infty = y$	$1 =$
$y = -\infty$	$y =$
Put $\theta = -\frac{\pi}{4}$	Put $\theta =$

$$\tan\left(-\frac{\pi}{4}\right) = y$$

$$-1 = y$$

$$y = -1$$

$$\tan\left(\frac{\pi}{2}\right) = y$$

$$+\infty = y$$

$$y = +\infty$$

FOR $y = \tan \theta$

θ	$-\frac{\pi}{2}$	$-\frac{\pi}{4}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$
$y = \tan \theta$	$-\infty$	-1	0	1	$+\infty$

Also $y = 2\theta$

$$\text{Put } \theta = -\frac{\pi}{2}$$

$$y = 2\left(-\frac{\pi}{2}\right) = -\frac{22}{7}$$

$$y = -3.142$$

$$\text{Put } \theta = \frac{\pi}{4}$$

$$y = 2\left(\frac{\pi}{4}\right) = \frac{\pi}{2}$$

$$y = 1.570$$

$$\text{Put } \theta = -\frac{\pi}{4}$$

$$y = 2\left(-\frac{\pi}{4}\right) = -\frac{\pi}{2}$$

$$y = -1.570$$

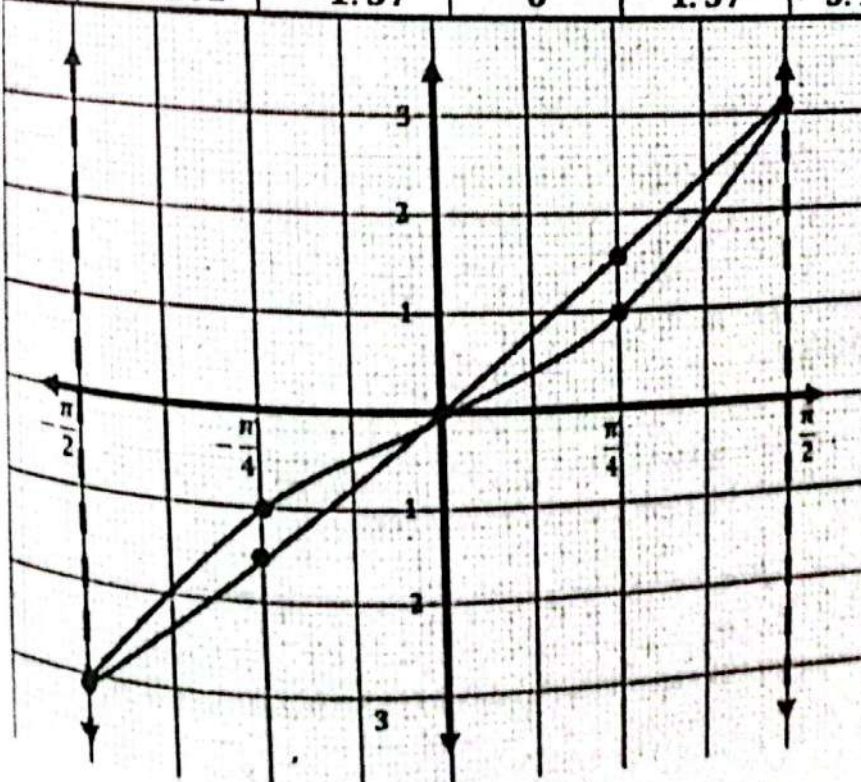
$$\text{Put } \theta = \frac{\pi}{2}$$

$$y = 2\left(\frac{\pi}{2}\right) = \frac{22}{7}$$

$$y = 3.142$$

FOR $y = 2\theta$

θ	$-\frac{\pi}{2}$	$-\frac{\pi}{4}$	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$
$y = 2\theta$	-3.142	-1.57	0	1.57	3.142



From the graph, we observe that the point of intersection of these equations is origin, Hence, the solution is $\theta = 0$

SECTION 'C' (DESCRIPTIVE-ANSWER QUESTIONS)
(Marks: $15 \times 2 = 30$)

NOTE: Attempt any TWO questions. All questions carry equal marks.

Q. 5 (a). Solve the non-homogeneous system of linear equations using Gauss Jordan method. (Marks 08)

$$\begin{aligned} x - y + 4z &= 4 & 2x + 2y - z &= 2 & 3x - 2y + 3z &= -3 \end{aligned}$$

ANSWER: (See Question No. 3, 2023)

Q. 5 (b). Show that the vectors $4\hat{i} - \hat{j} + \hat{k}$, $3\hat{i} - 2\hat{j} - \hat{k}$ and $\hat{i} + \hat{j} + 2\hat{k}$ are coplanar. (Marks 07)

SOLUTION:

Let $\vec{a} = 4\hat{i} - \hat{j} + \hat{k}$, $\vec{b} = 3\hat{i} - 2\hat{j} - \hat{k}$ $\vec{c} = \hat{i} + \hat{j} + 2\hat{k}$

Vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ will be coplanar if $[\vec{a} \ \vec{b} \ \vec{c}] = 0$

Now

$$\begin{aligned} [\vec{a} \ \vec{b} \ \vec{c}] &= \begin{vmatrix} 4 & -1 & 1 \\ 3 & -2 & -1 \\ 1 & 1 & 2 \end{vmatrix} \\ &= 4 \begin{vmatrix} -2 & -1 \\ 1 & 2 \end{vmatrix} + 1 \begin{vmatrix} 3 & -1 \\ 1 & 2 \end{vmatrix} + 1 \begin{vmatrix} 3 & -2 \\ 1 & 1 \end{vmatrix} \\ &= 4\{(-2)(2) - (-1)(1)\} + 1\{(3)(2) - (-1)(1)\} \\ &\quad + 1\{(3)(1) - (-2)(1)\} \\ &= 4(-4 + 1) + 1(6 + 1) + 1(3 + 2) \\ &= 4(-3) + 1(7) + 1(5) \\ &= -12 + 7 + 5 \\ &[\vec{a} \ \vec{b} \ \vec{c}] = 0 \end{aligned}$$

Hence Shwon the given vectors are coplanar

Q. 6 (a). If $\frac{y}{2} = \frac{1}{2} \left(\frac{4}{9}\right) + \frac{1 \cdot 3}{2^2 \cdot 2!} \left(\frac{4}{9}\right)^2 + \frac{1 \cdot 3 \cdot 5}{2^3 \cdot 3!} \left(\frac{4}{9}\right)^3 + \dots$, then show that $5y^2 - 20y - 16 = 0$. (Marks 08)

SOLUTION:

$$\frac{y}{2} = \frac{1}{2} \left(\frac{4}{9}\right) + \frac{1 \cdot 3}{2^2 \cdot 2!} \left(\frac{4}{9}\right)^2 + \frac{1 \cdot 3 \cdot 5}{2^3 \cdot 3!} \left(\frac{4}{9}\right)^3 + \dots$$

Adding 1 to both the sides of the given series,
We have

$$1 + \frac{y}{2} = 1 + \frac{1}{2} \left(\frac{4}{9}\right) + \frac{1 \cdot 3}{2^2 \cdot 2!} \left(\frac{4}{9}\right)^2 + \frac{1 \cdot 3 \cdot 5}{2^3 \cdot 3!} \left(\frac{4}{9}\right)^3 + \dots$$

Let the series on the right hand side of (i) be identified with

$$\therefore (1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots$$

By comparing above expression with equation (i)

$$1 + \frac{y}{2} = (1+x)^n \quad \text{--- (ii)}$$

And

$$n \frac{x}{2} = \frac{1}{2} \left(\frac{4}{9} \right)$$

$$nx = \frac{4}{9} \quad \text{--- (iii)}$$

Squaring both sides

$$(nx)^2 = \left(\frac{4}{9} \right)^2$$

$$n^2 x^2 = \frac{16}{81} \quad \text{--- (iv)}$$

Also

$$\frac{n(n-1)}{2!} x^2 = \frac{1 \cdot 3}{2 \cdot 2!} \left(\frac{4}{9} \right)^2$$

$$n(n-1)x^2 = \frac{3}{4} \left(\frac{16}{81} \right)$$

$$n(n-1)x^2 = \frac{4}{27} \quad \text{--- (v)}$$

Equation (v) divided by (iv)

$$\frac{n(n-1)x^2}{n^2 x^2} = \frac{\frac{4}{27}}{\frac{16}{81}}$$

$$\frac{(n-1)}{n} = \frac{4}{27} \times \frac{81}{4}$$

$$\frac{(n-1)}{n} = 3$$

$$n-1 = 3n$$

$$3n - n = -1$$

$$2n = -1$$

$$n = -\frac{1}{2}$$

Putting the value of "n" in equation (iii) we get

$$\left(-\frac{1}{2} \right) x = \frac{4}{9}$$

$$x = -\frac{8}{9}$$

By substituting the value of n and x in equation (II) we get

$$1 + \frac{y}{2} = \left(1 - \frac{4}{9}\right)^{-\frac{1}{2}}$$

$$\frac{2+y}{2} = \left(\frac{9-4}{9}\right)^{-\frac{1}{2}}$$

$$\frac{2+y}{2} = \left(\frac{5}{9}\right)^{-\frac{1}{2}}$$

Squaring both sides

$$\left(\frac{2+y}{2}\right)^2 = \left\{\left(\frac{5}{9}\right)^{-\frac{1}{2}}\right\}^2$$

$$\frac{4+4y+y^2}{4} = \left(\frac{5}{9}\right)^{-1}$$

$$y^2 + 4y + 4 = \frac{9}{5} \times 4$$

$$y^2 + 4y = \frac{36}{5} - 4$$

$$y^2 + 4y = \frac{36-20}{5}$$

$$y^2 + 4y = \frac{16}{5}$$

Whole equation multiply by '5'

$$5y^2 + 20y = 16$$

$$5y^2 + 20y - 16 = 0 \text{ Hence shown Answer}$$

Q. 6 (b). The sum of an infinite geometric series is half the sum of the squares of its terms. If the sum of its first two terms is $\frac{1}{2}$, find the series

SOLUTION:

Let

$$a + ar + ar^2 + ar^3 + \dots$$

is the given infinite geometric series in which 'a' as its first term and 'r' is its common ratio then, the sum of the square of its term will be

$$a^2 + a^2r^2 + a^2r^4 + a^2r^6 + \dots$$

According to the question first two terms is $4\frac{1}{2}$ i.e.

$$a + ar = 4\frac{1}{2} = \frac{8+1}{2}$$

$$a = \frac{9}{2(1+r)} \quad \text{--- (i)}$$

an infinite geometric series is half the sum of the squares of

of an infinite geometric series

$$= \frac{1}{2} (\text{Sum of the squares of its terms})$$

$$\therefore \text{Sum of an infinite geometric series} = \frac{a}{1-r}$$

$$\frac{a}{1-r} = \frac{1}{2} \left(\frac{a^2}{1-r^2} \right)$$

$$\therefore a^2 - b^2 = (a-b)(a+b)$$

$$\frac{a}{1-r} = \frac{1}{2} \left\{ \frac{a^2}{(1-r)(1+r)} \right\}$$

$$2 = \frac{a}{(1+r)}$$

$$a = 2(1+r) \quad \text{--- (ii)}$$

Equation (i) and (ii) we get

$$\frac{9}{2(1+r)} = 2(1+r)$$

$$(1+r)^2 = \frac{9}{4}$$

$$1 + 2r + r^2 = \frac{9}{4}$$

$$4 + 8r + 4r^2 = 9$$

$$4r^2 + 8r + 4 - 9 = 0$$

$$4r^2 + 8r - 5 = 0$$

$$4r^2 + 8r + 2r - 2r - 5 = 0$$

$$4r^2 + 10r - 2r - 5 = 0$$

$$2r(2r + 5) - 1(2r + 5) = 0$$

$$(2r + 5)(2r - 1) = 0$$

Or

$$2r - 1 = 0$$

$$2r = 1$$

$$r = \frac{1}{2}$$

$$2r + 5 = 0$$

$$2r = -5$$

$$r = -\frac{5}{2}$$

this value because

$$-\frac{5}{2} > 1$$

Now putting the value of 'r' in equation (i), we get

$$a = \frac{1}{2 \left(1 + \frac{1}{2}\right)} = \frac{1}{2 \left(\frac{2+1}{2}\right)}$$

$$a = \frac{1}{2 \left(\frac{3}{2}\right)}$$

$$a = \frac{1}{3}$$

Now the required series will be

$$3 + 3 \left(\frac{1}{2}\right) + 3 \left(\frac{1}{2}\right)^2 + 3 \left(\frac{1}{2}\right)^3 + \dots$$

$$3 + \frac{3}{2} + \frac{3}{4} + \frac{3}{8} + \dots$$

Answer

Q. 7 (a). Find and verify the general solution of the following trigonometric equation.

$$(\operatorname{cosec} x + 2)(2 \cos x - 1) = 0$$

SOLUTION:

$$(\operatorname{cosec} x + 2)(2 \cos x - 1) = 0$$

$$\operatorname{cosec} x + 2 = 0$$

$$\operatorname{cosec} x = -2$$

$$\therefore \operatorname{cosec} \theta = \frac{1}{\sin \theta}$$

$$\frac{1}{\sin x} = -2$$

$$\sin x = -\frac{1}{2}$$

$$x = \sin^{-1} \left(-\frac{1}{2}\right)$$

$$x = -\frac{\pi}{6}$$

Since $\sin \theta$ is negative in 3rd and 4th quadrants

FOR 3RD QUADRANTS

$$\therefore x = \pi + x$$

$$x = \pi - \frac{\pi}{6}$$

$$x = \frac{6\pi - \pi}{6}$$

$$x = \frac{5\pi}{6}$$

Also

$$\therefore x = \frac{3\pi}{2} - x$$

$$x = \frac{3\pi}{2} + \frac{\pi}{6}$$

$$x = \frac{9\pi + \pi}{6} = \frac{10\pi}{6}$$

$$x = \frac{5\pi}{3}$$

FOR 4th QUADRANTS

$$\therefore x = 2\pi - x$$

$$x = 2\pi + \frac{\pi}{6}$$

$$x = \frac{12\pi + \pi}{6}$$

$$x = \frac{13\pi}{6}$$

VERIFICATION:

FOR $x = -\frac{\pi}{6}$

$$\left\{ \operatorname{cosec}\left(-\frac{\pi}{6}\right) + 2 \right\} \left\{ 2\cos\left(-\frac{\pi}{6}\right) - 1 \right\} = 0$$

$$(-2 + 2)(\sqrt{3} - 1) = (0)(\sqrt{3} - 1) = 0 \text{ (True)}$$

FOR $x = \frac{5\pi}{6}$

$$\left\{ \operatorname{cosec}\left(\frac{5\pi}{6}\right) + 2 \right\} \left\{ 2\cos\left(\frac{5\pi}{6}\right) - 1 \right\} = 0$$

$$\left(-\frac{2}{\sqrt{3}} + 2\right)(1 - 1) = \left(\frac{-2 + 2\sqrt{3}}{\sqrt{3}}\right)(0) = 0 \text{ (True)}$$

FOR $x = \frac{5\pi}{3}$

$$\left\{ \operatorname{cosec}\left(\frac{5\pi}{3}\right) + 2 \right\} \left\{ 2\cos\left(\frac{5\pi}{3}\right) - 1 \right\} = 0$$

$$\left(-\frac{2}{\sqrt{3}} + 2\right)\left(2\left(\frac{1}{2}\right) - 1\right) = \left(\frac{-2 + 2\sqrt{3}}{\sqrt{3}}\right)(0) = 0 \text{ (True)}$$

FOR $x = \frac{7\pi}{6}$

$$\left\{ \operatorname{cosec}\left(\frac{7\pi}{6}\right) + 2 \right\} \left\{ 2\cos\left(\frac{7\pi}{6}\right) - 1 \right\} = 0$$

$$(-2 + 2)(\sqrt{3} - 1) = (0)(\sqrt{3} - 1) = 0 \text{ (True)}$$

FOR $x = \frac{\pi}{3}$

$$\left\{ \operatorname{cosec}\left(\frac{\pi}{3}\right) + 2 \right\} \left\{ 2\cos\left(\frac{\pi}{3}\right) - 1 \right\} = 0$$

$$\left(\frac{2}{\sqrt{3}} + 2\right)(1 - 1) = \left(\frac{2 + 2\sqrt{3}}{\sqrt{3}}\right)(0) = 0 \text{ (True)}$$

FOR $x = \frac{11\pi}{6}$

$$\left\{ \operatorname{cosec}\left(\frac{11\pi}{6}\right) + 2 \right\} \left\{ 2\cos\left(\frac{11\pi}{6}\right) - 1 \right\} = 0$$

$$(-2 + 2) \left\{ 2\left(\frac{\sqrt{3}}{2}\right) - 1 \right\} = (0)(\sqrt{3} - 1) = 0 \text{ (True)}$$

Hence,

$$\begin{aligned} \text{Solution set} = & \left\{ -\frac{\pi}{6} + 2n\pi \right\} \cup \left\{ \frac{\pi}{3} + 2n\pi \right\} \cup \left\{ \frac{5\pi}{6} + 2n\pi \right\} \\ & \cup \left\{ \frac{5\pi}{3} + 2n\pi \right\} \cup \left\{ \frac{11\pi}{6} + 2n\pi \right\} \cup \left\{ \frac{7\pi}{6} + 2n\pi \right\}, n \in \mathbb{Z} \end{aligned}$$

Q. 7 (b). Two planes start from Karachi International Airport at the same time and fly in directions that make an angle of 127° with each other. Their speeds are 525km/h . How far apart they are at the end of 2 hours of flying time? (Marks 07)

SOLUTION:

Let

Angle between directions of both planes = $\alpha = 127^\circ$

Speed of both planes = $v = 525\text{kmh}^{-1}$

Time of flying = 2 hours

How far apart both plans after 2 hours = ?

$$\because v = \frac{s}{t} \Rightarrow s = vt$$

$$s = (525\text{kmh}^{-1})(2 \text{ hours})$$

$$s = 1050\text{Km}$$

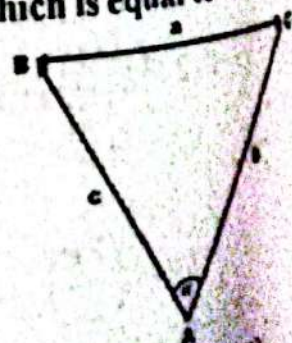
Since both planes covered an equal distance from Karachi airport in different directions at the end of two hours which is equal to 1025km .

If A represents Karachi airport and B and C represents the final position of both planes respectively and α is the angle between both directions as shown in the figure, then

Using law of cosine

$$a^2 = b^2 + c^2 - 2bc \cos \alpha$$

$$a^2 = (1050)^2 + (1050)^2 - 2(1050)(1050) \cos(127)$$



$$1102500 + 1102500 - 2205000 (-0.6018)$$

$$c^2 = 2205000 + 1327002.126$$

$$c^2 = 3532002.126$$

$$c = 11879.362 \text{ Km}$$

Planes will be 11879.362 Km apart after 2 hours of flying.

CBSE MODEL QUESTION PAPER 2

Annual Examination 2023

MATHEMATICS

CLASS XI (SCIENCE)

3 Hours

Total Marks: 100

SECTION 'A' (COMPULSORY)

MULTIPLE CHOICE QUESTIONS (M.C.Qs)

Minutes

(20 Marks)

Choose the correct answer for each of the given options:

If a complex number x , then its conjugate is:

☐ $-x$

☐ xi

☐ $-xi$

If a matrix $A = [a_{ij}]$ for which $a_{ij} = 0, i < j$, then A is

☐ Lower triangular ✓

☐ Hermitian

If vectors $\vec{a}, \vec{b}$ and $\vec{c}$ are coplanar, then $[\vec{a}, \vec{b}, \vec{c}] =$ _____

☐ 0 ✓

☐ $\vec{c}$

☐ $\vec{a}$

The harmonic mean between x and y then H is:

☐ $\frac{x+y}{2xy}$

☐ $\frac{2xy}{x+y}$ ✓

☐ $\frac{xy}{x+y}$

☐ $\frac{1}{n} + \frac{1}{n+1}$ ✓

☐ $\frac{2}{n} + \frac{1}{n+1}$

☐ $\frac{1}{n} - \frac{3}{n+1}$

The number of permutations of n objects according to some definite order is

☐ Permutation ✓

☐ none of these

The coefficient of x in the expansion of $(a + 2x)^n$ is:

☐ Middle term

☐ Last term

☐ 2nd last term

(viii) The function $f = \{(x, y) | y = mx + c\}$, m and c are re

is:

* Linear ✓

* Quadratic

* A circle

(ix) If $f(x, y) = 5x + 4y$ is an objective function and the co of the feasible region are $(5, 4)$, $(0, 0)$, $(4, 0)$ and $(0, 6)$ then is maximum at:

* $(4, 0)$

* $(0, 6)$

* $(4, 0)$

(x) $\sin(180^\circ + \theta) =$

* $-\cos\theta$

* $-\sin\theta$ ✓

* $-\tan\theta$

(xi) In any triangle ABC, $\sqrt{\frac{(s-a)(s-b)(s-c)}{ac}}$ is equal to:

* $\sin \frac{a}{2}$

* $\cos \frac{a}{2}$

* $\sin \frac{b}{2}$ ✓

(xii) $\cot^{-1} x =$

* $\frac{\pi}{2} - \tan^{-1} x$

* $\frac{\pi}{2} - \cot^{-1} x$

* $\frac{\pi}{2} + \tan^{-1} x$

(xiii) The function $y = \sin^{-1} x$, its domain is:

* $-1 \leq x \leq 1$

* $-1 < x < 1$

* $-1 \leq x \leq 1$ ✓

* $0 \leq x < 1$

(xiv) In $\triangle ABC$, $a + b - c =$

* $2s$

* $2(s - a)$

* $2(s - b)$

(xv) $1 - \cos \theta$ is:

* $2\cos^2 \frac{\theta}{2}$

* $2\tan^2 \frac{\theta}{2}$

* $2\sec^2 \frac{\theta}{2}$

(xvi) The value of $\frac{(n+1)!}{(n-1)!}$ is:

* $n(n+1)$ ✓

* $(n+1)!$

* $\frac{(n+1)}{(n-1)}$

(xvii) The harmonic mean between $2a$ and $2b$ is:

* $\pm\sqrt{ab}$

* $\frac{a+b}{2}$

* $\frac{4ab}{a+b}$ ✓

(xviii) If $n \in N$, then number of terms in $(a+b)^n$ is:

* n

* $(n-1)$

* $(n+1)$ ✓

(xix) If the coordinates of A and B are $(2, 3, 1)$ and $(3, 2, 1)$ then $\overline{AB}$ is:

* $(-1, 1, -2)$

* $(5, 5, 4)$

* $(1, -1, 2)$ ✓

(xx) If $\begin{bmatrix} 6 & \lambda \\ 3 & 2 \end{bmatrix}$ is singular matrix then $\lambda =$

* 4 ✓

* -4

* 12

SECTION 'B' (SHORT-ANSWER QUESTIONS)

(Marks: 10x5=50)

Attempt any TEN part questions from this section,
at least THREE part questions from each question.
Choose the tenth part question from any one sub-
questions carry equation marks.

NUMBER, MATRIX AND DETERMINANT, VECTORS AND FUNCTION & GRAPHS

the quadratic equation $z^2 - 4z + 5 = 0$ by completing
where z is a complex number.

See Question No. 2(i), 2023)

that the following matrix is Hermitian or Skew Hermitian

$$\begin{bmatrix} 3 & 1-2i & 4+7i \\ 1+2i & -4 & -2i \\ 4-7i & 2i & 5 \end{bmatrix}$$

$$\text{Let } A = \begin{bmatrix} 3 & 1-2i & 4+7i \\ 1+2i & -4 & -2i \\ 4-7i & 2i & 5 \end{bmatrix}$$

$$\bar{A} = \begin{bmatrix} 3 & 1-2i & 4+7i \\ 1+2i & -4 & -2i \\ 4-7i & 2i & 5 \end{bmatrix}$$

$$\bar{A} = \begin{bmatrix} 3 & 1+2i & 4-7i \\ 1-2i & -4 & 2i \\ 4+7i & -2i & 5 \end{bmatrix}$$

pose on both sides

$$(A)^t = \begin{bmatrix} 3 & 1+2i & 4-7i \\ 1-2i & -4 & 2i \\ 4+7i & -2i & 5 \end{bmatrix}$$

$$(A)^t = \begin{bmatrix} 3 & 1-2i & 4+7i \\ 1+2i & -4 & -2i \\ 4-7i & 2i & 5 \end{bmatrix}$$

$$(A)^t = A$$

matrix is Hermitian Matrix.

a, b, c are different and $\Delta = \begin{vmatrix} a & a^2 & 1+a^3 \\ b & b^2 & 1+b^3 \\ c & c^2 & 1+c^3 \end{vmatrix} = 0$, then

$$abc = 0.$$

$$\Delta = \begin{vmatrix} a & a^2 & 1+a^3 \\ b & b^2 & 1+b^3 \\ c & c^2 & 1+c^3 \end{vmatrix}$$

SOLUTION:

$$\Delta = \begin{vmatrix} a & a^2 & 1+a^3 \\ b & b^2 & 1+b^3 \\ c & c^2 & 1+c^3 \end{vmatrix}$$

According to property 7 (Corollary)

$$|A| = |B| + |C| = 0$$

So

$$\Delta_1 + \Delta_2 = \begin{vmatrix} a & a^2 & 1 \\ b & b^2 & 1 \\ c & c^2 & 1 \end{vmatrix} + \begin{vmatrix} a & a^2 & a^3 \\ b & b^2 & b^3 \\ c & c^2 & c^3 \end{vmatrix} = 0$$

Taking Common a from R_1 b from R_2 and c from R_3 of Δ_2

$$\begin{vmatrix} a & a^2 & 1 \\ b & b^2 & 1 \\ c & c^2 & 1 \end{vmatrix} + (abc) \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix} = 0$$

Interchange C_1 and C_2 of determinant Δ_2

$$\Delta_1 + \Delta_2 = \begin{vmatrix} a & a^2 & 1 \\ b & b^2 & 1 \\ c & c^2 & 1 \end{vmatrix} + \left\{ -(abc) \begin{vmatrix} a & 1 & a^2 \\ b & 1 & b^2 \\ c & 1 & c^2 \end{vmatrix} \right\}$$

Again Interchange C_2 and C_3 of determinant Δ_2

$$\Delta_1 + \Delta_2 = \begin{vmatrix} a & a^2 & 1 \\ b & b^2 & 1 \\ c & c^2 & 1 \end{vmatrix} + (abc) \begin{vmatrix} a & a^2 & 1 \\ b & b^2 & 1 \\ c & c^2 & 1 \end{vmatrix} =$$

Taking Common

$$\begin{vmatrix} a & a^2 & 1 \\ b & b^2 & 1 \\ c & c^2 & 1 \end{vmatrix} (1 + abc) = 0$$

Since a, b, c are different Numbers so

$$\begin{vmatrix} a & a^2 & 1 \\ b & b^2 & 1 \\ c & c^2 & 1 \end{vmatrix} \neq 0$$

Therefore

$$1 + abc = 0 \text{ Hence Shown}$$

Q. 2(iv). Using cross product, find the angle between

$$3\hat{i} - 2\hat{j} + 2\hat{k}$$

SOLUTION:

Let

$$\vec{a} = \hat{i} - 7\hat{j} + 7\hat{k},$$

$$\vec{b} = 3\hat{i} - 2\hat{j} + 2\hat{k}$$

the angle between two vector in cross product is

$$\sin \theta = \frac{|\vec{a} \times \vec{b}|}{|\vec{a}| |\vec{b}|}$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} i & j & k \\ 1 & -7 & 7 \\ 3 & -2 & 2 \end{vmatrix}$$

$$= i \begin{vmatrix} -7 & 7 \\ -2 & 2 \end{vmatrix} - j \begin{vmatrix} 1 & 7 \\ 3 & 2 \end{vmatrix} + k \begin{vmatrix} 1 & -7 \\ 3 & -2 \end{vmatrix}$$

$$= i\{(-7)(2) - (7)(-2)\} - j\{(1)(2) - (7)(3)\} + k\{(1)(-2) - (-7)(3)\}$$

$$= i(-14 + 14) - j(2 - 21) + k(-2 + 21)$$

$$= i(0) - j(-19) + k(19)$$

$$\vec{a} \times \vec{b} = 19j + 19k$$

$$|\vec{a} \times \vec{b}| = \sqrt{(0)^2 + (19)^2 + (19)^2}$$

$$|\vec{a} \times \vec{b}| = \sqrt{0 + 361 + 361}$$

$$|\vec{a} \times \vec{b}| = \sqrt{722}$$

$$|\vec{a} \times \vec{b}| = 19\sqrt{2}$$

$$|\vec{a}| = \sqrt{(1)^2 + (-7)^2 + (7)^2}$$

$$= \sqrt{1 + 49 + 49}$$

$$= \sqrt{99} = \sqrt{3 \times 3 \times 11}$$

$$|\vec{a}| = 3\sqrt{11}$$

$$|\vec{b}| = \sqrt{(3)^2 + (-2)^2 + (2)^2}$$

$$= \sqrt{9 + 4 + 4}$$

$$= \sqrt{17}$$

$$\sin \theta = \frac{|\vec{a} \times \vec{b}|}{|\vec{a}| |\vec{b}|} = \frac{19\sqrt{2}}{(3\sqrt{11})(\sqrt{17})}$$

$$\sin \theta = \frac{26.87}{(9.9498)(4.1231)} = \frac{26.87}{41.0}$$

$$\theta = \sin^{-1}(0.6553)$$

$$\theta = 40.94 \approx 41$$

Q. 2(v). Find the point of intersection where y is function of x
 $2x + y = 5$ and $x + y = 4$

SOLUTION:

$$2x + y = 5$$

$$y = 5 - 2x$$

$$\text{Put } x = 1$$

$$y = 5 - 2(1) = 5 - 2$$

$$x + y = 4$$

$$y = 4 - x$$

$$\text{Put } x = 1$$

$$y = 4 - 1$$

$$y = 3$$

$$\text{Put } x = 2$$

$$y = 5 - 2(2) = 5 - 4$$

$$y = 1$$

$$\text{Put } x = 3$$

$$y = 5 - 2(3) = 5 - 6$$

$$y = -1$$

x	1	2	3
y	3	1	-1

$$y = 3$$

$$\text{Put } x = 2$$

$$y = 4 - 2$$

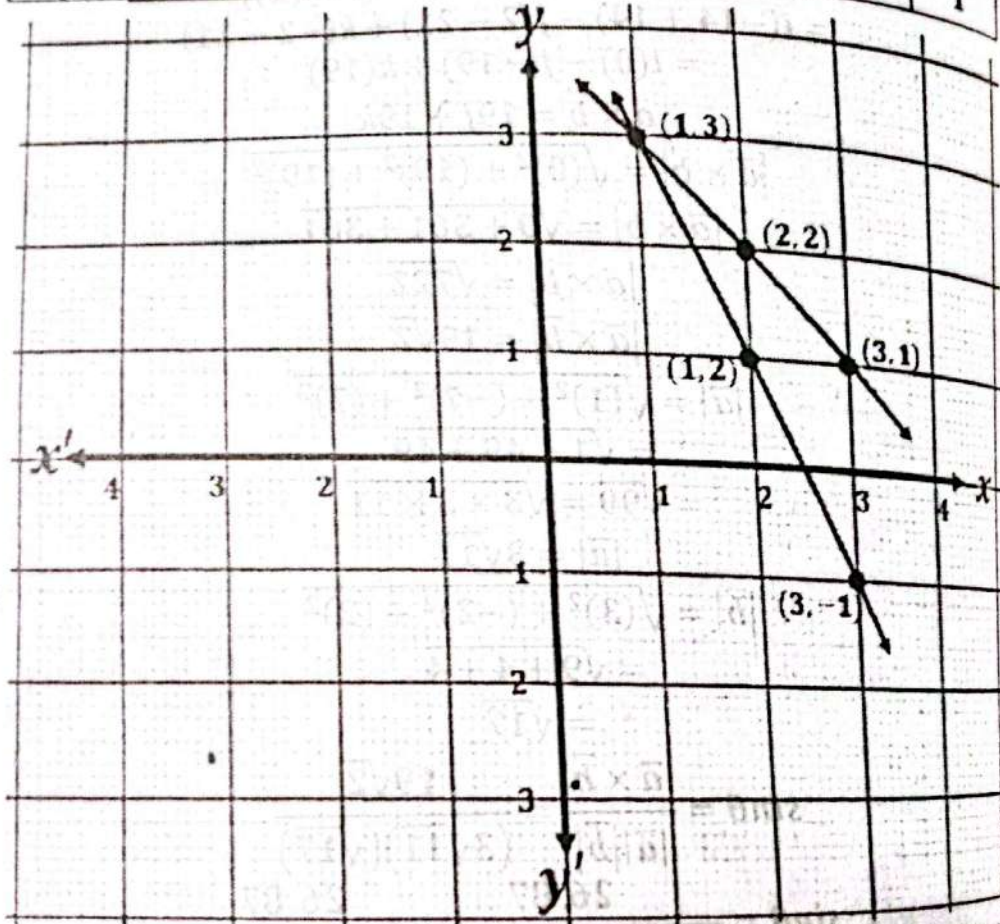
$$y = 2$$

$$\text{Put } x = 3$$

$$y = 4 - 3$$

$$y = 1$$

x	1	2	3
y	3	2	1



SEQUENCE & SERIES, PROBABILITY, MATHEMATICAL

INDUCTION & BINOMIAL THEOREM

Q. 3(i). Find the indicated terms in the following harmonic progression

$$\frac{1}{3}, \frac{1}{8}, \frac{1}{13}, \dots; 11\text{th term}$$

SOLUTION:

$$a = \frac{1}{3}$$

$$b = \frac{1}{8}$$

$$T_n = \frac{ab}{b + (n-1)(a-b)}$$

$$T_{11} = \frac{\left(\frac{1}{3}\right)\left(\frac{1}{8}\right)}{\frac{1}{8} + (11-1)\left(\frac{1}{3} - \frac{1}{8}\right)}$$

$$= \frac{\frac{1}{24}}{\frac{1}{8} + (10)\left(\frac{8-3}{24}\right)}$$

$$= \frac{\frac{1}{24}}{\frac{1}{8} + (10)\left(\frac{5}{24}\right)}$$

$$= \frac{\frac{1}{24}}{\frac{3+50}{24}}$$

$$T_{11} = \frac{1}{53} \text{ Answer}$$

By using partial fractions, Find the sum of the following series.

$$\frac{1}{3.5} + \frac{1}{5.7} + \dots \text{ to } n \text{ terms}$$

SOLUTION:

$$\frac{1}{1.3} + \frac{1}{3.5} + \frac{1}{5.7} + \dots \text{ to } n \text{ terms}$$

3, 5, 7,

Using the formula of Arithmetic Progression in the above series

$$a + (n-1)d$$

$$a = 1, d = 3 - 1 = 2$$

$$\therefore a_n = 1 + (n-1)2$$

$$a_n = 1 + 2n - 2$$

$$a_n = 2n - 1$$

$$a = 3, d = 5 - 3 = 2$$

$$\therefore a_n = 3 + (n-1)2$$

$$a_n = 3 + 2n - 2$$

$$a_n = 2n + 1$$

$$\frac{1}{1.3} + \frac{1}{3.5} + \frac{1}{5.7} + \dots \text{ to } n \text{ terms} = \frac{1}{(2n-1)(2n+1)}$$

$$\frac{1}{1.3} + \frac{1}{3.5} + \frac{1}{5.7} + \dots \text{ to } n \text{ terms} = \sum_{i=1}^n \frac{1}{(2i-1)(2i+1)}$$

Find Partial fractions of general term

$$\text{Let } \frac{1}{(2n-1)(2n+1)} = \frac{A}{2n-1} + \frac{B}{2n+1} \quad \text{--- (1)}$$

$$\text{Multiplying both sides by } (2n-1)(2n+1)$$

$$1 = A(2n+1) + B(2n-1) \quad \text{--- (2)}$$

Put $n = \frac{1}{2}$ in equation (2) to get the value of A.

$$1 = A \left\{ 2 \left(\frac{1}{2} \right) + 1 \right\} + B \left\{ 2 \left(\frac{1}{2} \right) - 1 \right\}$$

$$1 = A(1+1) + B(1-1)$$

$$1 = 2A + B(0)$$

$$1 = 2A$$

$$A = \frac{1}{2}$$

Put $n = -\frac{1}{2}$ in equation (2) to get the value of B

$$1 = A \left\{ 2 \left(-\frac{1}{2} \right) + 1 \right\} + B \left\{ 2 \left(-\frac{1}{2} \right) - 1 \right\}$$

$$1 = A(-1+1) + B(-1-1)$$

$$1 = A(0) - 2B$$

$$1 = -2B$$

$$B = -\frac{1}{2}$$

By using values of A and B in equation (1). We get

$$\frac{1}{(2n-1)(2n+1)} = \frac{\frac{1}{2}}{2n-1} + \frac{-\frac{1}{2}}{2n+1}$$

$$= \frac{1}{2(2n-1)} - \frac{1}{2(2n+1)}$$

$$= \frac{1}{2} \left[\frac{1}{(2n-1)} - \frac{1}{(2n+1)} \right]$$

Now

$$\sum_{r=1}^n \frac{1}{(2r-1)(2r+1)} = \frac{1}{2} \left[\sum_{r=1}^n \frac{1}{(2r-1)} - \sum_{r=1}^n \frac{1}{(2r+1)} \right]$$

By using theorem

$$\therefore \sum_{r=1}^n [f(r) - f(r+1)] = f(1) - f(n+1)$$

$$\therefore f(r) = \frac{1}{2r-1}$$

$$\begin{aligned}
 &= \frac{1}{2} [f(1) - f(n+1)] \\
 &= \frac{1}{2} \left[\frac{1}{2(1)-1} - \frac{1}{2n+1} \right] \\
 &= \frac{1}{2} \left[\frac{1}{2-1} - \frac{1}{2n+1} \right] \\
 &= \frac{1}{2} \left[\frac{1}{1} - \frac{1}{2n+1} \right] \\
 &= \frac{1}{2} \left[\frac{2n+1-1}{2n+1} \right] \\
 &= \frac{1}{2} \left[\frac{2n}{2n+1} \right] \\
 &= \frac{n}{2n+1} \text{ Answer}
 \end{aligned}$$

The members of a club are 12 boys and 8 girls. In how many

committee of 5 members be formed?

committee of 4 boys and 3 girls be formed?

SOLUTION:

Boys = 12

Girls = 8

No. of members = $12 + 8 = 20$

committee of 5 members be formed?

Members of committee be formed from total 20 members. i.e.

$$\text{No. of ways} = \binom{20}{5} = {}^{20}C_5$$

$$\text{No. of ways} = 15504 \text{ Answer}$$

committee of 4 boys and 3 girls be formed?

Boys can be chosen in $\binom{12}{4}$ ways and the girls in $\binom{8}{3}$ ways

By Product Principle, the required number of ways is:

$$\text{No. of ways} = \binom{12}{4} \cdot \binom{8}{3} = {}^{12}C_4 \times {}^8C_3$$

$$\text{No. of ways} = 495 \times 56$$

$$\text{No. of ways} = 27720 \text{ ways Answer}$$

A die is rolled. Find the probability using Venn diagram of

an number

(ii) A number greater than 4

number which is even and greater than 4.

SOLUTION:

Outcomes in a single rolled = $O(S) = \{1, 2, 3, 4, 5, 6\}$

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Possible outcomes of even numbers in a single rolled = $O(A) = \{2, 4, 6\} = 3$

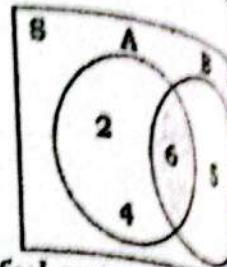
Possible outcomes of numbers greater than 4 in a single rolled = $O(B) = \{5, 6\} = 2$

Possible outcomes of even number greater than 4 in a single rolled = $O(C) = \{6\} = 1$

Probability = $P(A) = ?$

Probability = $P(B) = ?$

Probability = $P(A \cap B) = ?$



$$\therefore \text{Probability} = \frac{\text{Successful outcomes}}{\text{Possible outcomes}}$$

$$\therefore P(A) = \frac{O(A)}{O(S)} = \frac{3}{6}$$

$$P(A) = \frac{1}{2}$$

$$P(B) = \frac{O(B)}{O(S)} = \frac{2}{6}$$

$$P(B) = \frac{1}{3}$$

$$P(A \cap B) = \frac{O(A \cap B)}{O(S)} = \frac{1}{6}$$

$$P(A \cap B) = \frac{1}{6}$$

Q. 3(v). Prove that:

$2^{n+1} > (2n + 3)$, for all integral values of $n \geq 2$

SOLUTION:

$$2^{n+1} > (2n + 3)$$

Condition I:

$$2^{n+1} > (2n + 3)$$

Put $n = 2$

$$2^{2+1} > \{2(2) + 3\}$$

$$2^3 > 4 + 3$$

$$8 > 7$$

Since the given statement is true for $n = 2$

Condition II:

Suppose the given statement is true for $n = k$ i.e.

$$2^{k+1} > 2k + 3$$

Multiplying both sides of the above inequality by 2, we get

$$2 \cdot 2^{k+1} > (2k + 3) \cdot 2$$

$$2^{k+1+1} > 4k + 6$$

$$2^{k+1+1} > 2k + 2k + 4 + 2$$

$$2^{k+1+1} > 2k + 2 + 2k + 4$$

$$2^{k+1+1} > 2(k + 1) + 2(k + 2)$$

By putting $2(k + 2)$ from right side and keeping 3 instead, the statement holds and we have

$$2^{k+1+1} > 2(k + 1) + 3$$

∴ this statement is also true for $n = k + 1$

∴ by the principle of mathematical induction, it is true for all value of $n \geq 2$

NEAR PROGRAMING AND TRIGONOMETRY

Solve the following LP programming problems by graphical

when $x \geq 0, y \geq 0$.

$$Z(x, y) = 41x + 38y$$

$$\text{s.t. } 4x + 5y \leq 26; \quad 8x + 5y \leq 22; \quad 5x + 2y \leq 10$$

CONSTRAINT 1:

$$4x + 5y \leq 26$$

Consider it as an equation $4x + 5y = 26$

Intercept put $y = 0$

$$4x + 5(0) = 26 \Rightarrow 4x = 26 \Rightarrow x = \frac{26}{4}$$

$$x = \frac{13}{2}$$

Intersection point of line with x -axis is $(\frac{13}{2}, 0)$

Intercept put $x = 0$

$$4(0) + 5y = 26 \Rightarrow 5y = 26$$

$$y = \frac{26}{5}$$

Intersection point of line with y -axis is $(0, \frac{26}{5})$

put $x = 0$ and put $y = 0$ in $x + 2y \geq 10$

$$4(0) + 5(0) \leq 26$$

$$0 \leq 26 \text{ (true)}$$

CONSTRAINT 2:

$$8x + 5y \leq 22$$

Consider it as an equation $8x + 5y = 22$

For x -intercept put $y = 0$

$$8x + 5(0) = 22 \Rightarrow 8x = 22 \Rightarrow x = \frac{22}{8}$$

$$x = \frac{11}{4}$$

The intersection point of line with x -axis is $(\frac{11}{4}, 0)$

For y -intercept put $x = 0$

$$8(0) + 5y = 22 \Rightarrow 5y = 22$$

$$y = \frac{22}{5}$$

The intersection point of line with y -axis is $(0, \frac{22}{5})$

For origin put $x = 0$ and put $y = 0$ in $8x + 5y \leq 22$

$$8(0) + 5(0) \leq 22$$

$$0 \leq 22 \text{ (true)}$$

CONSTRAINT 3:

$$5x + 2y \leq 10$$

We consider it as an equation $5x + 2y = 10$

For x -intercept put $y = 0$

$$5x + 2(0) = 10 \Rightarrow 5x = 10 \Rightarrow x = \frac{10}{5}$$

$$x = 2$$

The intersection point of line with x -axis is $(2, 0)$

For y -intercept put $x = 0$

$$5(0) + 2y = 10 \Rightarrow 2y = 10 \Rightarrow y = \frac{10}{2}$$

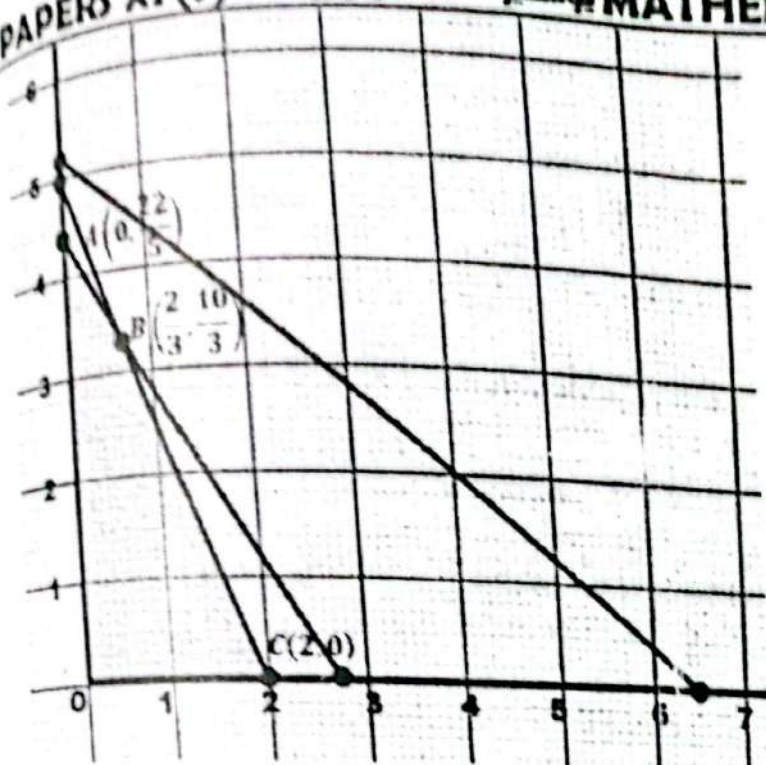
$$y = 5$$

The intersection point of line with y -axis is $(0, 5)$

For origin put $x = 0$ and put $y = 0$ in $5x + 2y \leq 10$

$$5(0) + 2(0) \leq 10$$

$$0 \leq 10 \text{ (true)}$$



Intersection of two linear inequalities is the required feasible region OABC which is the shaded area in the graph.

Solution points or corner points of the region OABC are

$O(0,0)$, $A(0, \frac{22}{5})$, $B(\frac{2}{3}, \frac{10}{3})$, and $C(2,0)$ in the graph

OBJECTIVE FUNCTION

SOLUTION POINTS OR CORNER POINTS	OBJECTIVE FUNCTION $f(x, y) = 41x + 38y$
$O(0,0)$	$f(x, y) = 41(0) + 38(0) = 0 + 0 = 0$
$A(0, \frac{22}{5})$	$f(x, y) = 41(0) + 38(\frac{22}{5})$ $= 0 + \frac{386}{5} = 167.2$
$B(\frac{2}{3}, \frac{10}{3})$	$f(x, y) = 41(\frac{2}{3}) + 38(\frac{10}{3}) = \frac{82}{3} + \frac{380}{3}$ $= \frac{462}{3} = 154$
$C(2,0)$	$f(x, y) = 41(2) + 38(0) = 82 + 0 = 82$

Maximum at $(0, \frac{22}{5})$; Maximum value = 167.2 Answer

(ii). Prove that: $\frac{\tan \alpha}{\cot(90^\circ - \alpha)} + \frac{\cos(90^\circ - \alpha)}{\sin \alpha} = 2$

SOLUTION:

$$\frac{\tan \alpha}{\cot(90^\circ - \alpha)} + \frac{\cos(90^\circ - \alpha)}{\sin \alpha} = 2$$

Taking L.H.S.:

$$\begin{aligned} &= \frac{\tan \alpha}{\cot(90^\circ - \alpha)} + \frac{\cos(90^\circ - \alpha)}{\sin \alpha} \\ \because \cot(90^\circ - \theta) &= \tan \theta \text{ and } \cos(90^\circ - \theta) = \sin \theta \\ &= \frac{\tan \alpha}{\tan \alpha} + \frac{\sin \alpha}{\sin \alpha} = 1 + 1 \\ &= 2 \end{aligned}$$

Hence Proved L.H.S. = R.H.S.

Q. 4(iii) Prove that:

(a). $\cos 4\theta = 8\cos^4\theta - 8\cos^2\theta + 1$

(b). $\frac{\sin 2\theta + \sin 4\theta + \sin 6\theta + \sin 8\theta}{\cos 2\theta + \cos 4\theta + \cos 6\theta + \cos 8\theta} = \tan 5\theta$

(a). $\cos 4\theta = 8\cos^4\theta - 8\cos^2\theta + 1$

SOLUTION:

$$\cos 4\theta = 8\cos^4\theta - 8\cos^2\theta + 1$$

Taking L.H.S:

$$\cos 4\theta = \cos(2\theta + 2\theta)$$

$$\begin{aligned} \because \cos(\alpha + \beta) &= \cos \alpha \cos \beta - \sin \alpha \sin \beta \\ &= \cos 2\theta \cos 2\theta - \sin 2\theta \sin 2\theta \\ &= (\cos 2\theta)^2 - (\sin 2\theta)^2 \end{aligned}$$

$$\because \sin 2\theta = 2\sin \theta \cos \theta$$

$$\because \cos 2\theta = 2\cos^2\theta - 1$$

$$\begin{aligned} &= (2\cos^2\theta - 1)^2 - (2\sin \theta \cos \theta)^2 \\ &= (2\cos^2\theta)^2 - 2(2\cos^2\theta)(1) + (1)^2 - (2\sin \theta \cos \theta)^2 \\ &= 4\cos^4\theta - 4\cos^2\theta + 1 - 4\sin^2\theta \cos^2\theta \end{aligned}$$

$$\because \sin^2\theta = 1 - \cos^2\theta$$

$$\begin{aligned} &= 4\cos^4\theta - 4\cos^2\theta + 1 - 4(1 - \cos^2\theta)\cos^2\theta \\ &= 4\cos^4\theta - 4\cos^2\theta + 1 - (4\cos^2\theta - 4\cos^4\theta) \\ &= 4\cos^4\theta - 4\cos^2\theta + 1 - 4\cos^2\theta + 4\cos^4\theta \\ &= 8\cos^4\theta - 8\cos^2\theta + 1 \end{aligned}$$

Hence Proved L.H.S. = R.H.S.

(b). $\frac{\sin 2\theta + \sin 4\theta + \sin 6\theta + \sin 8\theta}{\cos 2\theta + \cos 4\theta + \cos 6\theta + \cos 8\theta} = \tan 5\theta$

SOLUTION:

$$\frac{\sin 2\theta + \sin 4\theta + \sin 6\theta + \sin 8\theta}{\cos 2\theta + \cos 4\theta + \cos 6\theta + \cos 8\theta} = \tan 5\theta$$

Taking L.H.S:

$$\begin{aligned}
 &= \frac{\sin 2\theta + \sin 4\theta + \sin 6\theta + \sin 8\theta}{\cos 2\theta + \cos 4\theta + \cos 6\theta + \cos 8\theta} \\
 &= \frac{\sin 8\theta + \sin 2\theta + \sin 6\theta + \sin 4\theta}{\cos 8\theta + \cos 2\theta + \cos 6\theta + \cos 4\theta} \\
 &= \frac{(\sin 8\theta + \sin 2\theta) + (\sin 6\theta + \sin 4\theta)}{(\cos 8\theta + \cos 2\theta) + (\cos 6\theta + \cos 4\theta)}
 \end{aligned}$$

$$\because \sin u + \sin v = 2 \sin \frac{u+v}{2} \cos \frac{u-v}{2}$$

$$\because \cos u + \cos v = 2 \cos \frac{u+v}{2} \cos \frac{u-v}{2}$$

$$\begin{aligned}
 &= \frac{2 \sin \frac{8\theta+2\theta}{2} \cos \frac{8\theta-2\theta}{2} + (2 \sin \frac{6\theta+4\theta}{2} \cos \frac{6\theta-4\theta}{2})}{2 \cos \frac{8\theta+2\theta}{2} \cos \frac{8\theta-2\theta}{2} + (2 \cos \frac{6\theta+4\theta}{2} \cos \frac{6\theta-4\theta}{2})} \\
 &= \frac{2 \sin \frac{10\theta}{2} \cos \frac{6\theta}{2} + 2 \sin \frac{10\theta}{2} \cos \frac{2\theta}{2}}{2 \cos \frac{10\theta}{2} \cos \frac{6\theta}{2} + 2 \cos \frac{10\theta}{2} \cos \frac{2\theta}{2}}
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{2 \sin 5\theta \cos 3\theta + 2 \sin 5\theta \cos \theta}{2 \cos 5\theta \cos 3\theta + 2 \cos 5\theta \cos \theta} \\
 &= \frac{\sin 5\theta (2 \cos 3\theta + 2 \cos \theta)}{\cos 5\theta (2 \cos 3\theta + 2 \cos \theta)} \\
 &= \frac{\sin 5\theta}{\cos 5\theta}
 \end{aligned}$$

$$= \tan 5\theta$$

$$\because \frac{\sin \theta}{\cos \theta} = \tan \theta$$

Hence Proved L.H.S. = R.H.S.

Find the area of triangle ABC when

$$a = 41.34, b = 35.65, c = 56.81$$

ON:

$$a = 41.34, \quad b = 35.65, \quad c = 56.81$$

$$\because \Delta = \sqrt{s(s-a)(s-b)(s-c)}$$

$$\because s = \frac{1}{2}(a+b+c)$$

$$s = \frac{1}{2}(41.34 + 35.65 + 56.81)$$

$$s = \frac{1}{2}(133.8)$$

$$s = 66.9$$

$$66.9(66.9 - 41.34)(66.9 - 35.65)(66.9 - 56.81)$$

$$= \sqrt{66.9(25.56)(31.25)(10.09)}$$

$$= \sqrt{5391730.238}$$

$$\Delta = 734.284 \text{ Square units Answer}$$

Q. 4(v). Find the solution set of the equation $\sin x + \frac{1}{\sqrt{2}} = 0$, by using properties when $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$.

SOLUTION:

$$\sin x + \frac{1}{\sqrt{2}} = 0$$

$$\sin x = -\frac{1}{\sqrt{2}}$$

$$x = \sin^{-1}\left(-\frac{1}{\sqrt{2}}\right)$$

$$x = -\frac{\pi}{4}$$

∵ Sine function is negative in 3rd and 4th quadrants, hence

FOR 3RD QUADRANTS

$$\begin{aligned} \therefore x &= \pi + x \\ &= \pi + \left(-\frac{\pi}{4}\right) \\ &= \frac{4\pi - \pi}{4} \\ &= \frac{3\pi}{4} \\ x &= \frac{3\pi}{4} \end{aligned}$$

Neglecting this value because it lies in 2nd quadrant.

FOR 4TH QUADRANTS

$$\begin{aligned} \therefore x &= 2\pi - x \\ &= 2\pi - \left(-\frac{\pi}{4}\right) \\ &= \frac{4\pi + \pi}{4} \\ &= \frac{5\pi}{4} \\ x &= \frac{5\pi}{4} \end{aligned}$$

$$\begin{aligned} x &= \frac{3\pi}{2} - x \\ &= \frac{3\pi}{2} - \left(-\frac{\pi}{4}\right) \\ &= \frac{6\pi + \pi}{4} \\ &= \frac{7\pi}{4} \\ x &= \frac{7\pi}{4} \end{aligned}$$

$$\begin{aligned} x &= \frac{3\pi}{2} + x \\ &= \frac{3\pi}{2} + \left(-\frac{\pi}{4}\right) \\ &= \frac{6\pi - \pi}{4} \\ &= \frac{5\pi}{4} \\ x &= \frac{5\pi}{4} \end{aligned}$$

Thus

$$x = \frac{5\pi}{4} + 2n\pi \text{ and } x = \frac{7\pi}{4} + 2n\pi$$

Hence,

the solution set = $\left(\frac{5\pi}{4} + 2n\pi\right) \cup \left(\frac{7\pi}{4} + 2n\pi\right) n \in \mathbb{Z}$

SECTION 'C' (DESCRIPTIVE-ANSWER QUESTIONS)
(Marks: 15x2=30)

NOTE: Attempt any TWO questions. All questions carry equal marks.

Q.1. Solve the non-homogeneous system of linear equations using elimination method. (Marks 08)

2y + z = 8

2x - y + z = 3

x + y - z = 0

2x - y + z = 3

x + y - z = 0

SOLUTION:

2y + z = 8

Convert the system of the linear equation into matrix form of AX =

where $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & -1 & 1 \\ 1 & 1 & -1 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} 8 \\ 3 \\ 0 \end{bmatrix}$

Form the augmented matrix by including the constant elements as extra column in the coefficient matrix

Augmented matrix = $A_B = \begin{bmatrix} 1 & 2 & 1 & 8 \\ 2 & -1 & 1 & 3 \\ 1 & 1 & -1 & 0 \end{bmatrix}$

$A_B = \begin{bmatrix} 1 & 2 & 1 & 8 \\ 2 & -1 & 1 & 3 \\ 1 & 1 & -1 & 0 \end{bmatrix}$

Convert augmented matrix into echelon form by using elementary operations.

$A_B \sim \begin{bmatrix} 1 & 2 & 1 & 8 \\ 2 & -1 & 1 & 3 \\ 1 & 1 & -1 & 0 \end{bmatrix}$ by $R_2 + (-2)R_1$ and $R_3 - R_1$

$\sim \begin{bmatrix} 1 & 2 & 1 & 8 \\ 0 & -5 & -1 & -13 \\ 0 & -1 & -2 & -8 \end{bmatrix}$ by $\left(-\frac{1}{5}\right)R_2$

$$\sim \begin{bmatrix} 1 & 2 & 1 & 8 \\ 0 & 1 & \frac{1}{5} & \frac{13}{5} \\ 0 & -1 + 1 & -2 + \frac{1}{5} & -8 + \frac{13}{5} \end{bmatrix} \text{ by } R_3 + R_2$$

$$\sim \begin{bmatrix} 1 & 2 & 1 & 8 \\ 0 & 1 & \frac{1}{5} & \frac{13}{5} \\ 0 & 0 & -\frac{9}{5} \left(-\frac{5}{9}\right) & -\frac{27}{5} \left(-\frac{5}{9}\right) \end{bmatrix} \text{ by } \left(-\frac{5}{9}\right) R_3$$

$$\sim \begin{bmatrix} 1 & 2 & 1 & 8 \\ 0 & 1 & \frac{1}{5} & \frac{13}{5} \\ 0 & 0 & 1 & 3 \end{bmatrix}$$

From the above matrix, we get equivalent matrix equation

$$x + 2y + z = 8 \quad \text{--- (i)}$$

$$0x + y + \frac{1}{5}z = \frac{13}{5} \Rightarrow y + \frac{1}{5}z = \frac{13}{5} \quad \text{--- (ii)}$$

$$0x + 0y + z = 3 \Rightarrow z = 3 \quad \text{--- (iii)}$$

Putting the value of 'z' in equation (ii), we get

$$y + \frac{1}{5}(3) = \frac{13}{5}$$

$$\frac{5y + 3}{5} = \frac{13}{5} \Rightarrow 5y + 3 = 13$$

$$5y = 13 - 3 \Rightarrow y = \frac{10}{5}$$

$$y = 2$$

Now, putting the value of 'y' and 'z' in equation (i), we get

$$x + 2(2) + (3) = 8$$

$$x + 4 + 3 = 8 \Rightarrow x = 8 - 7$$

$$x = 1$$

Thus $x = 1$, $y = 2$, and $z = 3$ is the required solution **Answer**

Q. 5 (b). Find the volume of tetrahedron whose vertices are $A(2, 1, 7)$, $B(5, -1, 3)$, $C(4, 3, 5)$ and $D(0, 2, 3)$ (Marks 07)

SOLUTION:

$$A = (2, 1, 7)$$

$$B = (5, -1, 3)$$

$$C = (4, 3, 5)$$

$$D = (0, 2, 3)$$

$$\therefore \text{Volume of tetrahedron} = \frac{1}{6} [\overrightarrow{AB} \quad \overrightarrow{AC} \quad \overrightarrow{AD}]$$

$$\vec{B} - \vec{A} = (5, -1, 3) - (2, 1, 7) = 5 - 2, -1 - 1, 3 - 7$$

$$\vec{AB} = (3, -2, -4)$$

$$\vec{C} - \vec{A} = (4, 3, 5) - (2, 1, 7) = 4 - 2, 3 - 1, 5 - 7$$

$$\vec{AC} = (2, 2, -2)$$

$$\vec{D} - \vec{A} = (0, 2, 3) - (2, 1, 7) = 0 - 2, 2 - 1, 3 - 7$$

$$\vec{AD} = (-2, 1, -4)$$

$$\frac{1}{6} [\vec{AB} \quad \vec{AC} \quad \vec{AD}] = \frac{1}{6} \begin{vmatrix} 3 & -2 & -4 \\ 2 & 2 & -2 \\ -2 & 1 & -4 \end{vmatrix}$$

$$= \frac{1}{6} \left[3 \begin{vmatrix} 2 & -2 \\ 1 & -4 \end{vmatrix} + 2 \begin{vmatrix} 2 & -2 \\ -2 & -4 \end{vmatrix} - 4 \begin{vmatrix} 2 & 2 \\ -2 & 1 \end{vmatrix} \right]$$

$$= \frac{1}{6} [3[(2)(-4) - (-2)(1)] + 2[(2)(-4) - (-2)(-2)] - 4[(2)(1) - (2)(-2)]]$$

$$= \frac{1}{6} [3(-8 + 2) + 2(-8 - 4) - 4(2 + 4)]$$

$$= \frac{1}{6} [3(-6) + 2(-12) - 4(6)] = \frac{1}{6} (-18 - 24 - 24)$$

$$= \frac{1}{6} (-66) = \frac{1}{6} |-66|$$

$$= \frac{1}{6} (66)$$

Volume of tetrahedron = 11 Answer

4(a). Use binomial theorem to show that: (Marks 08)

$$\frac{1}{4} + \frac{13}{4 \cdot 6} + \frac{1 \cdot 3 \cdot 5}{4 \cdot 6 \cdot 8} = \sqrt[3]{4}$$

SOLUTION: (See Question No. 6, 2024 P.E.)

6(b). A business man earned Rs. 10,000 in the first month, Rs. 15,000 in the second month, Rs. 22,500 in the third month and so on. Find the total amount he will earn in 12 months. (Marks 07)

SOLUTION:

Amount earned in first month = $a_1 = \text{Rs. } 10,000$

Amount earned in second month = $a_2 = \text{Rs. } 15,000$

Amount earned in third month = $a_3 = \text{Rs. } 22,500$

Amount earned in third month = $a_n = 12$

Amount earned in 12 months = $S_n = ?$

As the series is in geometric series we get

$$10,000 + 15,000 + 22,500 + \dots a_n$$

$$a = a_1 = 10000 \text{ and } r = \frac{a_2}{a_1} = \frac{15000}{10000} = \frac{3}{2} > 1$$

Therefore

$$\begin{aligned} \therefore S_n &= \frac{a(r^n - 1)}{r - 1} \\ \therefore S_{12} &= \frac{10000 \left\{ \left(\frac{3}{2} \right)^{12} - 1 \right\}}{\frac{3}{2} - 1} = \frac{10000 \left(\frac{531441}{4096} - 1 \right)}{\frac{3-2}{2}} \\ &= \frac{10000 \left(\frac{531441 - 4096}{4096} \right)}{\frac{1}{2}} = 10000 \left(\frac{527345}{4096} \right) \times 2 \\ &= 20000(128.7463) \\ &= 2574926.75 \approx 2575,000 \text{ Rupees} \end{aligned}$$

Hence, the total amount in 12 months is 2,575,000 Rupees. Answer
Q. 7 (a). Find and verify the general solution of the following trigonometric equation.

$$4 \cos^2 x - 8 \sin x + 1 = 0$$

(Marks 08)

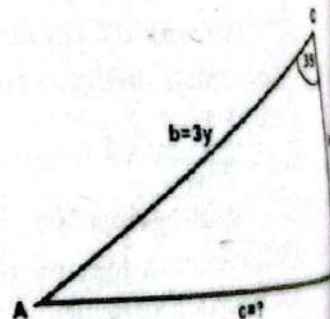
ANSWER: (See Question No. 10, 2024 G.S.)

Q. 7 (b). If one side of a triangle is y units long, another side is 3 times as long and the angle between the two sides measure 35° , find the measures of other two angles and the third side. (Marks 07)

SOLUTION:

Let a be the first side which is equal to y units and b be the second side which is equal to the $3y$, while the angle between these two sides is 35° as shown in figure, then

Using law of cosine



$$c^2 = a^2 + b^2 - 2ab \cos \gamma$$

Here

$$\begin{aligned} a &= y, \quad b = 3y, \quad \gamma = 35^\circ \\ c^2 &= (y)^2 + (3y)^2 - 2(y)(3y) \cos(35^\circ) \\ c^2 &= y^2 + 9y^2 - 6y^2 (0.8191) \\ c^2 &= 10y^2 - 4.914y^2 \\ c^2 &= 5.085y^2 \end{aligned}$$

Taking square root on both sides
 $c = 2.255 \text{ units}$ Answer